

New Techniques for Analyzing Differentials with Application to AES

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Abstract

We propose new techniques for estimating the probability that an input difference leads to an output difference in a block cipher (i.e., the probability of a differential) under the assumption of independent round-keys. We apply our techniques to AES, and show that the probability of every non-trivial differential in 8-round AES is within an additive factor of $2^{-128} \cdot \frac{1}{50}$ from the expected value of $\frac{1}{2^{128}-1}$.

We further apply our techniques to prove that 40-round AES is at most 2^{-135} -close to a pairwise independent permutation. This improves upon the work of Liu, Tessaro and Vaikuntanathan [CRYPTO 2021], who proved a similar bound for 9000-round AES.

To obtain our results, we develop and adapt a variety of techniques for analyzing differentials using functional analysis. We expect these techniques to be useful for analyzing differentials in additional block ciphers besides the AES.

1 Introduction

Differential cryptanalysis [BS91] is one of the most powerful attacks on modern block ciphers. It was originally applied to DES [Nat77] as well as to many other ciphers. Consequently, the Advanced Encryption Standard (AES) [DR99, DR20] was designed to resist this attack.

When applying differential cryptanalysis to an iterated block cipher, one starts by devising a distinguisher against a part of the cipher. A standard distinguisher exploits a high-probability *differential*, namely an input difference that leads to an output difference with higher probability than expected in a random keyed permutation. The probability of a differential is analyzed using *differential characteristics*, each specifying a particular way that the input difference may propagate to the output difference through the internal components of the cipher. Thus, estimating the probability of a differential typically involves summing up the probabilities of the most likely differential characteristics.

A standard simplifying assumption in the probability estimation of a differential characteristic for an iterated block cipher is that the round-keys are chosen independently, whereas in practice they are derived from the master-key according to the key-schedule. Under this assumption, the probability of a characteristic is simply the product of the propagation probabilities through all the internal rounds. Moreover, in order to analyze differential attacks, where the key is chosen once and fixed, one typically assumes that the actual probability of a characteristic for almost any fixed key is very close to the average probability (where the randomness is taken over the choice of plaintexts). This assumption is known as the *hypothesis of stochastic equivalence* of the *Markov cipher model*, formalized by Lai, Massey and Murphy [LMM91].

Through many years of research, we have gained a better understanding of differential attacks beyond the Markov cipher model. Examples include early works such as [DGV93] by Daemen, Govaerts and Vandewalle, who described a weakness in the key-schedule of the block cipher IDEA [LMM91]. The more recent work by Beyne and Rijmen [BR22] (which extends [DGV94]) develops a theory and investigates cases where the Markov cipher model (in particular, its hypothesis of stochastic equivalence) provides inaccurate results and may even lead to invalid attacks.

1.1 Security Arguments Against Differential Attacks

From the designer’s viewpoint (i.e., provable security), a logical first step towards arguing that a block cipher resists differential cryptanalysis is to prove that every differential has low probability assuming independent round-keys. This implies resistance against differential attacks in the Markov cipher model. Despite the inherent limitations of the Markov cipher model, this is still valuable, as it establishes a security argument for the cipher in resisting differential cryptanalysis independently of the key-schedule. Unfortunately, as detailed below, we are still far from accomplishing this first step for most block ciphers, and much more work is needed in this area.

A standard way to argue that a block cipher resists differential attacks is to show that all differential characteristics have very low probability assuming independent round-keys. This property is typically not difficult to show for a well-designed block cipher. Yet, this approach has the obvious flaw that it does not estimate the actual probability of any particular differential, which in principle, could be made large by a huge number of characteristics, each having negligible probability. This is not only a hypothetical possibility as such a significant “clustering effect” has been shown in several cases (c.f., [AK18, LPS21]).

In 1992, Nyberg and Knudsen [NK92, NK95] tackled this obstacle by initiating the line-of-work of formally proving security against differential attacks in the Markov cipher model assuming independent round-keys. Specifically, they built a specialized block cipher where the maximal probability of any differential (known as the maximal expected differential probability, or *MEDP*) was bounded by 2^{2-b} , where b is the block size.

A more recent example of such a dedicated block cipher (named BISON) was published in 2019 by Canteaut et al. [CLL⁺19], who proved a bound of 2^{1-b} on the MEDP of the cipher (with at least b internal rounds). They also proved that every non-zero output difference is possible. This is important in arguing resistance against impossible differential cryptanalysis [BBS05], which exploits a sufficiently large set of differentials with probability 0 (or very close to 0). Unfortunately, BISON is built differently from modern block ciphers and is inherently inefficient.

For modern block ciphers, one can typically bound the MEDP for a small number of rounds, obtaining a limited result.

The AES. Shortly after the AES became a standard, a series of works (such as [HLL⁺00, KMT01a, KMT01b, PSC⁺02, CKL⁺03, PSLL03, CSK⁺04]) investigated the MEDP of round-reduced AES variants (and related SP-network designs) under the common assumption that the round-keys are independent. The best known bound was obtained in 2005 by Keliher and Sui [KS07], who proved that the MEDP of 4-round AES is at most (about) 2^{-114} , slightly improving over the previously known bound obtained in [PSLL03].

While more research was dedicated to understanding differentials for round-reduced AES and related designs in various models (c.f., [DR06, DR07, DLP⁺09, CR15, BFL⁺23]), the bound by Keliher and Sui is still the best-known for any number of AES rounds (used in practice), assuming independent round-keys. In the Markov cipher model, this bound implies some resistance against

classical differential cryptanalysis (with one dominant differential) and no resistance against impossible differential cryptanalysis. Yet, impossible differential cryptanalysis is one of the most powerful attacks against round-reduced AES [LDKK08, MDRM10, BLNS18], hence it is particularly desirable to prove resistance against such attacks.

1.2 Pairwise Independence and General Provable Security

The most general security notion against an attacker that considers pairwise relations among the ciphertexts is closeness to pairwise (2-wise) independence. It asserts that for any pair of plaintexts, the distribution of the ciphertexts is close in statistical distance to the distribution of a uniformly chosen permutation, applied to the plaintexts.

Consequently, Liu, Tessaro and Vaikuntanathan [LTV21] studied the closeness to 2-wise independence of the AES, and proved that 9000 rounds of the cipher are at most 2^{-128} -close to 2-wise independence. Unfortunately, this is far from the number used in practice, which is at most 14 (for AES-256). The followup work [LPTV23] considered a modified variant of AES and showed that 192 rounds are sufficient for 2^{-128} -closeness. Yet, this result does not apply to the original AES, and the number of rounds is still not close to 14 rounds. We remark that [LTV21, LPTV23] also considered a stronger security notion of closeness to t -wise independence.

More generally, the provable security of theoretical constructions of block ciphers was first studied in [Vau03] and further explored in [MV15]. We also mention [HLLT21, BHL25], which studied a form of provable security against integral attacks assuming independent round-keys.

1.3 Our Results

Considering all the previous works, it seems that after more than 30 years of research, we have gained a better understanding of differential attacks. Yet, from a provable security perspective, we still do not have sufficiently good tools to tackle the basic problem of estimating the probability of a differential for modern block ciphers (and particularly for the AES), even when assuming independent round-keys. We also lack sufficiently strong techniques to address the related problem of proving closeness to 2-wise independence for practical block cipher constructions.

In this work, we attempt to narrow this gap by developing a set of techniques for addressing the two problems above. We apply our techniques to the AES, and prove two main results. First, for any integer $r \geq 2$, we show that any non-zero differential in $4r$ -round AES has probability in the range

$$\frac{1}{2^{128}-1} \pm 2^{-128} \cdot \frac{1}{50} \cdot 2^{-14.65 \cdot (r-2)}. \quad (1)$$

Specifically, for 8-round AES, any differential has probability in the range

$$\frac{1}{2^{128}-1} \pm 2^{-128} \cdot \frac{1}{50}. \quad (2)$$

Note that the probability of any non-zero differential in a 2-wise independent keyed permutation is $\frac{1}{2^{128}-1}$. This not only bounds the MEDP of 8-round AES and shows that all differentials are possible, it also proves that the probability of any differential is within the additive factor $2^{-128} \cdot \frac{1}{50}$ of the ideal probability. We are not aware of any previous result that proved such a strong bound for any block cipher (even for the dedicated ones [NK95, CLL⁺19]). In the Markov cipher model, this result rules out classical and impossible differential distinguishers for 8-round AES. Since all 3 AES variants have more than 8 rounds, we consider this as the main result of the paper.

Second, we prove that for any integer $r \geq 1$, $4r$ -round AES is

$$(2^{-3.8} \cdot 2^{-14.65 \cdot (r-1)})\text{-close} \quad (3)$$

to 2-wise independence. In particular, 8-round AES is 2^{-18} -close to 2-wise independence, 12-round AES is 2^{-33} -close, and 40-round AES is 2^{-135} -close. These numbers may be compared to those of [LTV21], which proved that 12-round AES is about 0.44-close to 2-wise independence, and 9000-round AES is 2^{-128} -close.

We note that if the 2-wise independence bound is sufficiently larger than 2^{-128} for a specific number of rounds, both the results (1) and (3) are of interest. Indeed, these bounds give different types of information about the difference distribution table (DDT), which can be used to lower bound the complexity of different types of attacks.

Finally, as we analyze differentials in the worst-case, we can improve the bound $2^{-113.09}$ of [KS07] on the MEDP of 4-round AES by a modest factor of about 11 to $2^{-116.54}$.

Independently, Beyne, Leander, and Schütt [BLS25] also analyzed the 2-wise independence of AES. In the next section, we give an overview of our techniques, and compare the results of [BLS25] to ours.

2 Technical Overview

Starting from some non-zero initial input difference, we treat the block cipher as a Markov chain, which repeatedly applies a fixed operation in each round. We view a distribution on differences at any intermediate state of the cipher as a vector over $\mathbb{R}$. Our goal is to show that this distribution converges quickly to the uniform distribution in some distance metric.

The convergence rate of a Markov chain typically changes as it approaches the stationary distribution. In order to obtain good bounds on the convergence rate, it is crucial to understand it at all stages.

The techniques developed in this paper focus on the convergence rate of a differential distribution through the first few rounds of the cipher, although they can be lifted to bound it at any stage. A good bound on the initial convergence rate is particularly important to obtain meaningful results for a small number of rounds that is used in practice.

In order to obtain our results, we develop and adapt a variety of tools and techniques for analyzing differentials using functional analysis. While none of these techniques is particularly innovative in its own right, the main novelty of this work is in the way the techniques are combined in differential analysis of the AES. We now summarize our techniques.

2.1 Starting Point

Given a b -bit block cipher E , suppose we wish to estimate the probability (over the choice of secret key) that a given non-zero input difference $y \in \mathbb{F}_2^b$ propagates to a given non-zero output difference $z \in \mathbb{F}_2^b$, denoted $\Pr[y \xrightarrow{E} z]$. Our goal is to bound $|\Pr[y \xrightarrow{E} z] - \frac{1}{2^{b-1}}|$, which measures the deviation of $\Pr[y \xrightarrow{E} z]$ from the ideal probability, $\frac{1}{2^{b-1}}$. The following is an adaptation of a “meet-in-the-middle” technique in the theory of Markov chains (c.f., [MT05, Appendix]). Suppose we can write E as a composition of two independent ciphers $E = E_2 \circ E_1$. Then, $\Pr[y \xrightarrow{E} z] = \sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[y \xrightarrow{E_1} x] \Pr[x \xrightarrow{E_2} z]$. Since E_2 is a permutation, then $\Pr[x \xrightarrow{E_2} z] = \Pr[z \xrightarrow{(E_2)^{-1}} x]$,

where $(E_2)^{-1}$ is the decryption function of E_2 . Hence,

$$\begin{aligned}
& \left| \Pr[y \xrightarrow{E} z] - \frac{1}{2^b-1} \right| = \left| \sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[y \xrightarrow{E_1} x] \Pr[z \xrightarrow{(E_2)^{-1}} x] - \frac{1}{2^b-1} \right| \\
& = \left| \sum_{x \in \mathbb{F}_2^b, x \neq 0} \left(\Pr[y \xrightarrow{E_1} x] - \frac{1}{2^b-1} \right) \left(\Pr[z \xrightarrow{(E_2)^{-1}} x] - \frac{1}{2^b-1} \right) \right| \\
& \leq \sqrt{\sum_{x \in \mathbb{F}_2^b, x \neq 0} \left(\Pr[y \xrightarrow{E_1} x] - \frac{1}{2^b-1} \right)^2} \sqrt{\sum_{x \in \mathbb{F}_2^b, x \neq 0} \left(\Pr[z \xrightarrow{(E_2)^{-1}} x] - \frac{1}{2^b-1} \right)^2} \\
& = \sqrt{\sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[y \xrightarrow{E_1} x]^2 - \frac{1}{2^b-1}} \sqrt{\sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[z \xrightarrow{(E_2)^{-1}} x]^2 - \frac{1}{2^b-1}}.
\end{aligned} \tag{4}$$

The second and last equalities are verified by opening up the product, and using the fact that $\sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[y \xrightarrow{E_1} x] = 1$ (and similarly for $(E_2)^{-1}$). The inequality is by Cauchy-Schwarz.¹

The expression $\sum_{x \in \mathbb{F}_2^b, x \neq 0} \Pr[y \xrightarrow{E_1} x]^2$ is the 2-norm squared, or the collision probability, of the distribution of differentials at the output of E_1 , where the input difference is y (and similarly for the second expression). Thus, we have reduced the problem of estimating the very fine-grained probability of a single rare event to the problem of bounding the 2-norm of two distributions.

2.2 General Approach

Recall that we view a distribution on differences at any intermediate state of the cipher as a vector over $\mathbb{R}$. Each internal operation of the cipher is considered as a linear operator (or a stochastic matrix) that acts on the vector via a matrix-vector product. In our case, each matrix is a (normalized) difference distribution table (DDT) that specifies how an input difference is mapped to a distribution on output differences by the corresponding operation of the cipher.

Our goal is to prove that each application of a round (or a few rounds) of the AES *contracts* the 2-norm of its input difference distribution. More accurately, in each round (or few rounds), the 2-norm of the input difference distribution becomes closer to the 2-norm of the uniform distribution over all non-zero differences,² which is exactly $\frac{1}{2^b-1}$ for a b -bit cipher ($b = 128$ for the AES). Thus, the expression in (4) becomes smaller.

We need to show that this contraction happens very quickly, starting from a distribution that is concentrated on a single input difference y (and has a 2-norm of 1). We achieve this by analyzing how the different components of the cipher contract this distribution.

2.3 Norm Bounds for the AES Sbox and Super-Sbox

We consider the keyed AES Sbox, which is a keyed permutation on a space of 8-bit words. We first compute its DDT, denoted by $\mathfrak{D}_{\text{AES}_S}$, and denote by $\mathfrak{D}_{\text{AES}_S}^-$ the 255×255 matrix obtained by truncating the row and column that correspond to the 0 input difference and 0 output difference. Using simple linear algebra, we compute the second largest eigenvalue of the matrix $(\mathfrak{D}_{\text{AES}_S}^-)^T \mathfrak{D}_{\text{AES}_S}^-$, denoted by λ (its value is $(\frac{18}{256})^2 \approx \frac{1}{202}$), which is known to bound the 2-norm contraction of $\mathfrak{D}_{\text{AES}_S}^-$. In more detail, denote the input difference distribution to the Sbox by the vector ζ (assuming it is

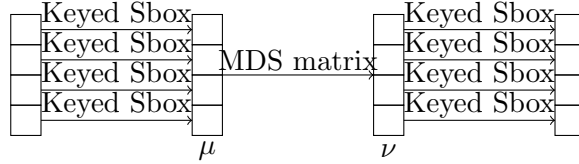
¹Alternatively, define $f(x) = (2^b - 1) \Pr[y \xrightarrow{E_1} x]$ and $g(x) = (2^b - 1) \Pr[z \xrightarrow{(E_2)^{-1}} x]$. Then (4) is reformulated as

$\text{Cov}[f, g] \leq \sqrt{\text{Var}[f] \text{Var}[g]}.$

²In Markov chain theory, this is known as contraction of the variance.

supported on non-zero differences and has 255 entries). As $\mathfrak{D}_{\text{AES}_S}^-$ is a stochastic matrix, a well-known result bounds its 2-norm contraction (relative to $\frac{1}{255}$, which is the 2-norm of the uniform distribution) by $\|\mathfrak{D}_{\text{AES}_S}^- \zeta\|_2^2 - \frac{1}{255} \leq \lambda(\|\zeta\|_2^2 - \frac{1}{255})$.

Next, we zoom out and analyze the AES super-Sbox, which is a keyed permutation (spanning “one and a half” AES rounds) over words of 32 bits, divided into 4 words of 8-bits. The super-Sbox is composed of a keyed layer of 4 Sboxes (applied independently to each of the words of the state), followed by a linear permutation over $(\mathbb{F}_2^8)^4$ and finally another keyed Sbox layer is applied to the state (see figure below). The linear permutation is an MDS matrix over $(\mathbb{F}_{2^8})^4 \cong (\mathbb{F}_2^8)^4$, which implements a mapping with special diffusion properties.



Denote the truncated DDT of the super-Sbox by $\mathfrak{D}_{\text{AES}_{\text{spS}}}^-$. Our next goal is to understand how the super-Sbox contracts the 2-norm of the input difference distribution. We classify input differences by the number of active Sboxes, which are Sboxes with a non-zero input difference. Generally, better contraction is obtained with more active Sboxes at the input, as non-active Sboxes do not modify their zero input difference.

Fix an input difference to the super-Sbox. Denote the distribution after the first Sbox layer by μ and the distribution after the application of the MDS matrix by ν (as in the figure). Since each Sbox in the first layer operates independently on its own input difference, μ is a product distribution on its 4 Sboxes (coordinates) that is easy to analyze. Its 2-norm mainly depends on the number of active Sboxes at the input. After the application of the MDS matrix, ν is no longer a product distribution (although its 2-norm is preserved). Consequently, it is no longer clear how to analyze the action of the second Sbox layer on ν , and bound the 2-norm of its output. We develop two tools for this purpose.

2.4 Tools used in the Analysis

One tool that we develop bounds the 2-norm of the output difference distribution of an Sbox layer, when applied to an arbitrary input difference distribution. The outcome is determined by various properties (in particular, the 2-norm) of the marginal input difference distributions of all subsets of Sboxes (i.e., coordinates). In our case, there are $2^4 = 16$ such coordinate subsets for ν . These marginal distributions are a result of applying the MDS matrix to the product distribution μ . Thus, our second tool studies how an MDS matrix contracts the 2-norm on its input (product) distribution, when considering the marginal distributions on coordinate subsets. We elaborate on these tools below.

2.4.1 Norm bounds for an Sbox layer.

We consider a general setting where an input difference distribution is input to a layer of k Sboxes ($k = 4$ for AES), and our goal is to bound the 2-norm of the output difference distribution. We first assume that the distribution is only supported on differences where all k Sboxes are active.

Basic result. The problem is purely linear-algebraic. As noted above we denote the input difference distribution by a vector ν (a distribution over k input elements from some domain), and denote the Sbox (truncated) DDT by a matrix A . The action of the Sbox layer on ν is described by a matrix-vector product with the matrix formed by the k -wise *Kronecker product* of A (denoted $A^{\otimes k}$).

We analyze how the Kronecker product contracts the 2-norm of a distribution in a fine-grained manner. This is done by decomposing ν into 2^k orthogonal components (each corresponding to a subset of coordinates) and studying how the Kronecker product operator $A^{\otimes k}$ acts on these components by utilizing its properties, and its spectral decomposition in particular. The outcome is a rather general result in linear algebra.

Denote the domain of every coordinate by Π and its size by $|\Pi|$ (for the AES Sbox, $\Pi = \mathbb{F}_2^8 \setminus \{0\}$ which is of size 255). For a set $\mathcal{J} \subseteq [k]$, denote by $\nu^{\subseteq \mathcal{J}}$ the marginal distribution of ν on the $\mathcal{J}$ -coordinates. The result bounds the 2-norm of the output difference distribution (vector) $A^{\otimes k}\nu$ by a sum of 2^k terms, each representing a marginal $\nu^{\subseteq \mathcal{J}}$ for $\mathcal{J} \subseteq [k]$. Specifically,

$$\|A^{\otimes k}\nu\|_2^2 \leq |\Pi|^{-k} \left(1 + \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} (|\Pi|^{|\mathcal{J}|} \|\nu^{\subseteq \mathcal{J}}\|_2^2 - 1) \right), \quad (5)$$

where $\lambda \leq 1$ is the second largest eigenvalue of $A^T A$. Roughly speaking, a subset $\mathcal{J}$ of size $|\mathcal{J}|$ contributes proportionally to the 2-norm of $\nu^{\subseteq \mathcal{J}}$, contracted by a factor of $\lambda^{|\mathcal{J}|}$. Thus, $A^{\otimes k}$ has a greater “contraction effect” on marginal input difference distributions on larger subsets. We remark that the trivial worst-case contraction bound is λ , namely $\|A^{\otimes k}\nu\|_2^2 - |\Pi|^{-k} \leq \lambda(\|\nu\|_2^2 - |\Pi|^{-k})$. It is obtained in case all singleton marginals $\nu^{\subseteq \{i\}}$ for $i \in [k]$ are independent, and $k-1$ of them are uniform (hence the Sbox layer only contracts one marginal). Yet, (5) gives a much better result than the worst-case bound in cases relevant to us.

Activity patterns. Arbitrary distributions may be supported on all *activity patterns*, $\mathcal{Z}_{\mathcal{D}}$ for $\mathcal{D} \subseteq [k]$. $\mathcal{Z}_{\mathcal{D}}$ is a set that includes all states where the Sboxes in the $\mathcal{D}$ -coordinates are non-active (have a zero difference), and the coordinates in $[k] \setminus \mathcal{D}$ are active. We thus decompose ν into its 2^k activity patterns, and apply (5) to each one with $k' = k - |\mathcal{D}|$ active Sboxes.

This decomposition does not require any novel ideas, but it is rather technical for two main reasons. First, the decomposition of the distribution ν into patterns is not determined by the 2-norm of its marginals. Consequently, we also consider the infinity norm of the marginal distributions.³ Second, we generally have 2^k different possible patterns, and for each pattern, our basic result considers the marginal distribution on all coordinate subsets of active Sboxes.

After some manipulations, we bound the 2-norm of the output difference distribution using a quadratic maximization program. It takes into account the 2-norm and infinity-norm of all marginal distributions on coordinate subsets ($\|\nu^{\subseteq \mathcal{T}}\|_2$ and $\|\nu^{\subseteq \mathcal{T}}\|_\infty$ for all $\mathcal{T} \subseteq [k]$). We solve this program using an algorithm described in Appendix A, whose code is included in Appendix B.

2.4.2 Norm bounds for an MDS matrix.

The MDS layer generates the distribution ν defined above, where its input is the product distribution μ on the marginals $\mu^{\subseteq \{i\}}$ for all $i \in [k]$. Our goal in analyzing the MDS matrix is to bound the 2-norm and infinity-norm of the marginal distributions $\nu^{\subseteq \mathcal{T}}$ for all $\mathcal{T} \subseteq [k]$, as required in the analysis of the final Sbox layer. For this purpose, we reuse a known result that the MDS matrix

³More precisely, we consider the maximal deviation of the marginal distributions from the uniform distribution.

applied to product distributions preserves the 2-norm and infinity-norm on marginal input and output coordinate sets that are of the same size (i.e., $\mu \subseteq \mathcal{S}$ and $\nu \subseteq \mathcal{T}$ for $|\mathcal{S}| = |\mathcal{T}|$).

However, preservation of the 2-norm and infinity-norm is not sufficient to obtain good contraction results for the super-Sbox. Recall that the final Sbox layer has a relatively small “contraction effect” on marginal input difference distributions $\nu \subseteq \mathcal{J}$ when $|\mathcal{J}|$ is small. Consequently, we prove that the MDS matrix actually contracts (rather than preserves) the 2-norm and infinity norm of output marginal distributions on coordinate subsets of size that is smaller than the number of active Sboxes at the input.

Thus, the Sbox layer and MDS matrix have complementary effects on marginal distributions, which together result in a significant overall “contraction effect”.

The analysis of the MDS layer utilizes the fact that, when applied to a product distribution, it is a convolution of its marginal distributions. We exploit properties of this convolution by considering additional norms of the marginal distributions, as well as their representation in the Fourier domain. The analysis can be extended to a non-product distribution, assuming bounds on Fourier coefficients of its marginals, but we do not use such extensions in this paper.

2.5 2-Norm Bounds for the 4-Round AES DDT

Next, we analyze how the 4-round AES DDT contracts the 2-norm of its input difference, depending on the number of active super-Sboxes at the input. In particular, this bounds the *1-to-2* norm of $\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}$, which is the 2-norm of the difference distribution after 4 rounds for the worst-case starting difference.

The analysis is based on the fact that the high-level structure of 4-round AES (excluding the final linear layer) is similar to the structure of the AES super-Sbox. Specifically, the Sbox layers are replaced with super-Sbox layers (and ShiftRows operations), and the MDS matrix is replaced with 4 MDS matrices applied in parallel to the columns of the AES state. Consequently, we use similar tools to those applied in the analysis of the super-Sbox.

Similar 2-norm bounds can be obtained for the decryption process using symmetric properties of AES. This allows to apply (4) and obtain the 8-round bound on the maximal deviation (2). Moreover, using the Cauchy-Schwarz inequality, we derive the 4-round 2-wise independence bound in (3).

2.6 2-Norm Bounds for the $4r$ -Round AES DDT

The 4-round bounds on individual differentials can be lifted to bound the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)})^T \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}$, which bounds the 2-norm contraction of 4-round AES for arbitrary input difference distributions. This is done via the *Frobenius norm* of the matrix $\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}$, denoted $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}\|_{\text{F}}$. On one hand, $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}\|_{\text{F}}^2 = \text{trace}((\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)})^T \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)})$, which is the sum of all of the eigenvalues. Hence, an upper bound on $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}\|_{\text{F}}^2 - 1$ upper bounds the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)})^T \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}$ (its largest eigenvalue is 1, so we subtract it).

We also have $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}\|_{\text{F}}^2 = \sum_{x,y} \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}[x,y]^2 = \sum_y \left(\sum_x \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(r)}[x,y]^2 \right)$, which sums the squared 2-norm contributions for all input differences, and is bounded using the tools above. By iterating the 4-round bound, we obtain the general results (1) and (3).

2.7 Previous and Independent Works

From a technical perspective, this work has superficial similarities to the previous work of Liu, Tessaro and Vaikuntanathan [LTV21]. Yet, the actual details differ significantly and lead to improvements by several orders of magnitude. First, both this work and [LTV21] show that several AES rounds contract a norm of the difference distribution, and then use this iteratively. However, we analyze the 2-norm contraction for 4-round AES, while [LTV21] analyzes the 1-norm contraction for 6-round AES.

At a lower level, both this work and [LTV21] analyze the action of an Sbox layer on its input difference distribution. However, the analysis of [LTV21] is only applicable to the specific Sbox construction used in AES, while our result is general. Moreover, even for the AES Sbox, our result proves better contraction by utilizing spectral analysis. Finally, both this work and [LTV21] analyze the action of an MDS matrix on its input difference distribution, assuming it is a product distribution. Specifically, [LTV21] uses the standard result that the matrix preserves the 2 and infinity norms when considering input and output coordinate subsets of the same size. We additionally analyze important cases where the output coordinate set size is smaller than that of the input.

We note, however, that [LTV21] also obtained results for more general AES-like ciphers and for t -wise independence for $t > 2$, while we obtain results only for pairwise independence of the AES (although some of our techniques are more widely applicable).

The independent work of Beyne, Leander, and Schütt [BLS25] also studied the 2-wise independence of AES using a more direct spectral analysis. Our techniques lead to stronger bounds for individual differentials, and in particular, a better bound on the 1-to-2 norm of the 4-round AES DDT. As described above, this bound is important for obtaining meaningful results for a small number of AES rounds used in practice. On the other hand, [BLS25] obtained a stronger bound on the general 2-norm contraction of the 4-round AES DDT,⁴ leading to better 2-wise independence bounds for a larger number of rounds. Combining our results gives improved bounds on the maximal deviation of the AES DDT starting from 12 rounds, and improved bounds on the 2-wise independence of AES starting from 8 rounds. Additional less trivial combinations of our techniques are likely to give even better bounds on the 2-wise independence of AES, as well as a better understanding of differentials in AES and in block ciphers in general.

Paper structure.

Next, we describe preliminaries. In Section 4 we describe our main tools as lemmas, and use them to prove our main results about differential analysis of the AES. In Section 5 we derive norm bounds for an MDS matrix, while in Section 6 and Section 7 we derive norm bounds for an Sbox layer.

3 Preliminaries

In this section, we describe preliminaries.

For a positive integer m , define $[m] = \{1, 2, \dots, m\}$, and for integers $0 < m_1 \leq m_2$ define $[m_1, m_2] = \{m_1, m_1 + 1, \dots, m_2\}$.

Given a subset $\mathcal{S}$, denote its size by $|\mathcal{S}|$, and denote by $\bar{\mathcal{S}}$ its complement when considering it as a subset of a larger set (that should be clear from the context). Throughout this paper n, k denote positive integers.

⁴Technically, we bound the Frobenius norm, which sums all the eigenvalues of $(\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)})^T \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}$, while [BLS25] is able to bound only the second largest eigenvalue.

Denote by $\mathbf{0}_n \in \mathbb{F}_2^n$ the element whose n bits are 0. Let $\mathbb{F}_2^{n(-)} = \mathbb{F}_2^n \setminus \{\mathbf{0}_n\}$, and $(\mathbb{F}_2^n)^{k(-)} = (\mathbb{F}_2^n)^k \setminus \{\mathbf{0}_{n \cdot k}\}$.

Given $x \in \mathbb{F}_{2^n}$, there is a standard linear encoding $enc : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_2^n$ that maps the 0 element over $\mathbb{F}_{2^n}$ to $\mathbf{0}_n$. In our context, this implies that we can also view x as an element of $\mathbb{F}_2^n$ (and vice-versa, view $y \in \mathbb{F}_2^n$ as an element of $\mathbb{F}_{2^n}$).

For $x \in (\mathbb{F}_{2^n})^k$, or $x \in (\mathbb{F}_2^n)^k$ denote its Hamming weight (over words of n bits) by $\text{wt}(x)$ ($0 \leq \text{wt}(x) \leq k$).

For a finite domain Π , $x = (x_1, \dots, x_k) \in \Pi^k$, and a subset $\mathcal{S} = \{i_1, i_2, \dots\} \subseteq [k]$, denote by $x_{\mathcal{S}} \in \Pi^{|\mathcal{S}|}$ the projection of x onto $\mathcal{S}$, whose j 'th element is x_{i_j} .

Let $\mathbb{1}$ be an indicator function that takes as input a predicate and returns 1 if it holds, and 0 otherwise.

MDS matrix and MDS mapping.

Definition 1 (MDS matrix). *A matrix $L \in (\mathbb{F}_{2^n})^{k \times k}$ is maximal distance separator (MDS) if for any non-zero input $x \in (\mathbb{F}_{2^n})^k$, $\text{wt}(x) + \text{wt}(Lx) \geq k + 1$.*

As noted above, $L \in (\mathbb{F}_{2^n})^{k \times k}$ can also be viewed as a linear mapping $L : (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$. It is a special case of an MDS mapping, defined below.

Definition 2 (MDS mapping). *A linear mapping $L : (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ is called an MDS mapping if for any non-zero input $x \in (\mathbb{F}_2^n)^k$, $\text{wt}(x) + \text{wt}(L(x)) \geq k + 1$.*

3.1 Linear Algebra over Functions

Let Π be a (finite) domain. Let $f : \Pi \rightarrow \mathbb{R}$ be a bounded function. We can view f as a vector with $|\Pi|$ entries, indexed by the elements of Π , namely $f \in \mathbb{R}^\Pi$. To emphasize that f is a function, we index its elements correspondingly (e.g., write $f(x)$ for $x \in \Pi$). The alternative view of f as a vector is important for applying linear operators on f , which in this view correspond to matrices. We will mostly be interested in linear operators mapping elements in $\mathbb{R}^\Pi$ to the same domain. Such a mapping A can be described as a matrix $A \in \mathbb{R}^{\Pi \times \Pi}$ that acts on f by the matrix-vector product Af .

We define additional linear algebra notation. For $x \in \Pi$, let $e_x \in \mathbb{R}^\Pi$ denote the standard basis vector such that $e_x(y) = 1$ for $y = x$, and $e_x(y) = 0$ for $y \neq x$. The domain Π (which determines the dimensions of e_x) should be clear from the context. For example, if $\Pi = \mathbb{F}_2^{n(-)}$, then e_x has $2^n - 1$ real entries. Denote by $\mathbf{1}_\Pi \in \mathbb{R}^\Pi$ the constant 1 vector in the domain Π . Further denote $\bar{\mathbf{1}}_\Pi = \frac{1}{|\Pi|} \mathbf{1}_\Pi$, and $\tilde{\mathbf{1}}_\Pi = \frac{1}{\sqrt{|\Pi|}} \mathbf{1}_\Pi$.

For a matrix $A \in \mathbb{R}^{\Pi \times \Pi}$, denote by $A[x, *] \in \mathbb{R}^\Pi$ the row indexed by $x \in \Pi$, and by $A[*, y] \in \mathbb{R}^\Pi$ the column indexed by $y \in \Pi$. Denote by A^T the transpose of A . We define the following linear operators.

Definition 3 (Summation projection operator). *Let $\mathcal{J} \subseteq [k]$. Given $f \in \mathbb{R}^{\Pi^k}$, the summation projection of f on coordinates $\mathcal{J}$ is the vector $f^{\subseteq \mathcal{J}} \in \mathbb{R}^{\Pi^{|\mathcal{J}|}}$ defined by $f^{\subseteq \mathcal{J}}(x) = \sum_{x' \in \Pi^{|\mathcal{J}|}} [f(x, x')]$, where in (x, x') , x is placed in $\mathcal{J}$ -coordinates (namely, $(x, x')_{\mathcal{J}} = x$), while x' is placed in the $\bar{\mathcal{J}}$ -coordinates.*

Definition 4 (Kronecker product). *Let $\Pi_1, \Pi_2, \Pi'_1, \Pi'_2$ be finite domains. Let $A \in \mathbb{R}^{\Pi_1 \times \Pi_2}, B \in \mathbb{R}^{\Pi'_1 \times \Pi'_2}$. The Kronecker product $A \otimes B \in \mathbb{R}^{(\Pi_1 \times \Pi'_1) \times (\Pi_2 \times \Pi'_2)}$ is the matrix defined by $(A \otimes B)[(x, x'), (y, y')] = A[x, y]B[x', y']$ for any $x \in \Pi_1, x' \in \Pi'_1, y \in \Pi_2, y' \in \Pi'_2$.*

The k -wise Kronecker product of $A \in \mathbb{R}^{\Pi_1 \times \Pi_2}$ (with itself) is denoted $A^{\otimes k}$ and defined inductively as $A^{\otimes 1} = A$, and for $k > 1$, $A^{\otimes k} = A^{\otimes k-1} \otimes A$.

Inner product and norms.

Let Π be a (finite) domain, and consider the vector space $\mathbb{R}^\Pi$. Let $f, g \in \mathbb{R}^\Pi$ be bounded. We define the standard inner product on this space by $\langle f, g \rangle = \sum_{x \in \Pi} f(x)g(x)$, and correspondingly, for any $p \geq 1$, the p -norm of f is defined as $\|f\|_p = \left(\sum_{x \in \Pi} |f(x)|^p \right)^{1/p}$, while $\|f\|_\infty = \max_{x \in \Pi} \{|f(x)|\}$.

Matrix norms.

For a finite domain Π , let $A \in \mathbb{R}^{\Pi \times \Pi}$. Let $p_1, p_2 \in \{1, 2, \infty\}$. Then $\|A\|_{p_1, p_2} = \max_{f \in \mathbb{R}^\Pi, 0 < \|f\|_{p_1} \leq 1} \{\|Af\|_{p_2}\}$. If $p_1 = p_2 = p$, we simply write $\|A\|_p$. We further define the Frobenius norm of A by $\|A\|_F = \sqrt{\sum_{x, y \in \Pi} A[x, y]^2}$. We state two basic properties of the norms that we use.

1. $\|A\|_{1,2} = \max_{y \in \Pi} \{\|A[*], y\|_2\} = \max_{y \in \Pi} \{\|Ae_y\|_2\}$.
2. Denote by $\lambda_1, \dots, \lambda_{|\Pi|}$ the eigenvalues of the symmetric matrix $A^T A$. Then, $\|A\|_F = \sqrt{\sum_{i=1}^{|\Pi|} \lambda_i}$.

3.2 Probability

Definition 5. Let Π be a finite domain. A probability distribution over Π is a vector (function) $\mu \in \mathbb{R}^\Pi$ such that $\mu(x) \geq 0$ for all $x \in \Pi$ and $\|\mu\|_1 = \sum_{x \in \Pi} |\mu(x)| = 1$.

The probability to obtain $x \in \Pi$ under μ is denoted by $\mu(x) = \Pr_{y \sim \mu}[y = x]$. Furthermore, given a set $\mathcal{S}$, denote by $x \sim \mathcal{S}$ a uniformly chosen $x \in \mathcal{S}$.

Note that $\bar{\mathbf{1}}_\Pi = \frac{1}{|\Pi|} \mathbf{1}_\Pi$ is the uniform distribution over Π , and for $x \in \Pi$, e_x is the distribution where the entire mass is concentrated on x . Moreover, any distribution μ satisfies $\langle \mu, \bar{\mathbf{1}}_\Pi \rangle = |\Pi|^{-1}$.

The maximal element probability under μ is $M[\mu] = \max_{x \in \Pi} \{\mu(x)\} = \|\mu\|_\infty$. The minimal probability is $MN[\mu] = \min_{x \in \Pi} \{\mu(x)\}$. We also define the maximal deviation of μ from its expectation $|\Pi|^{-1}$ as $DV[\mu] = \max_{x \in \Pi} \{|\mu(x) - |\Pi|^{-1}|\} = \|\mu - \bar{\mathbf{1}}_\Pi\|_\infty$. Observe that $M[\mu] \leq |\Pi|^{-1} + DV[\mu]$ and $MN[\mu] \geq |\Pi|^{-1} - DV[\mu]$.

The collision probability of a distribution μ is denoted by $\text{Col}[\mu]$ and defined as $\text{Col}[\mu] = \Pr_{\substack{x, x' \sim \mu \\ \text{independently}}} [x = x'] = \sum_{x \in \Pi} \mu(x)^2 = \|\mu\|_2^2$. Note that for any distributions $\mu, \nu \in \mathbb{R}^\Pi$,

$$\langle \mu - \bar{\mathbf{1}}_\Pi, \nu - \bar{\mathbf{1}}_\Pi \rangle = \langle \mu, \nu \rangle - \langle \mu, \bar{\mathbf{1}}_\Pi \rangle - \langle \bar{\mathbf{1}}_\Pi, \nu \rangle + \|\bar{\mathbf{1}}_\Pi\|_2^2 = \langle \mu, \nu \rangle - |\Pi|^{-1}. \quad (6)$$

$$\text{In particular,} \quad \|\mu - \bar{\mathbf{1}}_\Pi\|_2^2 = \|\mu\|_2^2 - |\Pi|^{-1} = \text{Col}[\mu] - |\Pi|^{-1}. \quad (7)$$

Assume that $\mu \in \mathbb{R}^{\Pi^k}$ is a probability distribution over Π^k . Its marginal distribution on the coordinates of $\mathcal{J}$ is the vector $\mu^{\subseteq \mathcal{J}} \in \mathbb{R}^{\Pi^{|\mathcal{J}|}}$ (defined using the summation projection operator). Observe that $\mu^{\subseteq \mathcal{J}}$ is a valid distribution.

Definition 6. The statistical distance between distributions $\mu, \nu \in \mathbb{R}^\Pi$ is defined as $\text{SD}(\mu, \nu) = \frac{1}{2} \|\mu - \nu\|_1$.

3.3 Doubly Stochastic Matrices

Definition 7. Let Π be a finite domain. A square matrix $A \in \mathbb{R}^{\Pi \times \Pi}$ is called doubly stochastic if for all $x, y \in \Pi$, $A[x, y] \geq 0$, and $\|A[*], y\|_1 = \|A[x, *]\|_1 = 1$.

Definition 8. For $A \in \mathbb{R}^{\Pi \times \Pi}$, define $\text{Dev}[A] = \max_{x, y \in \Pi} \{|A[x, y] - |\Pi|^{-1}|\}$.

Let $A, B \in \mathbb{R}^{\Pi \times \Pi}$ be doubly stochastic matrices. Then, AB is also doubly stochastic. If A is doubly stochastic, then A^T is also doubly stochastic and hence $A^T A$ is doubly stochastic. As $A^T A$ is symmetric, it is orthonormally diagonalizable, all of its eigenvalues are real and non-negative, and its largest eigenvalue is 1, corresponding to the eigenvector $\bar{\mathbf{1}}_\Pi$. If $A \in \mathbb{R}^{\Pi \times \Pi}$ is doubly stochastic, and $\mu \in \mathbb{R}^\Pi$ is a distribution, then $A\mu \in \mathbb{R}^\Pi$ is also a distribution.

We prove basic properties of doubly stochastic matrices. We start with the connection between the Frobenius norm and the second largest eigenvalue.

Proposition 1. Let $A \in \mathbb{R}^{\Pi \times \Pi}$ be a doubly stochastic matrix. Then, the second largest eigenvalue of $A^T A$ is at most $\|A\|_F^2 - 1$.

Proof. We have $\|A\|_F^2 = \sum_{i \in [\Pi]} \lambda_i$ where $\lambda_1, \dots, \lambda_{|\Pi|}$ are the sorted eigenvalues of $A^T A$. Since $A^T A$ is doubly stochastic, then $\lambda_1 = 1$, and moreover, all its eigenvalues are non-negative. Hence,

$$\lambda_2 \leq \left(\sum_{i \in [\Pi]} \lambda_i \right) - 1 = \|A\|_F^2 - 1.$$

■

We continue with several amplification results that stress the importance of the 1-to-2 norm of A and the second largest eigenvalue of $A^T A$.

Proposition 2. Let $A \in \mathbb{R}^{\Pi \times \Pi}$ be a doubly stochastic matrix. Denote the second largest eigenvalue of $A^T A$ by λ . Then, for any distribution $\mu \in \mathbb{R}^\Pi$,

$$\|A\mu\|_2^2 - |\Pi|^{-1} = \|A\mu - \bar{\mathbf{1}}_\Pi\|_2^2 = \|A(\mu - \bar{\mathbf{1}}_\Pi)\|_2^2 \leq \lambda \|\mu - \bar{\mathbf{1}}_\Pi\|_2^2 = \lambda(\|\mu\|_2^2 - |\Pi|^{-1}).$$

Proof. Since $A\mu$ is a distribution, by (7),

$$\|A\mu\|_2^2 - |\Pi|^{-1} = \|A\mu - \bar{\mathbf{1}}_\Pi\|_2^2.$$

Moreover, since A is doubly stochastic, then $A\bar{\mathbf{1}}_\Pi = \bar{\mathbf{1}}_\Pi$, hence

$$\|A(\mu - \bar{\mathbf{1}}_\Pi)\|_2^2 = \|A\mu - \bar{\mathbf{1}}_\Pi\|_2^2.$$

The last equality $\|\mu - \bar{\mathbf{1}}_\Pi\|_2^2 = \|\mu\|_2^2 - |\Pi|^{-1}$ is again by (7).

We now prove the inequality. Denote the sorted eigenvalues of $A^T A$ by $\lambda_1 = 1, \lambda_2 = \lambda, \dots, \lambda_{|\Pi|}$, and a corresponding set of orthonormal eigenvectors by $v_1 = \bar{\mathbf{1}}_\Pi, v_2, \dots, v_{|\Pi|}$ (if there are multiplicities among the eigenvalues, sort the corresponding eigenvectors arbitrarily). Note that $\|\bar{\mathbf{1}}_\Pi\|_2^2 = 1$.

Since $v_1, \dots, v_{|\Pi|}$ is an orthonormal basis, then we can write $\mu - \bar{\mathbf{1}}_\Pi = \sum_{i \in [\Pi]} \alpha_i v_i$, where $\alpha_i = \langle \mu - \bar{\mathbf{1}}_\Pi, v_i \rangle$. Hence,

$$\|\mu - \bar{\mathbf{1}}_\Pi\|_2^2 = \sum_{i \in [\Pi]} (\alpha_i)^2 \|v_i\|_2^2 = \sum_{i \in [\Pi]} (\alpha_i)^2. \quad (8)$$

Observe that since μ is a distribution, then

$$\alpha_1 = \langle \mu - \bar{\mathbf{1}}_\Pi, v_1 \rangle = \langle \mu - \bar{\mathbf{1}}_\Pi, \tilde{\mathbf{1}}_\Pi \rangle = \langle \mu, \tilde{\mathbf{1}}_\Pi \rangle - \langle \bar{\mathbf{1}}_\Pi, \tilde{\mathbf{1}}_\Pi \rangle = |\Pi|^{-1/2} - |\Pi|^{-1/2} = 0.$$

Hence,

$$A^T A(\mu - \bar{\mathbf{1}}_\Pi) = A^T A \left(\sum_{i=2}^{|\Pi|} \alpha_i v_i \right) = \sum_{i=2}^{|\Pi|} \alpha_i A^T A v_i = \sum_{i=2}^{|\Pi|} \alpha_i \lambda_i v_i.$$

Therefore,

$$\begin{aligned} \|A(\mu - \bar{\mathbf{1}}_\Pi)\|_2^2 &= \langle A(\mu - \bar{\mathbf{1}}_\Pi), A(\mu - \bar{\mathbf{1}}_\Pi) \rangle = \langle \mu - \bar{\mathbf{1}}_\Pi, A^T A(\mu - \bar{\mathbf{1}}_\Pi) \rangle = \left\langle \sum_{i=2}^{|\Pi|} \alpha_i v_i, \sum_{i=2}^{|\Pi|} \alpha_i \lambda_i v_i \right\rangle \\ &= \sum_{i=2}^{|\Pi|} (\alpha_i)^2 \lambda_i \|v_i\|_2^2 = \sum_{i=2}^{|\Pi|} (\alpha_i)^2 \lambda_i \leq \lambda \sum_{i=2}^{|\Pi|} (\alpha_i)^2 = \lambda \|\mu - \bar{\mathbf{1}}_\Pi\|_2^2, \end{aligned}$$

where the fourth and fifth equalities use the orthonormality of the eigenvectors and the final equality is by (8). ■

Proposition 3. *Let $A, B \in \mathbb{R}^{\Pi \times \Pi}$ be doubly stochastic matrices. Assume that the second largest eigenvalue of $A^T A$ is at most λ . Let $r \geq 0$ be an integer parameter. Then*

$$\|A^r B\|_{1,2}^2 - |\Pi|^{-1} \leq \lambda^r (\|B\|_{1,2}^2 - |\Pi|^{-1}).$$

Proof. The proof is by induction on r . For $r = 0$, it holds trivially.

For $r > 0$, we have

$$A^r B = A(A^{r-1} B).$$

Let $x \in \Pi$ be arbitrary and denote $\mu = (A^{r-1} B)e_x$. By the induction hypothesis we have

$$\|\mu\|_2^2 - |\Pi|^{-1} \leq \|A^{r-1} B\|_{1,2}^2 - |\Pi|^{-1} \leq \lambda^{r-1} (\|B\|_{1,2}^2 - |\Pi|^{-1}).$$

Hence,

$$\|A^r B e_x\|_2^2 - |\Pi|^{-1} = \|A\mu\|_2^2 - |\Pi|^{-1} \leq \lambda (\|\mu\|_2^2 - |\Pi|^{-1}) \leq \lambda^r (\|B\|_{1,2}^2 - |\Pi|^{-1}),$$

where the first inequality is by Proposition 2. Since $x \in \Pi$ is arbitrary, this completes the proof. ■

Proposition 4. *Let $A, B \in \mathbb{R}^{\Pi \times \Pi}$ be doubly stochastic matrices. Assume that the second largest eigenvalue of $A^T A$ is at most λ . Let $r \geq 0$ be an integer parameter. Let $x \in \Pi$ be arbitrary. Then $\|A^r B e_x - \bar{\mathbf{1}}_\Pi\|_2^2 \leq \lambda^r (\|B\|_{1,2}^2 - |\Pi|^{-1})$, and $\|A^r B e_x - \bar{\mathbf{1}}_\Pi\|_1 \leq |\Pi|^{1/2} \lambda^{r/2} \sqrt{\|B\|_{1,2}^2 - |\Pi|^{-1}}$.*

Proof. We have,

$$\|A^r B e_x - \bar{\mathbf{1}}_\Pi\|_2^2 = \|A^r B e_x\|_2^2 - |\Pi|^{-1} \leq \lambda^r (\|B\|_{1,2}^2 - |\Pi|^{-1}),$$

where the equality is by (7), and the inequality is by Proposition 3. This proves the first inequality.

For the second inequality, by Cauchy-Schwarz and the first inequality,

$$\|A^r B e_x - \bar{\mathbf{1}}_\Pi\|_1 \leq \sqrt{|\Pi| \|A^r B e_x - \bar{\mathbf{1}}_\Pi\|_2^2} \leq |\Pi|^{1/2} \lambda^{r/2} \sqrt{\|B\|_{1,2}^2 - |\Pi|^{-1}}.$$

■

Proposition 5. *Let $r \geq 0$ be an integer parameter. Let $A, B, C \in \mathbb{R}^{\Pi \times \Pi}$ be doubly stochastic matrices. Assume that the second largest eigenvalue of $B^T B$ is at most λ . Then, $\text{Dev}[AB^r C] \leq \lambda^{r/2} \sqrt{\|A^T\|_{1,2}^2 - |\Pi|^{-1}} \sqrt{\|C\|_{1,2}^2 - |\Pi|^{-1}}$.*

Proof. Denote $E = B^r C$. For any $x, y \in \Pi$,

$$\begin{aligned} \left| (AE)[x, y] - |\Pi|^{-1} \right| &= \left| \langle A[x, *], E[*], y \rangle - |\Pi|^{-1} \right| = \left| \langle A[x, *] - \bar{\mathbf{1}}_\Pi, E[*], y \rangle \right| \\ &\leq \sqrt{\|A[x, *] - \bar{\mathbf{1}}_\Pi\|_2^2} \sqrt{\|E[*], y - \bar{\mathbf{1}}_\Pi\|_2^2} = \sqrt{\|A[x, *]\|_2^2 - |\Pi|^{-1}} \sqrt{\|E[*], y\|_2^2 - |\Pi|^{-1}} \\ &= \sqrt{\|A^T[*], x\|_2^2 - |\Pi|^{-1}} \sqrt{\|E[*], y\|_2^2 - |\Pi|^{-1}} \leq \sqrt{\|A^T\|_{1,2}^2 - |\Pi|^{-1}} \sqrt{\|E\|_{1,2}^2 - |\Pi|^{-1}} \\ &\leq \lambda^{r/2} \sqrt{\|A^T\|_{1,2}^2 - |\Pi|^{-1}} \sqrt{\|C\|_{1,2}^2 - |\Pi|^{-1}}. \end{aligned}$$

The second and third equalities are by (6) and (7), respectively, which hold since $A[x, *]$ and $E[*], y$ are distributions, as A and E are doubly stochastic. The first inequality is by Cauchy-Schwarz, and the final inequality is by Proposition 3. Since $x, y \in \Pi$ are arbitrary, this completes the proof. ■

3.4 Boolean Functions and Fourier Analysis

We define the Fourier expansion for a Boolean function.

Definition 9 (Fourier expansion). *Given $\alpha \in \mathbb{F}_2^n$, define $\chi_\alpha : \mathbb{F}_2^n \rightarrow \{-1, 1\}$ by*

$$\chi_\alpha(x) = (-1)^{\sum_{i \in [n]} \alpha_i x_i} = \prod_{i \in [n]} (-1)^{\alpha_i x_i}.$$

The set $\{\chi_\alpha\}_{\alpha \in \mathbb{F}_2^n}$ is an orthonormal basis for the set of functions $\{f \mid f : \mathbb{F}_2^n \rightarrow \mathbb{R}\}$ with respect to the normalized inner product $\frac{1}{|\mathbb{F}_2^n|} \langle f, g \rangle_{\mathbb{R}} = \mathbb{E}_{x \sim \mathbb{F}_2^n} [f(x)g(x)]$. Hence any function in $\{f \mid f : \mathbb{F}_2^n \rightarrow \mathbb{R}\}$ can be decomposed to $f = \sum_{\alpha \in \mathbb{F}_2^n} \hat{f}(\alpha) \chi_\alpha$, where $\hat{f}(\alpha) = \mathbb{E}[\chi_\alpha f]$, and in particular, $\hat{f}(0) = \mathbb{E}[f]$. We call $\hat{f}(\alpha)$ the Fourier coefficient of f on α . To distinguish the Fourier domain from the input domain we write it as $\hat{\mathbb{F}}_2^n$, and thus $f(x) = \sum_{\alpha \in \hat{\mathbb{F}}_2^n} \hat{f}(\alpha) \chi_\alpha(x)$.

When dealing with Fourier analysis of distributions, it will typically be more convenient to work with normalized distributions (or density functions).

Definition 10 (Normalized Fourier transform). *For $f \in \mathbb{R}^{\mathbb{F}_2^n}$, define $\check{f} = 2^n \hat{f} = \widehat{2^n f}$.*

We stress that the normalization factor is determined by the range size of the function. The reason that this is useful is that for a distribution $\mu \in \mathbb{R}^{\mathbb{F}_2^n}$ and $\alpha \in \hat{\mathbb{F}}_2^n$, $\check{\mu}(\alpha) = \mathbb{E}_{x \sim \mu} [\chi_\alpha(x)]$ measures the correlation of a sample drawn from μ with χ_α . In particular $\check{\mu}(\mathbf{0}_n) = 1$.

We state a basic result that relates the Fourier transform and collision probability of a distribution (c.f., [O'D14, Exercise 1.23]).

Proposition 6 (Relation between Fourier transform and collision probability). *Let $\mu \in \mathbb{R}^{\mathbb{F}_2^n}$ be a distribution. Then,*

$$\text{Col}[\mu] = 2^{-n} + 2^n \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \hat{\mu}(\alpha)^2 = 2^{-n} + 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 = 2^{-n} \|\check{\mu}\|_2^2.$$

3.5 Difference Propagation

We define the difference distribution table for a keyed permutation and derive some of its basic properties.

Definition 11 (Difference distribution table (DDT)). *Let $F: \mathbb{F}_2^\ell \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, be a keyed permutation. Define the difference distribution table (DDT) as the matrix $\mathfrak{D}_F \in \mathbb{R}^{\mathbb{F}_2^n \times \mathbb{F}_2^n}$ where $\mathfrak{D}_F[x, y] = \Pr_{\kappa \sim \mathbb{F}_2^\ell, u \sim \mathbb{F}_2^n} [F(\kappa, u) \oplus F(\kappa, u \oplus y) = x]$. Further define the truncated difference distribution table as $\mathfrak{D}_F^- \in \mathbb{R}^{\mathbb{F}_2^{n(-)} \times \mathbb{F}_2^{n(-)}}$ as the sub-matrix of $\mathfrak{D}_F$ where $x, y \neq \mathbf{0}_n$.*

Our definition corresponds to the linear algebraic view of the matrix, where given an input difference $y \in \mathbb{F}_2^n$, the vector $\mathfrak{D}_F e_y$ is the distribution of the output difference. In this paper, F always has the property that it XORs an independent key at its entry (effectively randomizing u). In this case, the probability can be taken only over the key κ .

The following fact is standard.

Fact 1. *Let $F: \mathbb{F}_2^\ell \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, be a keyed permutation. Then:*

1. *Let F^{-1} denote the inverse of F . Then $\mathfrak{D}_{F^{-1}} = (\mathfrak{D}_F)^T$ and $\mathfrak{D}_{F^{-1}}^- = (\mathfrak{D}_F^-)^T$.*
2. *Both $\mathfrak{D}_F$ and $\mathfrak{D}_F^-$ are doubly stochastic.*
3. *If $\mu \in \mathbb{R}^{\mathbb{F}_2^n}$ is a distribution over the input differences to F , then the output difference distribution $\xi \in \mathbb{R}^{\mathbb{F}_2^n}$ is $\xi = \mathfrak{D}_F \mu$, namely*

$$\xi(x) = \Pr_{y \sim \mu, \kappa \sim \mathbb{F}_2^\ell, u \sim \mathbb{F}_2^n} [F(\kappa, u) \oplus F(\kappa, u \oplus y) = x] = \mathbb{E}_{y \sim \mu} [\mathfrak{D}_F[x, y]].$$

Definition 12 (Maximum expected differential probability deviation (MEDPD)). *Given a keyed permutation F over $\mathbb{F}_2^n$, define its maximum expected differential probability deviation (MEDPD) as $\text{Dev}[\mathfrak{D}_F^-]$.*

Keyed Sbox layer and Kronecker product.

We define the operation of an Sbox layer using the Kronecker product.

Fact 2 (Sbox layer operator). *Let $F: \mathbb{F}_2^\ell \times \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be a keyed permutation. Define the keyed permutation $F^*: (\mathbb{F}_2^\ell)^k \times (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ using k parallel independent applications of F , i.e., $F^*(\kappa, v) = (F(\kappa_1, v_1), \dots, F(\kappa_k, v_k))$.*

Let $\mu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be a distribution over the input differences to F^ . Then, the output difference distribution $\xi \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ is given by*

$$\xi(x) = \mathbb{E}_{y \sim \mu} \left[\prod_{i=1}^k \mathfrak{D}_F[x_i, y_i] \right] = \sum_{y \in (\mathbb{F}_2^n)^k} \mu(y_1, \dots, y_k) \prod_{i=1}^k \mathfrak{D}_F[x_i, y_i].$$
 The corresponding linear operator is the k -wise Kronecker product of $\mathfrak{D}_F$, namely $(\mathfrak{D}_F)^{\otimes k} \in \mathbb{R}^{(\mathbb{F}_2^n)^k \times (\mathbb{F}_2^n)^k}$. Therefore, $\xi = (\mathfrak{D}_F)^{\otimes k} \mu$, namely, $\mathfrak{D}_{F^} = (\mathfrak{D}_F)^{\otimes k}$.*

3.6 Key Alternating and AES-like Ciphers

3.6.1 Definition of key-alternating ciphers.

A key-alternating cipher is a block cipher F , parameterized as follows: let

$$\mathbf{prm} = \{b, F_{\text{RP}}, \mathbf{prm}_{F_{\text{RP}}}\},$$

where b is the block size, $F_{\text{RP}} : \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b$ is a (round) permutation and $\mathbf{prm}_{F_{\text{RP}}}$ are additional parameters for the permutation (defined later for AES-like ciphers). In addition, let r be a parameter, denoting the number of rounds (we do not include it in $\mathbf{prm}$ for technical reasons).

The r -round (encryption) function is denoted

$$(F_E^{(r)})_{\mathbf{prm}} : (\mathbb{F}_2^b)^{r+1} \times \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b.$$

The first parameter includes $r + 1$ round-keys, $\kappa_0, \kappa_1, \dots, \kappa_r$, where $\kappa_i \in \mathbb{F}_2^b$, and the second parameter is a plaintext $x \in \mathbb{F}_2^b$. In the following, we will drop the explicit specification of $\mathbf{prm}$ and simply denote the r -round function by $F_E^{(r)}$.

In order to define $F_E^{(r)}$, we first define the keyed round function,

$$F_{\text{RF}} : \mathbb{F}_2^b \times \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b,$$

using the round permutation $F_{\text{RP}} : \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b$. It is defined by masking the input to the round permutation as

$$F_{\text{RF}}(\kappa, x) = F_{\text{RP}}(x \oplus \kappa).$$

We will refer to the key addition as an AddRoundKey operation.

Similarly, the r -round function

$$F_{\text{RF}}^{(r)} : (\mathbb{F}_2^b)^r \times \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b,$$

is defined inductively as $F_{\text{RF}}^{(1)} = F_{\text{RF}}$, and for $r > 1$,

$$F_{\text{RF}}^{(r)}(\kappa_0, \dots, \kappa_{r-1}, x) = F_{\text{RF}}^{(1)}(\kappa_{r-1}, F_{\text{RF}}^{(r-1)}(\kappa_0, \dots, \kappa_{r-2}, x)).$$

Finally,

$$F_E^{(r)}(\kappa_0, \dots, \kappa_r, x) = F_{\text{RF}}^{(r)}(\kappa_0, \dots, \kappa_{r-1}, x) \oplus \kappa_r.$$

3.6.2 Basic results about key-alternating ciphers.

We state a simple fact about the DDT of key-alternating ciphers that is related to Fact 1.

Fact 3. *Let $F_E^{(r)}$ be the encryption function of a key-alternating cipher, where the round-keys are selected uniformly at random. Let F_{RF} be its round function. Then,*

$$\mathfrak{D}_{F_E^{(r)}} = \mathfrak{D}_{F_{\text{RF}}^{(r)}} = (\mathfrak{D}_{F_{\text{RF}}})^r.$$

Moreover, the DDT of the decryption function $(F_E^{(r)})^{-1}$ is $(\mathfrak{D}_{F_{\text{RF}}^{(r)}})^T = ((\mathfrak{D}_{F_{\text{RF}}})^T)^r$.

Definition 13 (Closeness to 2-wise independence). Let $F_E^{(r)}$ be a key-alternating cipher. For $x, y \in \mathbb{F}_2^b$ such that $x \neq y$, denote by $\mu_{x,y} \in \mathbb{R}^{\mathbb{F}_2^b} \times \mathbb{R}^{\mathbb{F}_2^b}$ the distribution defined by the process of sampling $(\kappa_0, \dots, \kappa_r) \sim (\mathbb{F}_2^b)^{r+1}$ and outputting $(F_E^{(r)}(\kappa_0, \dots, \kappa_r, x), F_E^{(r)}(\kappa_0, \dots, \kappa_r, y))$.

Let $\nu \in \mathbb{R}^{\mathbb{F}_2^b} \times \mathbb{R}^{\mathbb{F}_2^b}$ denote the distribution defined by the process of sampling two elements from $\mathbb{R}^{\mathbb{F}_2^b}$ uniformly without replacement.

Then, $F_E^{(r)}$ is ϵ -close to 2-wise independence if for any $x, y \in \mathbb{F}_2^b$ ($x \neq y$),

$$\text{SD}(\mu_{x,y}, \nu) \leq \epsilon.$$

The following result reduces the analysis of 2-wise independence to analyzing the distribution of differences.

Proposition 7 ([LTV21], Lemma 2.7). Let $F_E^{(r)}$ be a key-alternating cipher with round function F_{RF} . Assume that for any non-zero input difference $x \in \mathbb{F}_2^{b(-)}$,

$$\text{SD}(\mathfrak{D}_{F_{\text{RF}}}^{- (r)} e_x, \bar{\mathbf{1}}_{\mathbb{F}_2^{b(-)}}) \leq \epsilon.$$

Then, $F_E^{(r)}$ is ϵ -close to 2-wise independence.

This holds essentially because a key-alternating cipher with independent round-keys is trivially 1-wise independent, hence closeness to 2-wise independence is only determined by the output difference distribution.

3.7 The Round Permutation for AES-like Ciphers

Recall that a key-alternating cipher, F , is parameterized by $\mathbf{prm} = \{b, F_{\text{RP}}, \mathbf{prm}_{F_{\text{RP}}}\}$. We now define the round permutation $F_{\text{RP}} : \mathbb{F}_2^b \rightarrow \mathbb{F}_2^b$ for AES-like ciphers. First, we define the additional parameters

$$\mathbf{prm}_{F_{\text{RP}}} = \{w, k, F_{\text{PS}}, \{F_L^{(j)}\}_{j \in [k]}\},$$

where w is the word size, k is a parameter such that the state consists of k^2 words (i.e., $b = k^2 \cdot w$), $F_{\text{PS}} : \mathbb{F}_2^w \rightarrow \mathbb{F}_2^w$ is a public (unkeyed) permutation Sbox, and $F_L^{(j)} : (\mathbb{F}_{2^w})^k \rightarrow (\mathbb{F}_{2^w})^k$ are MDS matrices.

3.7.1 Indexing the square state.

The internal state of an AES-like cipher is described as a square $x \in (\mathbb{F}_2^w)^{k^2}$. We index its elements by pairs $(i, j) \in [k] \times [k]$, namely $x = (x_{(1,1)}, \dots, x_{(1,k)}, \dots, x_{(k,1)}, \dots, x_{(k,k)})$ (see Figure 1).

We define the following index pair sets: For every $j \in [k]$ define the column indices $\mathcal{C}_j = \{(1, j), \dots, (k, j)\}$. Further, for $\mathcal{J} \subseteq [k]$, define the column set $\mathcal{C}_{\mathcal{J}} = \cup_{j \in \mathcal{J}} \mathcal{C}_j$.

For every $j \in [k]$ define the diagonal indices $\mathcal{G}_j = \{(1, j), (2, j+1 \bmod k), \dots, (k, j-1 \bmod k)\}$, where the mod k operation works on the range $[k]$ (e.g., $k+1 \bmod k = 1$). Moreover, for every $\mathcal{J} \subseteq [k]$, define the diagonal set $\mathcal{G}_{\mathcal{J}} = \cup_{j \in \mathcal{J}} \mathcal{G}_j$.

Finally, for every $j \in [k]$ define the anti-diagonal indices $\mathcal{H}_j = \{(1, j), (2, j-1 \bmod k), \dots, (k, j+1 \bmod k)\}$, and for every $\mathcal{J} \subseteq [k]$, define the anti-diagonal set $\mathcal{H}_{\mathcal{J}} = \cup_{j \in \mathcal{J}} \mathcal{H}_j$ (see Figure 1).

For $x \in (\mathbb{F}_2^w)^{k^2}$, denote

$$\text{wt}_{\mathcal{G}}(x) = \sum_{j \in [k]} \mathbb{1}(\text{wt}(x_{\mathcal{G}_j}) > 0),$$

and

$$\text{wt}_{\mathcal{H}}(x) = \sum_{j \in [k]} \mathbb{1}(\text{wt}(x_{\mathcal{H}_j}) > 0),$$

which count the number of non-zero diagonals and non-zero anti-diagonals, respectively.

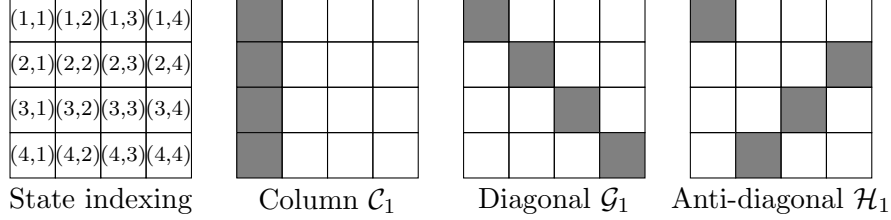


Figure 1: The AES state ($k = 4$)

3.7.2 Permutation specification.

F_{RP} is defined as a composition of the 3 operations below, each takes as input an internal state, and outputs an updated internal state.

1. SubWord consists of k^2 independent applications of the Sbox F_{PS} to the words of the state. Namely, on input x , the output state y is defined by $y_{(i,j)} = \text{F}_{\text{PS}}(x_{(i,j)})$ for every $(i, j) \in [k] \times [k]$.
2. ShiftRows is defined by setting $z_{\mathcal{C}_j} = y_{\mathcal{G}_j}$ for every $j \in [k]$. Namely, for every $(i, j) \in [k] \times [k]$, $z_{(i,j)} = y_{(i, j+i-1 \bmod k)}$. In other words, the words of the i 'th row are cyclically shifted by $i-1$ positions to the left.
3. MixColumns is applied to the k columns of the square using the MDS matrices as $u_{\mathcal{C}_j} = \text{F}_{\text{L}}^{(j)} z_{\mathcal{C}_j}$ for $j \in [k]$.

Our analysis does not make any use of properties of specific MDS matrices (besides the MDS property). Thus, to simplify notation, we assume throughout this paper that there is only one MDS matrix, F_{L} , that is applied to all columns, i.e., MixColumns is implemented by $u_{\mathcal{C}_j} = \text{F}_{\text{L}} z_{\mathcal{C}_j}$ for $j \in [k]$.

3.7.3 Alternative definition of AES-like ciphers using super-Sboxes.

We describe an alternative (and equivalent) definition of AES-like ciphers that is commonly used in the literature. First, it will also be convenient to combine the AddRoundKey and SubWord operations, applied to a word. Thus, we define the masked Sbox function

$$\text{F}_{\text{S}}: \mathbb{F}_2^w \times \mathbb{F}_2^w \rightarrow \mathbb{F}_2^w$$

by $\text{F}_{\text{S}}(\kappa, x) = \text{F}_{\text{PS}}(x \oplus \kappa)$.

Next, consider the consecutive operations of AddRoundKey, SubWord, ShiftRows, MixColumns, AddRoundKey, SubWord, ShiftRows. The observation behind the definition is that these operations can be described by independent applications of k keyed permutations (called super-Sboxes), each working on k words.

Denote the super-Sbox function by

$$F_{\text{spS}}: ((\mathbb{F}_2^w)^k)^2 \times (\mathbb{F}_2^w)^k \rightarrow (\mathbb{F}_2^w)^k,$$

where the first parameter consists of two partial round-keys (of length k words each), and the second parameter is a partial internal state (of length k words). F_{spS} returns an updated partial state of k words.

On inputs (partial) keys κ_1, κ_2 and (partial) internal state $x = (x_1, \dots, x_k) \in (\mathbb{F}_2^w)^k$, $F_{\text{spS}}(\kappa_1, \kappa_2, x)$ is defined by the operations below (see Figure 2).

1. Compute k independent applications of the masked Sbox function F_S to the words of the partial state. Namely, $y_i = F_S((\kappa_1)_i, x_i)$ for every $i \in [k]$.
2. Apply the MDS matrix to the partial state: $z = F_L y$.
3. Compute k independent applications of the masked Sbox function F_S to the words of the partial state. Namely, $u_i = F_S((\kappa_2)_i, z_i)$ for every $i \in [k]$.

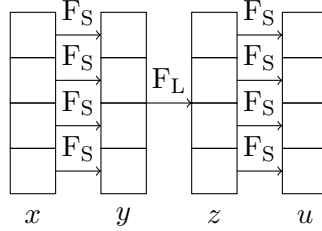


Figure 2: The AES super-Sbox ($k = 4$)

We now describe the alternative (encryption) round function $F_{\text{RF}}^{(2r)}$, which we redefine on pairs of rounds. For this purpose, observe that two ShiftRows applications are eliminated from the super-Sbox definition, and we reintroduce them. Specifically, $F_{\text{RF}}^{(2)}(\kappa_0, \kappa_1, x)$ is defined by the following operations (see Figure 3):

1. Set $y_{\mathcal{H}_j} = F_{\text{spS}}((\kappa_0)_{\mathcal{G}_j}, (\kappa_1)_{\mathcal{G}_j}, x_{\mathcal{G}_j})$ for all $j \in [k]$. Namely, the super-Sboxes are applied simultaneously to the diagonals of the state, and their outcomes form the anti-diagonals of the updated state. The diagonal shifts correspond to the two applications of ShiftRows, which map a diagonal to an anti-diagonal.
2. Apply MixColumns to the k columns of the square using the MDS matrix as $z_{\mathcal{C}_j} = F_L y_{\mathcal{C}_j}$ for $j \in [k]$.

For $r > 1$, the round function is iterated as

$$F_{\text{RF}}^{(2r)}(\kappa_0, \dots, \kappa_{2r-1}, x) = F_{\text{RF}}^{(2)}(\kappa_{2r-2}, \kappa_{2r-1}, F_{\text{RF}}^{(2(r-1))}(\kappa_0, \dots, \kappa_{2r-3}, x)).$$

Finally,

$$F_{\text{E}}^{(2r)}(\kappa_0, \dots, \kappa_{2r}, x) = F_{\text{RF}}^{(2r)}(\kappa_0, \dots, \kappa_{2r-1}, x) \oplus \kappa_{2r}.$$

The following standard fact asserts that decryption can also be described using the alternative description above.

Fact 4. *Up to a public transformation of the output, the inverse of the round function $F_{\text{RF}}^{(2r)}$ (applied in decryption) can also be described using the alternative description using super-Sboxes. The alternative description uses the inverse Sbox and MDS matrix operations, with the roles of the diagonals and anti-diagonals interchanged.*

The only observation required to obtain the alternative description of the inverse is that the order of the linear operations over $\mathbb{F}_2$, MixColumns and AddRoundKey, can be exchanged so that the inverse operations are performed in the same order as in $F_{\text{RF}}^{(2r)}$. Furthermore, after this exchange, the final MixColumns operation can be removed.

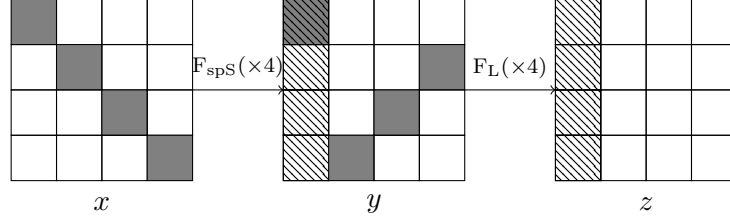


Figure 3: Two rounds of AES ($k = 4$) in the alternative representation

3.7.4 AES.

We give very partial details about the AES specification. The full details can be found in [DR20].

The AES is an AES-like cipher with the parameters $b = 128, k = 4, w = 8$.

AES has three variants depending on the master-key length (which is either 128, 192, or 256 bits). The master-key is used to compute all round-keys using a key-schedule. The number of rounds is $r \in \{10, 12, 14\}$, depending on the variant.

In this paper we assume that the round-keys $\kappa_0, \dots, \kappa_r$ are chosen independently and uniformly at random from $\mathbb{F}_2^{128}$, hence in our setting there is no distinction between the AES variants besides their number of rounds.

The AES Sbox is defined by $F_{\text{PS}}(x) = AF(x^{2^8-2})$, where the exponentiation is over $\mathbb{F}_{2^8}$, and AF is an affine transformation over $\mathbb{F}_2^8$, whose details are not important for this paper. The details of the MDS matrices $F_L^{(j)}$ for $j \in [4]$ are also not important for this paper.

In the following, we denote the r -round encryption function of AES by $\text{AES}_E^{(r)}$, its round function by $\text{AES}_{\text{RF}}^{(r)}$, its super-Sbox by AES_{spS} , its masked Sbox by AES_S , and its MDS matrix by AES_L .

4 Differential Analysis of AES

We prove our main results for AES. We start with the main tools that we use.

4.1 Tools Used in Analysis

We specify the tools that we use in the analysis as lemmas. These lemmas will be proved in the remainder of this paper.

4.1.1 Norm bounds for MDS mappings.

We derive useful norm bounds for MDS mappings, proved in Section 5. Consider the following setting.

Setting 1. Let $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ be an MDS mapping. Let $\mu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be a product distribution, namely $\mu(x) = \prod_{i \in [k]} \mu^{\subseteq \{i\}}(x_i)$ for all $x \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$. Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be the output distribution defined by $\nu(x) = \Pr_{x' \sim \mu}[L(x') = x]$.

The following well-known result (c.f., [LTV21]) asserts that when applied to product distributions, the MDS mapping preserves the 2 and ∞ norms of any input coordinate subset, when considering an output coordinate subset of the same size (or larger).

Lemma 1 (MDS diffusion). Assume Setting 1. Let $t \in [k]$. Then, for any $\mathcal{S} \subseteq [k]$ such that $t \geq |\mathcal{S}| \geq 1$,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq M[\mu^{\subseteq \mathcal{S}}] = \prod_{i \in \mathcal{S}} M[\mu^{\subseteq \{i\}}], \text{ and} \quad (9)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq \text{Col}[\mu^{\subseteq \mathcal{S}}] = \prod_{i \in \mathcal{S}} \text{Col}[\mu^{\subseteq \{i\}}]. \quad (10)$$

We examine Setting 1 in cases where t (the output coordinate subset size) is smaller than $|\mathcal{S}|$ (the input coordinate subset size), improving Lemma 1 in such cases. We prove such bounds via additional norms of the input distribution.

Lemma 2 (MDS contraction). Assume Setting 1. Let $q, m \geq 0$ be integers such that $t = q(m-1)$. Let $\mathcal{S} \subseteq [k]$ such that $|\mathcal{S}| = q \cdot m = t + q$. Then,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \{i\}}\|_m. \quad (11)$$

Remark 1 (Comparison between Lemma 1 and Lemma 2). In the setting of Lemma 2, assume for simplicity that $\|\mu^{\subseteq \{i\}}\|_m = \|\mu^{\subseteq \{i^*\}}\|_m$ and $\|\mu^{\subseteq \{i\}}\|_\infty = \|\mu^{\subseteq \{i^*\}}\|_\infty$ for some $i^* \in \mathcal{S}$ and all $i \in \mathcal{S}$.

Then, by Lemma 2,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq \|\mu^{\subseteq \{i^*\}}\|_m^{m \cdot q},$$

while by Lemma 1,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq \|\mu^{\subseteq \{i^*\}}\|_\infty^t = \|\mu^{\subseteq \{i^*\}}\|_\infty^{(m-1)q}.$$

Denote $\rho = \mu^{\subseteq \{i^*\}}$. Since,

$$\|\rho\|_m^{m \cdot q} = \left(\sum_x |\rho(x)|^m \right)^q \leq (\|\rho\|_\infty)^{(m-1)q} \left(\sum_x |\rho(x)| \right)^q = \|\rho\|_\infty^t,$$

Lemma 2 generally gives a better bound in this setting.

Next, we consider bounds via Fourier norms of the input distribution.

Lemma 3 (MDS Fourier contraction). Assume Setting 1. Let $t, m \in [k]$. Let $\mathcal{S} \subseteq [k]$ be a subset such that $|\mathcal{S}| = t \cdot m$. Then,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} \max_{j \in \mathcal{S}} \{\widetilde{\|\mu^{\subseteq \{j\}}\|_{2m}^{2m \cdot t}}\}, \text{ and} \quad (12)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} + 2^{-t \cdot n} (2^{t \cdot n} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \{j\}}]^t\} - 1)^m. \quad (13)$$

Moreover, assuming $m \geq 2$,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} (\max_{j \in \mathcal{S}} \{\|\mu^{\subseteq \{j\}}\|_m^{m \cdot t}\} - 1), \text{ and} \quad (14)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} (2^{t \cdot n} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \{j\}}]^t\} - 1)^{m/2}. \quad (15)$$

We note that (13) and (15) are obtained using Fourier analysis, even though this is not explicit in the bounds.

Remark 2 (Comparison between Lemma 2 and Lemma 3). *Consider the following setting: let $\mathcal{S} \subseteq [k]$ such that $|\mathcal{S}| = 2t$ (i.e., $m = 2$ in both lemmas). Assume that $\|\mu^{\subseteq \{j\}}\|_2^2 \leq \|\mu^{\subseteq \{j^*\}}\|_2^2$ for some $j^* \in \mathcal{S}$ and all $j \in \mathcal{S}$. Then, by (11) of Lemma 2,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{M}[\nu^{\subseteq \mathcal{T}}]\} \leq \|\mu^{\subseteq \{j^*\}}\|_2^{2t} = \text{Col}[\mu^{\subseteq \{j^*\}}]^t.$$

By (14) of Lemma 3 and Proposition 6,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{M}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} + 2^{-t \cdot n} (\|\mu^{\subseteq \{j^*\}}\|_2^{2t} - 1) = \text{Col}[\mu^{\subseteq \{j^*\}}]^t.$$

Hence, the two lemmas give identical bounds in this case. Generally, the lemmas are incomparable. However, assume that $|\mathcal{S}| > 2t$ and the input distribution μ has small Fourier coefficients, which is the case in several settings we consider. Then, Lemma 3 can give very strong results (the distance from uniform decays exponentially with $m = |\mathcal{S}|/t$), while Lemma 2 is limited as it does not directly depend on the Fourier transform of μ .

4.1.2 Norm bounds for an Sbox layer by spectral techniques.

We now state a lemma which bounds the collision probability for an Sbox layer using spectral techniques (as described in Section 2.4). We first define activity patterns.

Definition 14 (Activity pattern). *For every $\mathcal{D} \subseteq [k]$ the corresponding activity pattern over $(\mathbb{F}_2^n)^k$ is the set $\mathcal{Z}_{\mathcal{D}} = \{x \in (\mathbb{F}_2^n)^k : x_i = \mathbf{0}_n \iff i \in \mathcal{D}\}$.*

Lemma 4 (Spectral bound). *Let F be a keyed permutation over $\mathbb{F}_2^n$ such that the second largest eigenvalue of $(\mathfrak{D}_F^-)^T \mathfrak{D}_F^-$ is at most λ . Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a keyed Sbox layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$ (i.e., the 0 input difference has 0 probability). Moreover, assume that $\sum_{j=1}^k \binom{k}{j} \lambda^j \leq 1$. For every $1 \leq t \leq k$, let $\mathbf{C}_t, \mathbf{M}_t$ be constants such that:*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{C}_t \text{ and } \max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{M}[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{M}_t,$$

and $\mathbf{MN}_1$ be a constant such that $\min_{\mathcal{T} \subseteq [k], |\mathcal{T}|=1} \{\text{MN}[\nu^{\subseteq \mathcal{T}}]\} \geq \mathbf{MN}_1$.

For $1 \leq d \leq k-1$, let $\mathbf{Z}_d \in \{0, 1\}$ be an indicator such that if $\mathbf{Z}_d = 0$, then for any $\mathcal{D} \subset [k]$ with $|\mathcal{D}| = d$, $\Pr_{x \sim \nu}[x \in \mathcal{Z}_{\mathcal{D}}] = 0$.

Denote by $\xi = (\mathfrak{D}_F)^{\otimes k} \nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the keyed Sbox layer.

Then,

$$\begin{aligned} \text{Col}[\xi] \leq & (2^n - 1)^{-k} \left((\mathbf{P}_\emptyset)^2 + \sum_{j=1}^k \binom{k}{j} \lambda^j ((2^n - 1)^j \mathbf{C}_j - (\mathbf{P}_\emptyset)^2) \right) \\ & + \sum_{\mathcal{D} \subsetneq [k], \mathcal{D} \neq \emptyset} \mathbf{Z}_{|\mathcal{D}|} (2^n - 1)^{|\mathcal{D}|-k} \left((\mathbf{P}_{\mathcal{D}})^2 + \sum_{j=1}^{k-|\mathcal{D}|} \binom{k-|\mathcal{D}|}{j} \lambda^j ((2^n - 1)^j \mathbf{M}_{j+|\mathcal{D}|} \cdot \mathbf{M}_{|\mathcal{D}|} - (\mathbf{P}_{\mathcal{D}})^2) \right), \end{aligned} \quad (16)$$

where $\{\mathbf{P}_{\mathcal{D}}\}_{\mathcal{D} \subsetneq [k]}$ are variables that satisfy the following constraints:

$$\mathbf{P}_\emptyset + \sum_{\mathcal{D} \subsetneq [k], \mathcal{D} \neq \emptyset} \mathbf{Z}_{|\mathcal{D}|} \mathbf{P}_{\mathcal{D}} = 1, \quad (17)$$

$$0 \leq \mathbf{P}_\emptyset \leq \min\{1 - k \cdot \mathbf{M}_1 + \binom{k}{2} \mathbf{M}_2, 1\}, \quad (18)$$

for every $\mathcal{D} \neq \emptyset$,

$$0 \leq \mathbf{P}_{\mathcal{D}} \leq \mathbf{M}_{|\mathcal{D}|}, \quad (19)$$

and moreover for every $d \in [k]$

$$\mathbf{P}_{\{d\}} \geq \mathbf{M}_1 - \sum_{j=2}^{k-1} \binom{k-1}{j-1} \mathbf{M}_j. \quad (20)$$

The lemma bounds $\text{Col}[\xi]$ using a quadratic maximization problem, whose variables $\{\mathbf{P}_{\mathcal{D}}\}_{\mathcal{D} \subsetneq [k]}$ correspond to the probability of the output difference to be in each activity pattern. Appendix A describes an algorithm that solves this quadratic maximization problem efficiently, while the code that implements this algorithm is given in Appendix B.

The lemma will be used several times - in analysis of the super-Sbox DDT (with the AES Sbox layer), and in computation of the Frobenius norm of 4-round AES DDT (with the AES super-Sbox layer).

Remark 3. We can avoid solving the quadratic maximization problem by deriving a simplified and slightly weaker variant of Lemma 4. This is done by ignoring constraints (17) and (20) (which can only increase the maximal value of the right-hand-side of (16)), and setting all the variables to their maximal possible values according to constraints (18) and (19). In our applications, the simplified lemma gives slightly weaker results than with Lemma 4.

Remark 4. We can strengthen Lemma 4 by defining $\mathbf{MN}_t$ for any $1 \leq t \leq k$ such that

$$\min_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{MN}[\nu^{\subseteq \mathcal{T}}]\} \geq \mathbf{MN}_t,$$

and using these values to derive additional constraints on the pattern probabilities $\mathbf{P}_{\mathcal{D}}$. However, such strengthened variants do not seem to have a substantial effect on our results.

4.2 Differential Analysis of the AES Super-Sbox

We derive properties of the AES super-Sbox DDT.

4.2.1 Properties of the AES Sbox DDT (and its inverse).

Recall that the masked AES Sbox function is denoted by $\text{AES}_S: \mathbb{F}_2^8 \times \mathbb{F}_2^8 \rightarrow \mathbb{F}_2^8$. We list properties of $\mathfrak{D}_{\text{AES}_S}$ (and $(\mathfrak{D}_{\text{AES}_S})^T$) along with constants that we use. These properties include the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_S}^-)^T \mathfrak{D}_{\text{AES}_S}^-$ and the norms and Fourier norms of each column and row of $\mathfrak{D}_{\text{AES}_S}$.

The norms are derived from the following fact: any column $\mathfrak{D}_{\text{AES}_S}[:, y]$, and row $\mathfrak{D}_{\text{AES}_S}[x, :]$ (where $x, y \neq \mathbf{0}_8$) has 129 entries with value 0, 126 entries with value $\frac{2}{256}$ and a single entry with value $\frac{4}{256}$.

The Fourier norms are derived from the following fact: the absolute value of the (normalized) Fourier transform of any column $\mathfrak{D}_{\text{AES}_S}[:, y]$, and row $\mathfrak{D}_{\text{AES}_S}[x, :]$ (where $x, y \neq \mathbf{0}_8$) has 32 entries with value 0, 84 entries with value $\frac{8}{256}$, 74 entries with value $\frac{16}{256}$, 52 entries with value $\frac{24}{256}$ and 13 entries with value $\frac{32}{256}$. In addition, the trivial coefficient on $\widehat{\mathbf{0}}_8$ is 1.

The properties are:

1. The second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_S}^-)^T \mathfrak{D}_{\text{AES}_S}^-$ is $\lambda_{\text{AES}_S} \triangleq \left(\frac{18}{256}\right)^2$.
2. For any input difference $x \in \mathbb{F}_2^{8(-)}$, $\mathcal{N}_{\text{AES}_S}(\infty) \triangleq \|\mathfrak{D}_{\text{AES}_S} e_x\|_\infty = \frac{4}{256}$, and for any $\ell \in \{2, 3, 4\}$, $\mathcal{N}_{\text{AES}_S}(\ell) \triangleq \|\mathfrak{D}_{\text{AES}_S} e_x\|_\ell^\ell = \frac{126 \cdot 2^\ell + 4^\ell}{256^\ell} = \frac{126 + 2^\ell}{128^\ell}$.
3. For any input difference $x \in \mathbb{F}_2^{8(-)}$, and $\ell \in \{2, 3, 4, 6, 8\}$, $\mathcal{F}_{\text{AES}_S}(\ell) \triangleq \|\widetilde{\mathfrak{D}_{\text{AES}_S} e_x}\|_\ell^\ell = 1 + \frac{84 \cdot 8^\ell + 74 \cdot 16^\ell + 52 \cdot 24^\ell + 13 \cdot 32^\ell}{256^\ell}$.

The code used for deriving these properties is given in Appendix B.1.

4.2.2 Differential analysis of the AES super-Sbox.

Our main technical result regarding the AES super-Sbox bounds the 2-norm of its output difference distribution as a function of the number of active Sboxes at the input.

Lemma 5. *Let $x \in (\mathbb{F}_2^8)^4$ be a non-zero input difference to the AES super-Sbox, AES_{spS} . Denote by $\xi \in \mathbb{R}^{(\mathbb{F}_2^8)^4}$ the output difference distribution. Then,*

$$\text{Col}[\xi] = \text{Col}[\mathfrak{D}_{\text{AES}_{\text{spS}}} e_x] \leq \mathcal{N}_{\text{AES}_{\text{spS}}}^{(\text{wt}(x))}(2) \triangleq 2^{-32}(1 + f(\text{wt}(x)))$$

where $f(1) = \frac{1}{5.05}$, $f(2) = \frac{1}{24}$, $f(3) = \frac{1}{283.4}$, $f(4) = \frac{1}{438}$. Hence,

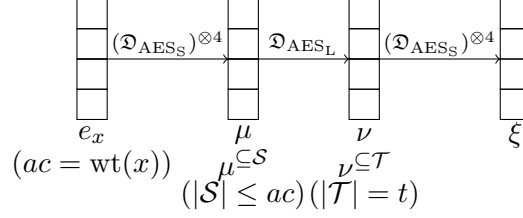
$$\|\mathfrak{D}_{\text{AES}_{\text{spS}}}^-\|_{1,2}^2 \leq 2^{-32} \left(1 + \frac{1}{5.05}\right).$$

Moreover, all of these properties also hold for the inverse $(\text{AES}_{\text{spS}})^{-1}$.

Proof. We will prove the result for AES_{spS} . It also holds for the inverse AES super-Sbox by symmetry (as shown in Fact 4) and by observing that every property of an AES super-Sbox that we use in the proof is also shared by the inverse.

Define the following two distributions over $(\mathbb{F}_2^8)^4$: μ is the distribution after first key addition and Sbox layer, and ν is the distribution after the MixColumns operation (see figure below).

Note that ξ is obtained after a masked Sbox layer applied to ν . Thus, our goal is to apply Lemma 4 to ν in order to bound $\text{Col}[\xi]$. Hence, we need estimates for the values $\mathbf{C}_t = \mathbf{C}_t(\nu)$, $\mathbf{M}_t = \mathbf{M}_t(\nu)$ for $1 \leq t \leq 4$ and $\mathbf{MN}_1 = \mathbf{MN}_1(\nu)$. In addition, we need values for the indicators $\mathbf{Z}_d = \mathbf{Z}_d(\nu)$ for $1 \leq d \leq 3$.



Bounds depending on $\text{wt}(x)$.

Denote $ac = \text{wt}(x)$, namely there are ac active Sboxes at the input of the MixColumns operation. We condition the analysis on the value of ac , and write the values we need to analyze depending on ac and t, d as $\mathbf{Z}_d^{(ac)}, \mathbf{C}_t^{(ac)}, \mathbf{M}_t^{(ac)}$ and $\mathbf{MN}_1^{(ac)}$.

By Definition 1, the output always has weight (at least) $5 - ac$. Namely, there are at most $4 - (5 - ac) = ac - 1$ non-active Sboxes at the output, so the probability of having at least ac non-active Sboxes at the output is 0. Hence, we set $\mathbf{Z}_d^{(ac)} = 0$ for all $d \geq ac$, and $\mathbf{Z}_d^{(ac)} = 1$ otherwise.

The parameter t of $\mathbf{C}_t^{(ac)}$ and $\mathbf{M}_t^{(ac)}$ in Lemma 4 specifies the size of the coordinate subset $\mathcal{T}$ at the MDS layer output. We bound $\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\}$ and $\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{M}[\nu^{\subseteq \mathcal{T}}]\}$, thus bounding $\mathbf{C}_t^{(ac)}$ and $\mathbf{M}_t^{(ac)}$, respectively. Similarly, we lower bound $\min_{\mathcal{T} \subseteq [k], |\mathcal{T}|=1} \{\mathbf{MN}[\nu^{\subseteq \mathcal{T}}]\}$ to obtain $\mathbf{MN}_1^{(ac)}$.

For this purpose, observe that μ is a product distribution at the entry of the MDS matrix (which acts deterministically on differences over $\mathbb{F}_2$), as in Setting 1. Hence, Lemma 1, Lemma 2 and Lemma 3 are applicable.

Based on t , and on the specific result we apply (one of the lemmas 1, 2 and 3), we select a set of active Sboxes $\mathcal{S}$ with $|\mathcal{S}| \leq ac$ at the input of the MDS layer and consider the distributions $\mu^{\subseteq \{i\}} = \mathcal{D}_{\text{AES}_S} e_{x_i}$ for $i \in \mathcal{S}$ ($x_i \in \mathbb{F}_2^8$). As stated at the beginning of this section, all such distributions have the same properties (in particular, norms and Fourier norms). We give the details below.

Bounds by Lemma 1. We first apply Lemma 1 using the AES Sbox constants defined above. To apply this lemma, we select a set $\mathcal{S}$ of active Sboxes at the input to the MDS layer of size $s \triangleq |\mathcal{S}| = \min\{t, ac\}$.

By (10), $\mathbf{C}_t^{(ac)} \leq \max_{y \in \mathbb{F}_2^8, y \neq \mathbf{0}_8} \{(\|\mathcal{D}_{\text{AES}_S} e_y\|_2^2)^s\} \leq (\mathcal{N}_{\text{AES}_S}(2))^s$.

Similarly, by (9), $\mathbf{M}_t^{(ac)} \leq \max_{y \in \mathbb{F}_2^8, y \neq \mathbf{0}_8} \{(\|\mathcal{D}_{\text{AES}_S} e_y\|_\infty)^s\} \leq (\mathcal{N}_{\text{AES}_S}(\infty))^s$.

Bounds by Lemma 2. We apply Lemma 2 (in particular, (11)) to improve the results in cases where the set of size t is smaller by 1 than the number of active Sboxes ac . Specifically, in all the cases below we select the set $\mathcal{S}$ containing all the active Sboxes (i.e., $|\mathcal{S}| = ac$), and we have $t = |\mathcal{S}| - 1$. For Lemma 2, this implies that $q = 1$ and $m = |\mathcal{S}| = ac$. With these parameters,

$$\mathbf{M}_t^{(ac)} \leq \max_{y \in \mathbb{F}_2^8, y \neq \mathbf{0}_8} \{\|\mathcal{D}_{\text{AES}_S} e_y\|_{ac}^{ac}\} \leq \mathcal{N}_{\text{AES}_S}(ac).$$

In particular, we obtain for $(ac = 2, t = 1)$, $\mathbf{M}_1^{(2)} \leq \mathcal{N}_{\text{AES}_S}(2)$, for $(ac = 3, t = 2)$, $\mathbf{M}_2^{(3)} \leq \mathcal{N}_{\text{AES}_S}(3)$, and for $(ac = 4, t = 3)$, $\mathbf{M}_3^{(4)} \leq \mathcal{N}_{\text{AES}_S}(4)$.

Bounds by Lemma 3. We apply Lemma 3 (in particular (12) and (14)) to improve the results. Specifically, we apply it with $t = 1$ and $ac \in \{2, 3, 4\}$ by selecting $\mathcal{S}$ to contain all the active Sboxes. Hence, in Lemma 3, $|\mathcal{S}| = ac$, $t = 1$ and $m = |\mathcal{S}| = ac$.

We obtain, $\mathbf{C}_1^{(ac)} \leq 2^{-8} \cdot \mathcal{F}_{\text{AES}_S}(2 \cdot ac)$ and $\mathbf{M}_1^{(ac)} \leq 2^{-8} \cdot \mathcal{F}_{\text{AES}_S}(ac)$.

We further apply Lemma 3 for $t = 2$ and $ac = 4$ (hence $m = 2$). We obtain $\mathbf{C}_2^{(4)} \leq 2^{-16}(\mathcal{F}_{\text{AES}_S}(4))^2$ and $\mathbf{M}_2^{(4)} \leq 2^{-16}(\mathcal{F}_{\text{AES}_S}(2))^2$.

Finally, by (14) with $t = 1$ and $m = ac$, $\mathbf{MN}_1^{(ac)} \geq 2^{-8} - 2^{-8}(\mathcal{F}_{\text{AES}_S}(ac) - 1)$.

Plugging these into Lemma 4 proves the result. The code used for the calculations is given in Appendix B.2. We give more details below for the simple case of a single active Sbox, namely $ac = \text{wt}(x) = 1$.

4.2.3 $\text{wt}(x) = 1$.

We have $\mathbf{Z}_d^{(1)} = 0$ for all $d \in \{1, 2, 3\}$ and hence $\mathbf{P}_\emptyset = 1$. Therefore, we only need bounds on $\mathbf{C}_t^{(1)}$ for $t \in [4]$ in order to apply Lemma 4. As above, by Lemma 1, $\mathbf{C}_t^{(1)} \leq \mathcal{N}_{\text{AES}_S}(2)$ for $t \in [4]$. Plugging these into Lemma 4, we obtain

$$\text{Col}[\xi] \leq 255^{-4} \left(1 + \sum_{j=1}^4 \binom{4}{j} (\lambda_{\text{AES}_S})^j (\mathcal{N}_{\text{AES}_S}(2) \cdot 255^j - 1) \right) \leq 2^{-32} \left(1 + \frac{1}{5.05} \right). \quad \blacksquare$$

We now bound $\|\text{AES}_{\text{SPS}}^-\|_{\text{F}}^2$.

Lemma 6. *The Frobenius norm of the AES super-Sbox DDT satisfies*

$$\|\mathfrak{D}_{\text{AES}_{\text{SPS}}}^-\|_{\text{F}}^2 \leq 1 + \frac{1}{426}.$$

Consequently, by Proposition 1, the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{SPS}}}^-)^{\text{T}} \mathfrak{D}_{\text{AES}_{\text{SPS}}}^-$ is at most $\frac{1}{426}$. Moreover, this holds for the inverse AES super-Sbox.

Proof. We prove the result for AES_{SPS} . The proof for the inverse is similar.

Conditioning on $\text{wt}(x) \in [4]$, we obtain

$$\begin{aligned} \|\mathfrak{D}_{\text{AES}_{\text{SPS}}}^-\|_{\text{F}}^2 - 1 &= \sum_{x, y \in (\mathbb{F}_2^8)^{4(-)}} \mathfrak{D}_{\text{AES}_{\text{SPS}}}^-[x, y]^2 - 1 = \sum_{y \in (\mathbb{F}_2^8)^{4(-)}} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y] - 1 \\ &\leq \sum_{i=1}^4 \left| \{x \in (\mathbb{F}_2^8)^{4(-)} \mid \text{wt}(x) = i\} \right| \left(\max_{\{y \mid \text{wt}(y)=i\}} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y]\} - 2^{-32} \right) \\ &\leq \sum_{i=1}^3 \left| \{x \in (\mathbb{F}_2^8)^{4(-)} \mid \text{wt}(x) = i\} \right| \left(\max_{\{y \mid \text{wt}(y)=i\}} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y]\} - 2^{-32} \right) \\ &\quad + 2^{32} \left(\max_{\{y \mid \text{wt}(y)=4\}} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y]\} - 2^{-32} \right). \end{aligned} \quad (21)$$

The final inequality holds since

$$\max_{\{y \mid \text{wt}(y)=4\}} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y]\} - 2^{-32} \geq \frac{1}{2^{32} - 1} - 2^{-32} > 0.$$

For $i \in [4]$, the expressions $\max_{\{y \mid \text{wt}(y)=i\}} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}}^- e_y]\}$ are bounded in Lemma 5, while

$$\left| \{x \in (\mathbb{F}_2^8)^{4(-)} \mid \text{wt}(x) = i\} \right| \leq \binom{4}{i} \cdot 2^{8 \cdot i}.$$

Plugging these into (21) and using the results of Lemma 5, we deduce

$$\|\mathfrak{D}_{\text{AES}_{\text{spS}}}^{-}\|_{\mathbb{F}}^2 - 1 \leq 2^{-32} \left(\left(4 \cdot 2^8 \cdot \frac{1}{5} + 6 \cdot 2^{16} \cdot \frac{1}{24} + 4 \cdot 2^{24} \cdot \frac{1}{283} \right) + 2^{32} \cdot \frac{1}{438} \right) \leq \frac{1}{426}.$$

■

4.3 Overview – Bounds on the 1-to-2 and Frobenius Norms of 4-Round AES DDT

We give an overview of the bounds on the 1-to-2 and Frobenius Norms of 4-Round AES DDT. The general approach is as follows. We omit the application of the final linear layer, analyzing “3.5 rounds”. We view MixColumns applied to the full AES state as a linear mapping from anti-diagonals to diagonals, acting on $k = 4$ words of $n' = k \cdot n = 4 \cdot 8 = 32$ bits. Thus, MixColumns is an MDS mapping (as in Definition 2) from anti-diagonals to diagonals, and analogs of lemmas 1, 2 and 3 are derived for MixColumns applied to the full state. Overall, 3.5-round AES is composed of a super-Sbox layer (placing the results in the anti-diagonals), followed by an MDS mapping (from anti-diagonals to diagonals) and a final super-Sbox layer applied to the diagonals. This is the same high-level structure as the AES super-Sbox.

4.3.1 The 1-to-2 norm of 4-round (3.5-round) AES DDT.

We overview the bound on the 1-to-2 norm of 4-round AES DDT. Fix a non-zero difference at the input of 3.5-round AES and denote the output distribution by ξ' . Denote the difference distributions after the first super-Sbox layer and after 2 rounds by μ and ν , respectively. Note that ν is obtained by applying MixColumns to μ , while $\xi' = (\mathfrak{D}_{\text{AES}_{\text{spS}}})^{\otimes 4} \nu$ is obtained after a second super-Sbox layer (as in Figure 4).

Assume that every input difference in the support of ν has at least 3 active Sboxes in each of the $k = 4$ diagonals. By Lemma 5, the output difference distribution ξ' has 2-norm at most $(2^{-32}(1 + f(3)))^4 = 2^{-128}(1 + \frac{1}{283.4})^4 \leq 2^{-128}(1 + \frac{1}{70})$. While our assumption does not hold, it holds approximately: first, as derived in [KS07], any output difference of an active super-Sbox has probability at most $2^{-28.2}$. Therefore, μ has at least one anti-diagonal with this property. By the analog of Lemma 1 for MixColumns, each diagonal of ν attains every input difference with probability at most $2^{-28.2}$. As there are at most $1 + \binom{4}{1}2^8 + \binom{4}{2}2^{16} < 2^{18.6}$ possible diagonal values with less than 3 active Sboxes, we conclude (by a union bound over the 4 diagonals) that with probability at least $1 - 4 \cdot 2^{18.6} \cdot 2^{-28.2} \geq 1 - 2^{-7.6}$, a difference drawn from ν has at least 3 active Sboxes in each of the 4 diagonals. Thus, our simplifying assumption holds approximately, and the final bound is $2^{-128}(1 + \frac{1}{50})$ rather than $2^{-128}(1 + \frac{1}{70})$.

4.3.2 The Frobenius norm of 4-round (3.5-round) AES DDT.

We give an overview of the bound on the Frobenius norm of the 4-round AES DDT. As 3.5-round AES has the same high-level structure as the AES super-Sbox, the proof resembles the proof of Lemma 6.

First, we derive an analog of Lemma 5 for 3.5-round AES. This is done by dividing the non-zero input differences to 3.5-round AES into 5 subsets $i \in \{0, 1, 2, 3, 4\}$ according to the minimal number of active Sboxes in a diagonal (which is between 0 and 4), and bounding the maximal output difference collision probability for an input difference in each subset. Similarly to the proof of Lemma 5, Lemma 4 is used to bound the output difference collision probability, while relying on

the analogs of lemmas 1, 2 and 3 to bound the constants required for application of Lemma 4 to the final super-Sbox layer.

Recall that the proof of Lemma 5 used exact bounds on several norms of the AES Sbox. On the other hand, we only have estimates for a few norms of the super-Sbox: the output 2-norm depending on the number of active input difference Sboxes (Lemma 5), the output infinity norm (derived in [KS07]), as well as the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{spS}}}^-)^T \mathfrak{D}_{\text{AES}_{\text{spS}}}^-$ (Lemma 6). Consequently, we apply somewhat rougher bounds, using the analog of Lemma 1, as well as (13), (15) of Lemma 3.

Finally, an analog of Lemma 6 bounds the Frobenius norm of 3.5-round AES DDT by summing the contribution of each subset of input differences $i \in \{0, 1, 2, 3, 4\}$ to the output collision probability. For input differences in subset 0 (with a non-active diagonal), the worst-case bound of $2^{-128}(1 + \frac{1}{50})$ (derived by bounding the 1-to-2 norm) is used, while the analog of Lemma 5 is used to bound the contribution of input differences in each subset $i \in \{1, 2, 3, 4\}$.

Calculation sketch. We sketch the calculation for input differences in subset 4, which has 4 active Sboxes in each input diagonal. This is the largest subset that contributes the most to our bound on the Frobenius norm. Fix a difference at the input of 3.5-round AES with 4 active Sboxes in each diagonal. As previously, denote the difference distributions after the first super-Sbox layer and after 2 rounds by μ and ν , respectively. In the application of Lemma 4 to ν , the dominant term is the first one, i.e., $(2^{32} - 1)^{-4} \left((P_\emptyset)^2 + \sum_{j=1}^4 \binom{4}{j} \lambda^j ((2^{32} - 1)^j C_j - (P_\emptyset)^2) \right)$. This term accounts for the output collision probability among differences drawn from ν where all diagonals are active. For simplicity, here we use the trivial bound $P_\emptyset = 1$ and approximate $2^{32} - 1 \approx 2^{32}$. As noted above, by Lemma 6 we have $\lambda \leq \frac{1}{426}$. Hence we (approximately) bound the dominant term by

$$2^{-128} \left(1 + \sum_{j=1}^4 \binom{4}{j} \frac{1}{426^j} (2^{32 \cdot j} C_j - 1) \right). \quad (22)$$

It remains to bound C_j for $j \in \{1, 2, 3, 4\}$, which is done by applying the analog of Lemma 1, and (13) of Lemma 3 to μ . Since the input difference to 3.5-round AES has 4 active Sboxes in each diagonal, by Lemma 5 each anti-diagonal of μ has collision probability at most $2^{-32}(1 + \frac{1}{438})$.

In order to bound C_1 , we apply the analog of (13) of Lemma 3 with $n = 32$, $S = [4]$, $t = 1$ (hence $m = 4$) and obtain $C_1 \leq 2^{-32} + 2^{-32}(2^{32} \cdot 2^{-32}(1 + \frac{1}{438}) - 1)^4 = 2^{-32} + 2^{-32} \cdot \frac{1}{438^4} < 2^{-32}(1 + 2^{-35})$. The contribution of $j = 1$ to (22) is thus at most $2^{-128}(4 \cdot \frac{1}{426} \cdot 2^{-35}) \leq 2^{-128} \cdot 2^{-41}$. Thus, the contraction of the 2-norm for marginals on single coordinates (diagonals) is mostly a result of the MDS layer (the analog of (13) of Lemma 3) applied to μ , rather than the term $\frac{1}{426}$ contributed by the super-Sbox layer applied to ν .

In order to bound C_2 , we apply the analog of (13) of Lemma 3 with $n = 32$, $S = [4]$, $t = 2$ (hence $m = 2$) and obtain $C_2 \leq 2^{-64} + 2^{-64}(2^{64} \cdot 2^{-64}(1 + \frac{1}{438})^2 - 1)^2 < 2^{-64}(1 + \frac{1}{218^2}) < 2^{-64} \cdot (1 + 2^{-15.5})$. The contribution of $j = 2$ to (22) is thus at most $2^{-128}(6 \cdot \frac{1}{426^2} \cdot 2^{-15.5}) < 2^{-128} \cdot 2^{-30.38}$. Hence, the contraction of the 2-norm for marginals on 2 coordinate (diagonal) subsets is a result of both the MDS layer applied to μ and the term $\frac{1}{426^2}$ contributed by the super-Sbox layer applied to ν .

In order to bound C_3 and C_4 , we simply apply the analog of Lemma 1. For C_3 , we apply it with $|S| = 3$ (the choice of S is not important) and $t = 3$, obtaining $C_3 \leq 2^{-32 \cdot 3}(1 + \frac{1}{438})^3 \leq 2^{-96}(1 + 2^{-7.18})$. The contribution of $j = 3$ to (22) is at most $2^{-128}(4 \cdot \frac{1}{426^3} \cdot 2^{-7.18}) < 2^{-128} \cdot 2^{-31.38}$.

For C_4 , we apply the analog of Lemma 1 with $S = [4]$ and $t = 4$, obtaining $C_4 \leq 2^{-32 \cdot 4}(1 + \frac{1}{438})^4 \leq 2^{-128}(1 + 2^{-6.76})$. The contribution of $j = 4$ to (22) is at most $2^{-128}(\frac{1}{426^4} \cdot 2^{-6.76}) < 2^{-128} \cdot 2^{-41.69}$. Hence, the contraction of the 2-norm for marginals on 3 and 4 coordinate subsets is a result of the terms $\frac{1}{426^3}$ and $\frac{1}{426^4}$, respectively, contributed by the super-Sbox layer applied to ν .

Summing up the contributions of $j \in \{1, 2, 3, 4\}$, we bound the dominant term (22) by $2^{-128}(1 + 2^{-41} + 2^{-30.38} + 2^{-31.38} + 2^{-41.69}) < 2^{-128}(1 + 2^{-29.7})$. Thus (after subtracting 1), the contribution of input differences in subset 4 (which is of size less than 2^{128}) to the Frobenius norm of 3.5-round AES DDT is at most $2^{-29.7}$. This is close to our precise bound on the Frobenius norm, which is $1 + 2^{-29.3}$. Thus, by Proposition 1, the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^T \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-$ is at most $2^{-29.3}$.

4.4 The 1-to-2 Norm of 4-Round AES DDT

In this section, we bound the 1-to-2 Norm of 4-Round AES DDT. We begin by specifying additional tools used in the analysis.

4.4.1 Norm bounds for square MDS mappings.

We consider the setting where MDS mappings are applied in parallel to a square AES-like state, and list similar results to those of Section 4.1 in this setting. We use the square indexing defined in Section 3.7.

Define the following additional notation: given a linear mapping $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$, denote the column-wise application of L by $LC = LC_L: (\mathbb{F}_2^n)^{k^2} \rightarrow (\mathbb{F}_2^n)^{k^2}$ for $x \in (\mathbb{F}_2^n)^{k^2}$ by $(LC(x))_{c_j} = L(x_{c_j})$ for $j \in [k]$. The following property is well-known.

Lemma 7 (Square MDS basic property). *Let $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ be an MDS mapping. Then, for any non-zero $x \in (\mathbb{F}_2^n)^{k^2}$,*

$$\text{wt}_{\mathcal{H}}(x) + \text{wt}_{\mathcal{G}}(LC(x)) \geq k + 1,$$

$$\text{wt}_{\mathcal{G}}(x) + \text{wt}_{\mathcal{H}}(LC(x)) \geq k + 1.$$

Consequently, LC is an MDS mapping from anti-diagonals $\{\mathcal{H}_i\}_{i \in [k]}$ to diagonals $\{\mathcal{G}_i\}_{i \in [k]}$ acting on words of size $n \cdot k$ bits (and also vice-versa from the diagonals to the anti-diagonals).

Proof. Since x is non-zero, there exists $j \in [k]$ such that $\text{wt}(x_{c_j}) > 0$. Moreover,

$$\text{wt}(x_{c_j}) \leq \text{wt}_{\mathcal{H}}(x),$$

since each column intersects each anti-diagonal on one element. In addition,

$$\text{wt}(L(x_{c_j})) \leq \text{wt}_{\mathcal{G}}(LC(x))$$

holds similarly. Combining with Definition 2 we obtain

$$k + 1 \leq \text{wt}(x_{c_j}) + \text{wt}(L(x_{c_j})) \leq \text{wt}_{\mathcal{H}}(x) + \text{wt}_{\mathcal{G}}(LC(x)).$$

The proof of the second inequality is similar. ■

Considering Lemma 7, the analysis applied to arbitrary MDS mappings in Section 3.7 also holds for LC with appropriate adjustments. We explicitly state the analogous results that we use.

We first define the following setting.

Setting 2. *Let $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ be an MDS mapping. Define the column-wise application of L by $LC = LC_L: (\mathbb{F}_2^n)^{k^2} \rightarrow (\mathbb{F}_2^n)^{k^2}$ for $x \in (\mathbb{F}_2^n)^{k^2}$ by $(LC(x))_{c_j} = L(x_{c_j})$ for $j \in [k]$.*

Let $\mu \in \mathbb{R}^{(\mathbb{F}_2^n)^{k^2}}$ be an input distribution to LC . Assume that μ is a product distribution on the anti-diagonals $(\mathcal{H}_j)_{j \in [k]}$, namely $\mu(x) = \prod_{j \in [k]} \mu^{\subseteq \mathcal{H}_j}(x_{\mathcal{H}_j})$ for all $x \in (\mathbb{F}_2^n)^{k^2}$. Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^{k^2}}$ be the output distribution defined by $\nu(x) = \Pr_{x' \sim \mu}[LC(x') = x]$.

The following proposition is analogous to Lemma 1.

Lemma 8 (Square MDS diffusion). *Assume Setting 2. Let $t \in [k]$. Then, for any $\mathcal{S} \subseteq [k]$ such that $t \geq |\mathcal{S}| \geq 1$,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\} \leq \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \mathcal{H}_i}\|_{\infty} = \prod_{i \in \mathcal{S}} M[\mu^{\subseteq \mathcal{H}_i}], \text{ and} \quad (23)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\} \leq \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \mathcal{H}_i}\|_2^2 = \prod_{i \in \mathcal{S}} \text{Col}[\mu^{\subseteq \mathcal{H}_i}]. \quad (24)$$

In addition, the lemma holds if we interchange the roles of the diagonals $(\mathcal{G}_j)_{j \in [k]}$ and the anti-diagonals $(\mathcal{H}_j)_{j \in [k]}$ (in particular, μ is a product distribution on the sets $(\mathcal{G}_j)_{j \in [k]}$).

The following lemma is analogous to (a part of) Lemma 3.

Lemma 9 (Square MDS Fourier contraction). *Assume Setting 2. Let $t, m \in [k]$. Denote $n' = n \cdot k$. Let $\mathcal{S} \subseteq [k]$ be a subset such that $|\mathcal{S}| = t \cdot m$. Then,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\} \leq 2^{-t \cdot n'} + 2^{-t \cdot n'} (2^{t \cdot n'} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \mathcal{H}_j}]^t\} - 1)^m.$$

Moreover, assuming that $m \geq 2$,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\} \leq 2^{-t \cdot n'} (2^{t \cdot n'} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \mathcal{H}_j}]^t\} - 1)^{m/2}.$$

Furthermore, the lemma holds if we interchange the roles of the diagonals $(\mathcal{G}_j)_{j \in [k]}$ and the anti-diagonals $(\mathcal{H}_j)_{j \in [k]}$.

4.4.2 Norm bounds for an Sbox layer by expected collision probability.

We will prove a lemma that bounds the norm for an Sbox layer by expected collision probability. It will be applied to the AES super-Sbox layer (as described in Section 2.4). The lemma is technical and followed by discussion.

Lemma 10 (Bound by expected collision probability). *Let F be a keyed permutation over the space $\mathbb{F}_2^n$ such that*

$$\|\mathfrak{D}_F^-\|_{1,2}^2 = \max_{y \in \mathbb{F}_2^{n(-)}} \{\text{Col}[\mathfrak{D}_F^- e_y]\} \leq \mathbf{C}$$

for some constant $\mathbf{C} \geq 0$.

Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$. For every $1 \leq t \leq k$, let $\mathbf{M}_t$ be a constant such that

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{M}_t.$$

Moreover, for $1 \leq d \leq k-1$, let $\mathbf{Z}_d \in \{0,1\}$ be an indicator that satisfies the following constraint: if $\mathbf{Z}_d = 0$, then for any $\mathcal{D} \subseteq [k], |\mathcal{D}| = d$

$$\Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}] = 0.$$

Let $\text{ColF}: \mathbb{R} \rightarrow \mathbb{R}$ be a function (that depends on $\mathfrak{D}_F$) with the following property: if $\rho \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ is a distribution that is supported only on $\mathcal{Z}_{\emptyset}$ such that $M[\rho^{\subseteq \{i\}}] \leq \mathbf{M}$ for all $i \in [k]$, then the output

difference distribution satisfies

$$\mathbb{E}_{z \sim \rho} [\text{Col}[(\mathfrak{D}_F)^{\otimes k} e_z]] \leq \text{ColF}(\mathbf{M}).$$

Denote by $\xi = (\mathfrak{D}_F)^{\otimes k} \nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the Sbox layer. Then,

$$\text{Col}[\xi] \leq \text{ColF}\left(\frac{\mathbf{M}_1}{1 - k \cdot \mathbf{M}_1}\right) + \sum_{d=1}^{k-1} \binom{k}{d} \mathbf{Z}_d(\mathbf{M}_d)^2 \cdot \mathbf{C}^{k-d}, \quad (25)$$

assuming $k \cdot \mathbf{M}_1 < 1$.

Not including the ColF function, the right-hand-side of (25) has $2^k - 2$ terms. Each term bounds the collision probability within a pattern $\mathcal{Z}_{\mathcal{D}}$ with $|\mathcal{D}| = d$ for $1 \leq d \leq k-1$ by $(\mathbf{M}_d)^2 \cdot \mathbf{C}^{k-d}$, assuming $\mathbf{Z}_d = 1$ (we consider $\binom{k}{d}$ as a sum over all subsets of size d). This is a simple bound that is easy to use. It only takes into account the basic 1-to-2 norm of the Sbox, as well as a bound on the ∞ norm of the marginal input difference distribution on all subsets, depending on their size (in addition to the pattern indicators).

On the other hand, the ColF function that is used to obtain a refined bound for the collision probability inside the pattern $\mathcal{Z}_{\emptyset}$ is somewhat ad-hoc. We will use this bound to optimize our analysis for AES. We note, however, that the collision probability inside $\mathcal{Z}_{\emptyset}$ generally dominates $\text{Col}[\xi]$ when the Sbox is large enough, since this pattern contains most of the states. Thus, it makes sense to separately optimize the collision probability inside $\mathcal{Z}_{\emptyset}$.

4.4.3 Bound on entries of $\mathfrak{D}_{\text{AES}_{\text{spS}}}$ (and its inverse).

We will use a bound on the entries of $\mathfrak{D}_{\text{AES}_{\text{spS}}}$ (derived in [KS07]), and define a corresponding constant. Specifically, for any non-zero input difference $x \in (\mathbb{F}_2^8)^4$,

$$\mathbf{M}[\mathfrak{D}_{\text{AES}_{\text{spS}}} e_x] \leq \mathcal{N}_{\text{AES}_{\text{spS}}}(\infty) \triangleq 53 \cdot 2^{-34} \approx 1.656 \cdot 2^{-29}. \quad (26)$$

We remark that it is easy to analytically prove a slightly weaker bound that would have a negligible impact on our results.

4.4.4 The 1-to-2 norm of 4-round AES DDT.

We begin with an auxiliary result that will be used to define the ColF function of Lemma 10.

Proposition 8. *Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^8)^{16}}$ be an input difference distribution to the AES super-Sbox layer $(\text{AES}_{\text{spS}})^{\otimes 4}$ such that $\text{wt}_{\mathcal{G}}(x) = 4$ for all x in the support of ν . Assume that for each $j \in [4]$, $\mathbf{M}[\nu|_{\mathcal{G}_j}] \leq \mathbf{M}$ for some $\mathbf{M} > 0$. Then,*

$$\mathbb{E}_{x \sim \nu} [\text{Col}[(\mathfrak{D}_{\text{AES}_{\text{spS}}})^{\otimes 4} e_x]] = \mathbb{E}_{x \sim \nu} \left[\prod_{j \in [4]} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{spS}}} e_{x_{\mathcal{G}_j}}] \right] \leq 2^{-128} (2^{20.83} \mathbf{M} + 1.0142).$$

Moreover, this also holds for the inverse $(\text{AES}_{\text{spS}})^{-1}$.

Proof. We prove the result for AES_{spS} . The proof for the inverse is similar using Fact 4.

Partition the support of ν into 4 sets. For $i \in [4]$, the set

$$\mathcal{Y}_i = \{y \in (\mathbb{F}_2^8)^{16} \mid \min_{j \in [4]} \{\text{wt}(y_{\mathcal{G}_j})\} = i\}$$

includes all the AES states such that the diagonal with the minimal number of active Sboxes has exactly i active Sboxes.

Hence,

$$\begin{aligned}
\mathbb{E}_{x \sim \nu} \left[\prod_{j \in [4]} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_{x_{\mathcal{G}_j}}] \right] &\leq \sum_{i=1}^4 \Pr_{x \sim \nu} [x \in \mathcal{Y}_i] \max_{x \in \mathcal{Y}_i} \left\{ \prod_{j \in [4]} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_{x_{\mathcal{G}_j}}] \right\} \\
&\leq \sum_{i=1}^4 \Pr_{x \sim \nu} [x \in \mathcal{Y}_i] (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(i)}(2))^4 \\
&\leq \Pr_{x \sim \nu} [x \in \mathcal{Y}_1] (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(1)}(2))^4 + \Pr_{x \sim \nu} [x \in \mathcal{Y}_2] (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(2)}(2))^4 + (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(3)}(2))^4,
\end{aligned} \tag{27}$$

where the second inequality is by Lemma 5, and the third inequality holds since $\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(3)}(2) > \mathcal{N}_{\text{AES}_{\text{SPS}}}^{(4)}(2)$.

For every diagonal index $j \in [4]$, the number of diagonal values with weight $i \in [2]$ is at most $\binom{4}{i} \cdot 2^{8 \cdot i}$, hence by the assumption

$$\Pr_{x \sim \nu} [\text{wt}(x_{\mathcal{G}_j}) = i] \leq \mathbf{M} \cdot \binom{4}{i} \cdot 2^{8 \cdot i}.$$

By a union bound over all 4 diagonals, for $i \in [2]$

$$\Pr_{x \sim \nu} [x \in \mathcal{Y}_i] \leq 4 \cdot \mathbf{M} \cdot \binom{4}{i} \cdot 2^{8 \cdot i}.$$

Plugging these into (27), we deduce

$$\begin{aligned}
&\mathbb{E}_{x \sim \nu} \left[\prod_{j \in [4]} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_{x_{\mathcal{G}_j}}] \right] \\
&\leq 4 \cdot \mathbf{M} \cdot 4 \cdot 2^8 (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(1)}(2))^4 + 4 \cdot \mathbf{M} \cdot 6 \cdot 2^{16} (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(2)}(2))^4 + (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(3)}(2))^4 \\
&\leq \mathbf{M} \cdot (2^{12} \cdot (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(1)}(2))^4 + 3 \cdot 2^{19} (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(2)}(2))^4 + (\mathcal{N}_{\text{AES}_{\text{SPS}}}^{(3)}(2))^4) \\
&\leq 2^{-128} \left(\mathbf{M} (2^{12} (1 + f(1))^4 + 3 \cdot 2^{19} (1 + f(2))^4) + (1 + f(3))^4 \right) \\
&\leq 2^{-128} (2^{20.83} \mathbf{M} + 1.0142),
\end{aligned}$$

where the function f is defined in Lemma 5. This completes the proof. ■

We now bound the 1-to-2 norm of the 4-round AES DDT.

Theorem 1. *Let $x \in (\mathbb{F}_2^8)^{16}$ be any input non-zero difference to the 4-round AES round function $\text{AES}_{\text{RF}}^{(4)}$. Denote by $\xi \in \mathbb{R}^{(\mathbb{F}_2^8)^{16}}$ the output difference distribution. Then, $\text{Col}[\xi] = \|\xi\|_2^2 \leq 2^{-128} \left(1 + \frac{1}{50} \right)$. Hence, $\left| \|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-} \|_{1,2}^2 - \frac{1}{2^{128}-1} \right| \leq 2^{-128} \cdot \frac{1}{50}$. The same also holds for the inverse $(\text{AES}_{\text{RF}}^{(4)})^{-1}$.*

Proof. We will prove the theorem for $\text{AES}_{\text{RF}}^{(4)}$. It holds for the inverse by Fact 4 and by observing that every property of $\text{AES}_{\text{RF}}^{(4)}$ that we use also holds for the inverse.

Define the distributions $\nu, \mu \in \mathbb{R}^{(\mathbb{F}_2^8)^{16}}$, where μ is the distribution obtained after the application of the AES super-Sboxes layer to the initial input difference (and placing the results in the anti-

diagonals of the state), and ν is the distribution obtained after applying the MixColumns operation to μ (i.e., after the application of $\text{AES}_{\text{RF}}^{(2)}$ to the initial input difference).

Denote by ξ' the distribution before the final MixColumns application (see Figure 4). Clearly,

$$\text{Col}[\xi] = \text{Col}[\xi'],$$

hence we analyze ξ' .

Note that ξ' is obtained after an additional invocation of the AES super-Sbox layer to ν . Thus, our goal is to apply Lemma 10 to ν in order to bound $\text{Col}[\xi']$. For this purpose, we need to estimate several values. First, we use Lemma 5, which bounds

$$\|\mathfrak{D}_{\text{AES}_{\text{spS}}}^-\|_{1,2}^2 \leq 2^{-32} \left(1 + \frac{1}{5.05}\right).$$

Second, by Proposition 8, we define

$$\text{ColF}(\mathbf{M}) = 2^{-128} (2^{20.83} \mathbf{M} + 1.0142).$$

In addition, we need values for the indicators $\mathbf{Z}_d = \mathbf{Z}_d(\nu)$ for $1 \leq d \leq 3$. Finally, we need estimates for the values $\mathbf{M}_t = \mathbf{M}_t(\nu)$ for $1 \leq t \leq 4$.

We condition the remainder of the analysis on $\text{wt}_{\mathcal{G}}(x)$, which is the number of active diagonals in the input difference, denoted by ac .



Figure 4: Distributions considered in proofs of Theorem 1 and Lemma 11

Bounds depending on $ac = \text{wt}_{\mathcal{G}}(x)$.

Since there are ac active diagonal super-Sboxes at the input to the first super-Sbox layer, there are ac active anti-diagonals at the input to the MixColumns operation in the second round (i.e., in every state in the support of μ).

Consider the diagonals of ν , which is obtained by applying MixColumns to μ . By Lemma 7, every state in the support of ν , has at least $5 - ac$ active diagonals, namely, there are at most $4 - (5 - ac) = ac - 1$ non-active (diagonal) super-Sboxes in every state in the support of ν . In other words, $\mathbf{Z}_d^{(ac)} = 0$ for all $d \geq ac$, and $\mathbf{Z}_d^{(ac)} = 1$ otherwise.

In our context, the parameter t of $\mathbf{M}_t^{(ac)}$ in Lemma 10 specifies the size of the subset $\mathcal{T}$ of diagonals at the output of the MDS layer. Our goal is to bound $\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{M}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\}$ as tightly as possible, thus bounding $\mathbf{M}_t^{(ac)}$.

For this purpose, observe that μ is a product distribution on the anti-diagonals at the input of the MixColumns layer, as in Setting 2. Hence Lemma 8 is applicable. In order to apply it, we use the bound $\mathcal{N}_{\text{AES}_{\text{spS}}}(\infty)$ defined in (26).

More specifically, we apply (23) of Lemma 8 by selecting a set $\mathcal{S}$ of active anti-diagonals at the

input to the MDS layer of size $s \triangleq |\mathcal{S}| = \min\{t, ac\}$. For $t, ac \in [4]$ we obtain

$$\mathbf{M}_t^{(ac)} \leq \max_{y \in (\mathbb{F}_2^8)^{4(-)}} \{(\|\mathfrak{D}_{\text{AES}_{\text{spS}}}^- e_y\|_\infty)^s\} \leq (\mathcal{N}_{\text{AES}_{\text{spS}}}(\infty))^s.$$

Plugging these into Lemma 10 and evaluating the expression for each $ac \in \{1, 2, 3, 4\}$ completes the proof. The code used for the calculations is given in Appendix B.3. ■

4.5 The Frobenius Norm of 4-Round AES DDT

We bound the Frobenius Norm of the 4-Round AES DDT. We begin with the following lemma.

Lemma 11. *Let $x \in (\mathbb{F}_2^8)^{16}$ be a non-zero input difference to the 4-round AES round function $\text{AES}_{\text{RF}}^{(4)}$, and denote by $\mathbf{U} = \min_{j \in [4]} \{\text{wt}(xg_j)\}$ the minimal number of active Sboxes in a diagonal of x . Assume that $\mathbf{U} \in [4]$, implying that all diagonals are active (non-zero).*

Denote by $\xi \in \mathbb{R}^{(\mathbb{F}_2^8)^{16}}$ the output difference distribution. Then,

$$\text{Col}[\xi] \leq 2^{-128}(1 + g(\mathbf{U})),$$

where

$$g(1) = 2^{-15.5}, g(2) = 2^{-21.7}, g(3) = 2^{-28.7}, g(4) = 2^{-29.7}.$$

Proof. The proof starts similarly to the proof of Theorem 1. Specifically, define the distributions $\nu, \mu \in \mathbb{R}^{(\mathbb{F}_2^8)^{16}}$, where μ is the distribution obtained after the application of the AES super-Sbox layer to the initial input difference and ν is the distribution obtained after applying the MixColumns operation to μ .

Denote by ξ' the distribution before the final MixColumns application such that $\text{Col}[\xi] = \text{Col}[\xi']$. Note that ξ' is obtained after an additional invocation of the AES super-Sbox layer to ν (see Figure 4). Our goal is to apply Lemma 4 (rather than Lemma 10, as in the proof of Theorem 1) to ν in order to bound $\text{Col}[\xi']$.

For the purpose of applying Lemma 4, we use Lemma 6, which bounds the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{spS}}}^-)^T \mathfrak{D}_{\text{AES}_{\text{spS}}}^-$ by $\frac{1}{426}$.

Since we assume that the input difference x has 4 active diagonals, every state in the support of ν potentially has between 0 and 3 non-active diagonals. Hence, we set the indicators $\mathbf{Z}_d = \mathbf{Z}_d(\nu) = 1$ for all $1 \leq d \leq 3$.

Finally, we need estimates for the values $\mathbf{M}_t = \mathbf{M}_t(\nu)$, $\mathbf{C}_t = \mathbf{C}_t(\nu)$ for $1 \leq t \leq 4$ and $\mathbf{MN}_1 = \mathbf{MN}_1(\nu)$. These bounds will be a function of the parameter $\mathbf{U}$, hence we write $\mathbf{M}_t^{(\mathbf{U})}$, $\mathbf{C}_t^{(\mathbf{U})}$ and $\mathbf{MN}_1^{(\mathbf{U})}$.

In our context, the parameter t of $\mathbf{C}_t^{(\mathbf{U})}$ and $\mathbf{M}_t^{(\mathbf{U})}$ in Lemma 4 specifies the size of the coordinate subset $\mathcal{T}$ of diagonals at the output of the MDS layer. Our goal is to bound $\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\}$ and $\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{M}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\}$ as tightly as possible, thus bounding $\mathbf{C}_t^{(\mathbf{U})}$ and $\mathbf{M}_t^{(\mathbf{U})}$, respectively. Similarly, $\min_{\mathcal{T} \subseteq [k], |\mathcal{T}|=1} \{\mathbf{MN}[\nu^{\subseteq \mathcal{G}\mathcal{T}}]\}$ lower bounds $\mathbf{MN}_1^{(\mathbf{U})}$.

Since μ is a product distribution on the anti-diagonals as in Setting 2, Lemma 8 and Lemma 9 are applicable.

Bounds by Lemma 8. We first bound $\mathbf{C}_t^{(\mathbf{U})}$ by applying Lemma 8 using the AES super-Sbox 2-norm bounds obtained in Lemma 5. More specifically, we apply (24) of Lemma 8 by selecting the set $\mathcal{S}$ of active anti-diagonals at the input to the MDS layer of size $|\mathcal{S}| = t$ (as we have 4 active

anti-diagonals, we can indeed select such a set for $t \in [4]$). For $\mathbf{U} \in [4]$ we obtain

$$\mathbf{C}_t^{(\mathbf{U})} \leq (\mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2))^t.$$

For $\mathbf{M}_t^{(\mathbf{U})}$, similarly to the proof of Theorem 1, we use the bound $\mathcal{N}_{\text{AES}_{\text{spS}}}(\infty)$ defined in (26). As above, we select a set $\mathcal{S}$ of active anti-diagonals with $|\mathcal{S}| = t$. For $\mathbf{U} \in [4]$ we obtain

$$\mathbf{M}_t^{(\mathbf{U})} \leq (\mathcal{N}_{\text{AES}_{\text{spS}}}(\infty))^t.$$

Bounds by Lemma 9. We further apply Lemma 9 in the cases $t = 1$ and $t = 2$. The value of m is $|\mathcal{S}|/t = 4/t$. Once again, we use the AES super-Sbox 2-norm bounds obtained in Lemma 5. We obtain

$$\begin{aligned} \mathbf{C}_1^{(\mathbf{U})} &\leq 2^{-32} + 2^{-32}(2^{32} \cdot \mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2) - 1)^4, \\ \mathbf{C}_2^{(\mathbf{U})} &\leq 2^{-64} + 2^{-64}(2^{64}(\mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2))^2 - 1)^2, \\ \mathbf{M}_1^{(\mathbf{U})} &\leq 2^{-32} + 2^{-32}(2^{32} \cdot \mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2) - 1)^2, \\ \mathbf{M}_2^{(\mathbf{U})} &\leq 2^{-64} + 2^{-64}(2^{64}(\mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2))^2 - 1) = (\mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2))^2, \\ \mathbf{MN}_1^{(\mathbf{U})} &\geq 2^{-32} - 2^{-32}(2^{32} \cdot \mathcal{N}_{\text{AES}_{\text{spS}}}^{(\mathbf{U})}(2) - 1)^2. \end{aligned}$$

Plugging these values into Lemma 4 completes the proof. The code used for the calculations is given in Appendix B.2. ■

We bound the Frobenius norm of 4-round AES DDT.

Theorem 2. *The Frobenius norm of the 4-round AES DDT satisfies $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}\|_{\text{F}}^2 \leq 1 + 2^{-29.3}$. Consequently, by Proposition 1, the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)})^{\text{T}} \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}$ is at most $2^{-29.3}$.*

Proof. We have

$$\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}\|_{\text{F}}^2 = \sum_{x, y \in (\mathbb{F}_2^8)^{16(-)}} \mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}[x, y]^2 = \sum_{y \in (\mathbb{F}_2^8)^{16(-)}} \text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)} e_y].$$

By Theorem 1, for any $y \in (\mathbb{F}_2^8)^{16(-)}$,

$$\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)} e_y] \leq 2^{-128} \left(1 + \frac{1}{50}\right).$$

This gives a bound of $\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-(4)}\|_{\text{F}}^2 \leq 1 + \frac{1}{50}$, which is significantly weaker than what we need to prove.

In order to improve the bound, we partition the set of non-zero differences in $(\mathbb{F}_2^8)^{16}$ into 5 subsets: For $i \in \{0, 1, 2, 3, 4\}$, $\mathcal{Y}_i$ contains all states such that the minimal number of active Sboxes in a diagonal is i . Formally,

$$\mathcal{Y}_i = \{y \in (\mathbb{F}_2^8)^{16} \mid y \neq \mathbf{0}_{8 \cdot 16} \wedge \min_{j \in [4]} \{\text{wt}(y_{\mathcal{G}_j})\} = i\}.$$

We have

$$\begin{aligned}
\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-}\|_{\text{F}}^2 - 1 &< \sum_{y \in (\mathbb{F}_2^8)^{16}(-)} (\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-} e_y] - 2^{-128}) = \sum_{y \in (\mathbb{F}_2^8)^{16}, y \neq \mathbf{0}_{8 \cdot 16}} (\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y] - 2^{-128}) \\
&= \sum_{i=0}^4 \sum_{y \in \mathcal{Y}_i} (\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y] - 2^{-128}) \leq \sum_{i=0}^4 |\mathcal{Y}_i| (\max_{y \in \mathcal{Y}_i} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\} - 2^{-128}) \\
&\leq \sum_{i=0}^3 |\mathcal{Y}_i| (\max_{y \in \mathcal{Y}_i} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\} - 2^{-128}) + 2^{128} \max_{y \in \mathcal{Y}_4} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\} - 2^{128}.
\end{aligned} \tag{28}$$

The final inequality holds since

$$\max_{y \in \mathcal{Y}_4} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\} - 2^{128} \geq 0.$$

Note that $\mathcal{Y}_0$ contains all non-zero states with a non-active diagonal. This is a relatively small subset of size at most

$$|\mathcal{Y}_0| \leq \binom{4}{1} \cdot 2^{32 \cdot 3} = 2^{98}.$$

For states in this set we use the bound of Theorem 1,

$$\max_{y \in \mathcal{Y}_0} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\} - 2^{-128} \leq \max_{y \in (\mathbb{F}_2^8)^{16}(-)} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-} e_y]\} - 2^{-128} \leq 2^{-128} \cdot \frac{1}{50}.$$

For subsets $\mathcal{Y}_i$ with $i \in \{1, 2, 3\}$ (which have $4 - i$ non-active Sboxes in some diagonal), we have

$$|\mathcal{Y}_i| \leq \binom{4}{1} \binom{4}{i} \cdot 2^{(16 - (4 - i)) \cdot 8} = 2^{128} \cdot 4 \cdot \binom{4}{i} \cdot 2^{8i - 32}.$$

For all $i \in [4]$, we bound $\max_{y \in \mathcal{Y}_i} \{\text{Col}[\mathfrak{D}_{\text{AES}_{\text{RF}}}^{(4)} e_y]\}$ by Lemma 11 (using the function g).

Plugging all the values into (28), we obtain

$$\begin{aligned}
\|\mathfrak{D}_{\text{AES}_{\text{RF}}}^{-}\|_{\text{F}}^2 - 1 &\leq 2^{98} \cdot 2^{-128} \cdot \frac{1}{50} + 4 \cdot 4 \cdot 2^{-24} \cdot g(1) + 4 \cdot 6 \cdot 2^{-16} \cdot g(2) + 4 \cdot 4 \cdot 2^{-8} \cdot g(3) + g(4) \\
&= 2^{98} \cdot 2^{-128} \cdot \frac{1}{50} + 4 \cdot 4 \cdot 2^{-24} \cdot 2^{-15.5} + 4 \cdot 6 \cdot 2^{-16} \cdot 2^{-21.7} + 4 \cdot 4 \cdot 2^{-8} \cdot 2^{-28.7} + 2^{-29.7} \\
&= 2^{-30} \cdot \frac{1}{50} + 16 \cdot 2^{-39.5} + 24 \cdot 2^{-37.7} + 16 \cdot 2^{-36.7} + 2^{-29.7} \leq 1.543 \cdot 2^{-30} \leq 2^{-29.3}.
\end{aligned}$$

■

4.6 The MEDP of 4-round AES

We improve the bound on the MEDP of 4-round AES.

Theorem 3. *The MEDP of 4-round AES is at most $2^{-116.54}$.*

Proof. We apply [PSLL03, Thm. 1], which bounds the MEDP of a 2-round AES-like cipher. In our case, by Lemma 7, we can apply it to 4-round AES (i.e., 3.5-round AES) with the AES super-Sbox. By exploiting symmetric properties of the super-Sbox, the bound reduces to bounding the 5-norm of the super-Sbox output distribution (to the fifth power), for the worst-case non-zero

input difference $x \in \mathbb{F}_2^{32}$. Thus, the MEDP is bounded by

$$\|\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_x\|_5^5 \leq \|\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_x\|_\infty^3 \cdot \|\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_x\|_2^2.$$

By [KS07], $\|\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_x\|_\infty^3 < 2^{-3 \cdot 28.27}$, while by Lemma 5, $\|\mathfrak{D}_{\text{AES}_{\text{SPS}}} e_x\|_2^2 \leq 2^{-31.73}$. Hence, we obtain the claimed bound of $2^{-116.54}$. $\blacksquare$

4.7 The MEDPD and 2-Wise Independence of $4r$ -Round AES

We derive our results that bound the MEDPD and closeness to 2-wise independence of $4r$ -round AES.

4.7.1 The MEDPD of $4r$ -round AES for $r \geq 2$.

Theorem 4. *For any integer $r \geq 2$, the MEDPD of $4r$ -round AES satisfies*

$$\text{Dev}[\mathfrak{D}_{\text{AES}_E^{(4r)}}^-] \leq 2^{-128} \cdot \frac{1}{50} \cdot 2^{-14.65(r-2)} \leq 2^{-128} \cdot 2^{-5.6} \cdot 2^{-14.65(r-2)}.$$

Proof. By Fact 3,

$$\mathfrak{D}_{\text{AES}_E^{(4r)}}^- = \mathfrak{D}_{\text{AES}_E^{(4)}}^- (\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^{r-2} \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-.$$

Applying Proposition 5 with this decomposition, we obtain

$$\text{Dev}[\mathfrak{D}_{\text{AES}_E^{(4r)}}^-] \leq \lambda^{(r-2)/2} \sqrt{\|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-\|_{1,2}^2 - \frac{1}{2^{128}-1}} \sqrt{\|(\mathfrak{D}_{\text{AES}_E^{(4)}}^-)^T\|_{1,2}^2 - \frac{1}{2^{128}-1}}, \quad (29)$$

where λ is the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^T \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-$.

By Theorem 2, $\lambda \leq 2^{-29.3}$. Moreover by Theorem 1,

$$\|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-\|_{1,2}^2 - \frac{1}{2^{128}-1} \leq 2^{-128} \cdot \frac{1}{50}.$$

By Fact 3, $(\mathfrak{D}_{\text{AES}_E^{(4)}}^-)^T = (\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^T$ is the DDT of the inverse, $(\text{AES}_E^{(4)})^{-1}$, and by Theorem 1 (applied to the inverse) a similar inequality holds for $\|(\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^T\|_{1,2}^2$. Plugging the above results into (29), we obtain

$$\text{Dev}[\mathfrak{D}_{\text{AES}_E^{(4r)}}^-] \leq 2^{(-29.3/2)(r-2)} \sqrt{2^{-128} \cdot \frac{1}{50}} \sqrt{2^{-128} \cdot \frac{1}{50}} = 2^{-128} \cdot \frac{1}{50} \cdot 2^{-14.65(r-2)}.$$

$\blacksquare$

4.7.2 The 2-wise independence of $4r$ -round AES.

Theorem 5. *Let $r \geq 1$ be an integer. Let $x \in (\mathbb{F}_2^8)^{16(-)}$ be an input difference to $4r$ -round AES. Then,*

$$\|\mathfrak{D}_{\text{AES}_E^{(4r)}}^- e_x - \bar{\mathbf{1}}_{(\mathbb{F}_2^8)^{16(-)}}\|_2^2 = \|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4r)}}^- e_x - \bar{\mathbf{1}}_{(\mathbb{F}_2^8)^{16(-)}}\|_2^2 \leq 2^{-128} \cdot 2^{-5.6} \cdot 2^{-29.3(r-1)}.$$

Moreover, $4r$ -round AES is $(2^{-3.8} \cdot 2^{-14.65(r-1)})$ -close to 2-wise independence.

Proof. By Fact 3,

$$\mathfrak{D}_{\text{AES}_E^{(4r)}}^- = \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4r)}}^- = (\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^{r-1} \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-.$$

Applying Proposition 4 with this decomposition, we obtain

$$\|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4r)}}^- e_x - \bar{\mathbf{1}}_{(\mathbb{F}_2^8)^{16(-)}}\|_2^2 \leq \lambda^{r-1} \left(\|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^- \|_{1,2}^2 - \frac{1}{2^{128} - 1} \right), \quad (30)$$

where λ is the second largest eigenvalue of $(\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-)^T \mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^-$.

By Theorem 2, $\lambda \leq 2^{-29.3}$. Moreover, by Theorem 1,

$$\|\mathfrak{D}_{\text{AES}_{\text{RF}}^{(4)}}^- \|_{1,2}^2 - \frac{1}{2^{128} - 1} \leq 2^{-128} \cdot \frac{1}{50} \leq 2^{-128} \cdot 2^{-5.6}.$$

Plugging the above results into (30), we obtain

$$\|\mathfrak{D}_{\text{AES}_E^{(4r)}}^- e_x - \bar{\mathbf{1}}_{(\mathbb{F}_2^8)^{16(-)}}\|_2^2 \leq 2^{-128} \cdot 2^{-5.6} \cdot 2^{-29.3(r-1)}.$$

The bound on the 2-wise independence follows similarly by the second part of Proposition 4 and by Proposition 7. $\blacksquare$

5 Norm Bounds for MDS Mappings

In this section we bound the 2 and infinity norms of the marginal output distributions of an MDS matrix (or more generally, an MDS mapping), when applied to a product distribution, proving Lemma 1, Lemma 2 and Lemma 3.

We now give additional preliminaries used in the analysis of MDS mappings.

5.1 Norms, Boolean Functions and Convolution

5.1.1 Norms.

We start by proving a basic fact about norms.

Proposition 9. *Let $f \in \mathbb{R}^{\Pi_1 \times \Pi_2}$ have a product structure, namely, $f(x_1, x_2) = f^{\subseteq\{1\}}(x_1) f^{\subseteq\{2\}}(x_2)$ for all $x_1 \in \Pi_1, x_2 \in \Pi_2$. Then, for any $p \geq 1$,*

$$\|f\|_p = \|f^{\subseteq\{1\}}\|_p \|f^{\subseteq\{2\}}\|_p \text{ and } \|f\|_\infty = \|f^{\subseteq\{1\}}\|_\infty \|f^{\subseteq\{2\}}\|_\infty.$$

Proof. We have

$$\begin{aligned} \|f\|_p^p &= \sum_{(x_1, x_2) \in \Pi_1 \times \Pi_2} |f(x_1, x_2)|^p = \sum_{(x_1, x_2) \in \Pi_1 \times \Pi_2} |f^{\subseteq\{1\}}(x_1)|^p |f^{\subseteq\{2\}}(x_2)|^p \\ &= \left(\sum_{x \in \Pi_1} |f^{\subseteq\{1\}}(x)|^p \right) \left(\sum_{x \in \Pi_2} |f^{\subseteq\{2\}}(x)|^p \right) = \|f^{\subseteq\{1\}}\|_p^p \|f^{\subseteq\{2\}}\|_p^p. \end{aligned}$$

Similarly,

$$\begin{aligned}\|f\|_\infty &= \max_{(x_1, x_2) \in \Pi_1 \times \Pi_2} \{|f(x_1, x_2)|\} = \max_{(x_1, x_2) \in \Pi_1 \times \Pi_2} \{|f^{\subseteq \{1\}}(x_1)| |f^{\subseteq \{2\}}(x_2)|\} \\ &= \max_{x \in \Pi_1} \{|f^{\subseteq \{1\}}(x)|\} \max_{x \in \Pi_2} \{|f^{\subseteq \{2\}}(x)|\} = \|f^{\subseteq \{1\}}\|_\infty \|f^{\subseteq \{2\}}\|_\infty.\end{aligned}$$

■

5.1.2 Convolution.

We define the convolution for Boolean Functions and then describe its properties. More details can be found in the first chapter of [O'D14].

Definition 15 (Convolution). *Let $f, g \in \mathbb{R}^{\mathbb{F}_2^n}$. Their convolution $f * g \in \mathbb{R}^{\mathbb{F}_2^n}$ is defined by $(f * g)(x) = \sum_{y \in \mathbb{F}_2^n} f(y)g(x \oplus y) = \sum_{y \in \mathbb{F}_2^n} g(y)f(x \oplus y) = (g * f)(x)$.*

Fact 5 (Convolution of distributions). *If $\mu, \nu \in \mathbb{R}^{\mathbb{F}_2^n}$ are distributions, then $\mu * \nu$ is a distribution on $x \in \mathbb{F}_2^n$ obtained by choosing $y \sim \mu$, $z \sim \nu$ independently and setting $x = y \oplus z$.*

Proposition 10. *Let $\mu, \nu \in \mathbb{R}^{\mathbb{F}_2^n}$ be two distributions. Then $M[\mu * \nu] \leq M[\mu]$ and $\text{Col}[\mu * \nu] \leq \text{Col}[\mu]$.*

Proof.

$$\begin{aligned}\text{Col}[\mu * \nu] &= \sum_{x \in \mathbb{F}_2^n} \left(\sum_{y \in \mathbb{F}_2^n} \nu(y) \mu(x \oplus y) \right)^2 = \sum_{x \in \mathbb{F}_2^n} \left(\mathbb{E}_{y \sim \nu} [\mu(x \oplus y)] \right)^2 \\ &\leq \sum_{x \in \mathbb{F}_2^n} \mathbb{E}_{y \sim \nu} [\mu(x \oplus y)^2] = \mathbb{E}_{y \sim \nu} \left[\sum_{x \in \mathbb{F}_2^n} \mu(x \oplus y)^2 \right] = \text{Col}[\mu],\end{aligned}$$

where we have used Jensen's inequality for the convex function $z \rightarrow z^2$.

For the second inequality, fix any $x \in \mathbb{F}_2^n$. Then,

$$(\mu * \nu)(x) = \sum_{y \in \mathbb{F}_2^n} \mu(y) \nu(x \oplus y) \leq \max_{y \in \mathbb{F}_2^n} \{\mu(y)\} \sum_{y \in \mathbb{F}_2^n} \nu(x \oplus y) = \max_{y \in \mathbb{F}_2^n} \{\mu(y)\} = M[\mu].$$

■

5.1.3 Basic results in Fourier analysis.

We derive basic results in Fourier analysis.

Proposition 11 (Fourier coefficients of convolution). *Let $f, g : \mathbb{F}_2^n \rightarrow \mathbb{R}$. Then for all $\alpha \in \hat{\mathbb{F}}_2^n$, $\widehat{f * g}(\alpha) = 2^n \widehat{f}(\alpha) \widehat{g}(\alpha)$. Alternatively, $\widetilde{f * g}(\alpha) = \check{f}(\alpha) \check{g}(\alpha)$.*

Proposition 12. *Let $\mu \in \mathbb{R}^{\mathbb{F}_2^n}$ be a distribution. Then, for any integer $p \geq 2$, $\|\check{\mu}\|_p^p - 1 \leq (2^n \text{Col}[\mu] - 1)^{p/2}$.*

Proof. We have

$$\begin{aligned}\|\check{\mu}\|_p^p - 1 &= \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} |\check{\mu}(\alpha)|^p \leq \left(\max_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \{|\check{\mu}(\alpha)|\} \right)^{p-2} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 \\ &\leq \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 \right)^{(p-2)/2} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 = \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 \right)^{p/2} = (2^n \text{Col}[\mu] - 1)^{p/2},\end{aligned}$$

where the final equality is by Proposition 6. ■

Proposition 13. *Let $\mu_1, \dots, \mu_p \in \mathbb{R}^{\mathbb{F}_2^n}$ be distributions, and let $\mu = \mu_1 * \dots * \mu_p$. Then, $\text{Col}[\mu] \leq 2^{-n} + 2^{-n} \prod_{i=1}^p (\|\check{\mu}_i\|_{2p}^{2p} - 1)^{1/p}$, and $\text{DV}[\mu] \leq 2^{-n} \prod_{i=1}^p (\|\check{\mu}_i\|_p^p - 1)^{1/p}$.*

Proof. For any $\alpha \in \hat{\mathbb{F}}_2^n$, by Proposition 11, $\check{\mu}(\alpha) = \prod_{i=1}^p \check{\mu}_i(\alpha)$. Hence, by Proposition 6,

$$\begin{aligned} \text{Col}[\mu] &= 2^{-n} + 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \check{\mu}(\alpha)^2 = 2^{-n} + 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \prod_{i=1}^p \check{\mu}_i(\alpha)^2 \\ &\leq 2^{-n} + 2^{-n} \prod_{i=1}^p \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} |\check{\mu}_i(\alpha)|^{2p} \right)^{1/p} = 2^{-n} + 2^{-n} \prod_{i=1}^p \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^n} |\check{\mu}_i(\alpha)|^{2p} - 1 \right)^{1/p} \\ &= 2^{-n} + 2^{-n} \prod_{i=1}^p (\|\check{\mu}_i\|_{2p}^{2p} - 1)^{1/p}. \end{aligned}$$

where the inequality is by Hölder's inequality applied p times (or by the Brascamp–Lieb inequality). This proves the first inequality.

Next, for any $x \in \mathbb{F}_2^n$, by similar derivation,

$$\mu(x) = \sum_{\alpha \in \hat{\mathbb{F}}_2^n} \chi_\alpha(x) \hat{\mu}(\alpha) = 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n} \chi_\alpha(x) \check{\mu}(\alpha) = 2^{-n} + 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \chi_\alpha(x) \check{\mu}(\alpha).$$

Hence,

$$|\mu(x) - 2^{-n}| \leq 2^{-n} \sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} \prod_{i=1}^p |\check{\mu}_i(\alpha)| \leq 2^{-n} \prod_{i=1}^p \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^n, \alpha \neq \mathbf{0}_n} |\check{\mu}_i(\alpha)|^p \right)^{1/p} = 2^{-n} \prod_{i=1}^p (\|\check{\mu}_i\|_p^p - 1)^{1/p}.$$

■

Proposition 14. *Let $\mu \in \mathbb{R}^{\mathbb{F}_2^{n_1} \times \mathbb{F}_2^{n_2}}$ be a product distribution, namely, $\mu(x_1, x_2) = \mu^{\subseteq \{1\}}(x_1) \mu^{\subseteq \{2\}}(x_2)$ for all $x_1 \in \mathbb{F}_2^{n_1}, x_2 \in \mathbb{F}_2^{n_2}$. Then, for any $p \geq 1$,*

$$\|\check{\mu}\|_p^p = \|\widetilde{\mu^{\subseteq \{1\}}}\|_p^p \|\widetilde{\mu^{\subseteq \{2\}}}\|_p^p.$$

Proof. First,

$$\begin{aligned} \check{\mu}(\alpha_1, \alpha_2) &= \mathbb{E}_{(x_1, x_2) \sim \mathbb{F}_2^{n_1} \times \mathbb{F}_2^{n_2}} [\chi_{\alpha_1, \alpha_2}(x_1, x_2) 2^{n_1 + n_2} \mu(x_1, x_2)] \\ &= \mathbb{E}_{(x_1, x_2) \sim \mathbb{F}_2^{n_1} \times \mathbb{F}_2^{n_2}} [\chi_{\alpha_1}(x_1) 2^{n_1} \mu^{\subseteq \{1\}}(x_1) \chi_{\alpha_2}(x_2) 2^{n_2} \mu^{\subseteq \{2\}}(x_2)] \\ &= \left(\mathbb{E}_{x_1 \sim \mathbb{F}_2^{n_1}} [\chi_{\alpha_1}(x_1) 2^{n_1} \mu^{\subseteq \{1\}}(x_1)] \right) \left(\mathbb{E}_{x_2 \sim \mathbb{F}_2^{n_2}} [\chi_{\alpha_2}(x_2) 2^{n_2} \mu^{\subseteq \{2\}}(x_2)] \right) = \widetilde{\mu^{\subseteq \{1\}}}(\alpha_1) \widetilde{\mu^{\subseteq \{2\}}}(\alpha_2). \end{aligned}$$

Similarly to the proof of Proposition 9, we obtain

$$\begin{aligned} \|\check{\mu}\|_p^p &= \sum_{(\alpha_1, \alpha_2) \in \hat{\mathbb{F}}_2^{n_1} \times \hat{\mathbb{F}}_2^{n_2}} |\check{\mu}(\alpha_1, \alpha_2)|^p = \sum_{(\alpha_1, \alpha_2) \in \hat{\mathbb{F}}_2^{n_1} \times \hat{\mathbb{F}}_2^{n_2}} |\widetilde{\mu^{\subseteq \{1\}}}(\alpha_1)|^p |\widetilde{\mu^{\subseteq \{2\}}}(\alpha_2)|^p \\ &= \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^{n_1}} |\widetilde{\mu^{\subseteq \{1\}}}(\alpha)|^p \right) \left(\sum_{\alpha \in \hat{\mathbb{F}}_2^{n_2}} |\widetilde{\mu^{\subseteq \{2\}}}(\alpha)|^p \right) = \|\widetilde{\mu^{\subseteq \{1\}}}\|_p^p \|\widetilde{\mu^{\subseteq \{2\}}}\|_p^p. \end{aligned}$$

■

We conclude by describing how the Fourier transform behaves under invertible linear transformations.

Proposition 15. *Let $L: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ be an invertible linear transformation (matrix). Let $\mu \in \mathbb{R}^{\mathbb{F}_2^n}$ be an input distribution to L . Consider the output distribution $\nu \in \mathbb{R}^{\mathbb{F}_2^n}$, defined by $\nu(x) = \Pr_{y \sim \mu}[L(y) = x] = \Pr_{y \sim \mu}[y = L^{-1}(x)] = \mu(L^{-1}(x))$. Then, for any $\alpha \in \hat{\mathbb{F}}_2^n$, $\check{\nu}(\alpha) = \check{\mu}(L^T(\alpha))$.*

Proof. We have

$$\begin{aligned} \hat{\nu}(\alpha) &= \mathbb{E}_{x \sim \mathbb{F}_2^n} [\chi_\alpha(x) \nu(x)] = \mathbb{E}_{x \sim \mathbb{F}_2^n} [\chi_\alpha(x) \mu(L^{-1}(x))] = \mathbb{E}_{y \sim \mathbb{F}_2^n} [\chi_\alpha(L(y)) \mu(y)] = \mathbb{E}_{y \sim \mathbb{F}_2^n} [\chi_{L^T(\alpha)}(y) \mu(y)] \\ &= \hat{\mu}(L^T(\alpha)). \end{aligned}$$

■

5.2 Norm Bounds for MDS Mappings

We use the following additional notation in this section. Given $\mathcal{S} \subseteq [k]$ and $y \in (\mathbb{F}_2^n)^{|\mathcal{S}|}$, define $\text{zc}(y, \mathcal{S}) \in (\mathbb{F}_2^n)^k$ as the vector that is obtained by placing y in the $\mathcal{S}$ -coordinates, and filling the remaining coordinates with zeros. Namely, $\text{zc}(y, \mathcal{S})_{\mathcal{S}} = y$, and $\text{zc}(y, \mathcal{S})_{\bar{\mathcal{S}}} = \mathbf{0}_{(k-|\mathcal{S}|)n}$.

Definition 16 (Restricted mapping and distribution.). *Let $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$ be a linear mapping. Let $\mathcal{S}, \mathcal{T} \subseteq [k]$ be subsets such that $\mathcal{T}$ is non-empty.*

The linear mapping $L_{\mathcal{S} \rightarrow \mathcal{T}}: (\mathbb{F}_2^n)^{|\mathcal{S}|} \rightarrow (\mathbb{F}_2^n)^{|\mathcal{T}|}$ is defined by $y \mapsto (L(\text{zc}(y, \mathcal{S})))_{\mathcal{T}}$. Namely, $L_{\mathcal{S} \rightarrow \mathcal{T}}$ is the restricted linear mapping from the indices of $\mathcal{S}$ (where indices not in $\mathcal{S}$ are set to 0) to those of $\mathcal{T}$.

Furthermore, let $\mu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be a distribution at the input of L , and denote by $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output distribution defined by $\nu(x) = \Pr_{x' \sim \mu}[L(x') = x]$. Define the distribution $\nu_{\mathcal{S} \rightarrow \mathcal{T}} \in \mathbb{R}^{(\mathbb{F}_2^n)^{|\mathcal{T}|}}$ by $\nu_{\mathcal{S} \rightarrow \mathcal{T}}(x) = \Pr_{x' \sim \mu}[\text{zc}(x', \mathcal{S})_{\mathcal{T}} = x]$.

We will use the following standard characterization of an MDS matrix.

Proposition 16 (MDS mapping characterization). *Let $L \in (\mathbb{F}_{2^n})^{k \times k}$ be an MDS matrix, and consider it as a linear mapping $L: (\mathbb{F}_2^n)^k \rightarrow (\mathbb{F}_2^n)^k$. Then, for any subsets $\mathcal{S}, \mathcal{T} \subseteq [k]$ such that $|\mathcal{S}| = |\mathcal{T}| > 0$, the mapping $L_{\mathcal{S} \rightarrow \mathcal{T}}: (\mathbb{F}_2^n)^{|\mathcal{S}|} \rightarrow (\mathbb{F}_2^n)^{|\mathcal{T}|}$ defines a permutation (i.e., is invertible).*

The following fact follows by the product structure of μ and linearity of L .

Fact 6 (Convolution of restricted mappings). *Assume Setting 1. Let $\mathcal{S}_1, \dots, \mathcal{S}_m$ be a partition of $[k]$ into subsets. Then, for any non-empty $\mathcal{T} \subseteq [k]$, $\nu^{\subseteq \mathcal{T}} = \nu_{\mathcal{S}_1 \rightarrow \mathcal{T}} * \dots * \nu_{\mathcal{S}_m \rightarrow \mathcal{T}}$.*

We now restate and prove Lemma 1.

Lemma 1 (MDS diffusion). *Assume Setting 1. Let $t \in [k]$. Then, for any $\mathcal{S} \subseteq [k]$ such that $t \geq |\mathcal{S}| \geq 1$,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq M[\mu^{\subseteq \mathcal{S}}] = \prod_{i \in \mathcal{S}} M[\mu^{\subseteq \{i\}}], \text{ and} \quad (9)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq \text{Col}[\mu^{\subseteq \mathcal{S}}] = \prod_{i \in \mathcal{S}} \text{Col}[\mu^{\subseteq \{i\}}]. \quad (10)$$

Proof. Fix $\mathcal{T} \subseteq [k]$ with $|\mathcal{T}| = t$. We may assume without loss of generality that $|\mathcal{S}| = |\mathcal{T}|$, as adding indices to $\mathcal{T}$ cannot increase $M[\nu^{\subseteq \mathcal{T}}]$ and $\text{Col}[\nu^{\subseteq \mathcal{T}}]$. For an MDS mapping L , the mapping $L_{\mathcal{S} \rightarrow \mathcal{T}}$ is a permutation by Proposition 16, and hence $M[\nu_{\mathcal{S} \rightarrow \mathcal{T}}] = M[\mu^{\subseteq \mathcal{S}}]$ and $\text{Col}[\nu_{\mathcal{S} \rightarrow \mathcal{T}}] = \text{Col}[\mu^{\subseteq \mathcal{S}}]$.

Furthermore, since μ is a product distribution, by Fact 6 and Proposition 10,

$$M[\nu^{\subseteq \mathcal{T}}] = M[\nu_{\mathcal{S} \rightarrow \mathcal{T}} * \nu_{\bar{\mathcal{S}} \rightarrow \mathcal{T}}] \leq M[\nu_{\mathcal{S} \rightarrow \mathcal{T}}] = M[\mu^{\subseteq \mathcal{S}}] = \|\mu^{\subseteq \mathcal{S}}\|_{\infty} = \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \{i\}}\|_{\infty},$$

where the final equality is by Proposition 9.

Similarly, $\text{Col}[\nu^{\subseteq \mathcal{T}}] \leq \text{Col}[\mu^{\subseteq \mathcal{S}}] = \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \{i\}}\|_2^2 = \prod_{i \in \mathcal{S}} \text{Col}[\mu^{\subseteq \{i\}}]$. ■

We restate and prove Lemma 2.

Lemma 2 (MDS contraction). *Assume Setting 1. Let $q, m \geq 0$ be integers such that $t = q(m-1)$. Let $\mathcal{S} \subseteq [k]$ such that $|\mathcal{S}| = q \cdot m = t + q$. Then,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{M[\nu^{\subseteq \mathcal{T}}]\} \leq \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \{i\}}\|_m. \quad (11)$$

Proof. Fix $\mathcal{T} \subseteq [k]$ with $|\mathcal{T}| = t$. As in the proof of Lemma 1,

$$M[\nu^{\subseteq \mathcal{T}}] = M[\nu_{\mathcal{S} \rightarrow \mathcal{T}} * \nu_{\bar{\mathcal{S}} \rightarrow \mathcal{T}}] \leq M[\nu_{\mathcal{S} \rightarrow \mathcal{T}}].$$

Assume without loss of generality that $\mathcal{T} = [q(m-1)]$. Fix $x \in (\mathbb{F}_2^n)^{q(m-1)}$. Our goal is to bound $\nu_{\mathcal{S} \rightarrow \mathcal{T}}(x)$, thus bounding $M[\nu^{\subseteq \mathcal{T}}]$. Observe that

$$\begin{aligned} \nu_{\mathcal{S} \rightarrow \mathcal{T}}(x) &= \nu_{\mathcal{S} \rightarrow [q(m-1)]}(x) = \sum_{z \in (\mathbb{F}_2^n)^q} \nu_{\mathcal{S} \rightarrow [q \cdot m]}(x, z) \\ &= \sum_{z \in (\mathbb{F}_2^n)^q} \Pr_{y \sim \mu^{\subseteq \mathcal{S}}} [L_{\mathcal{S} \rightarrow [q \cdot m]}(y) = (x, z)]. \end{aligned}$$

Since $|\mathcal{S}| = q \cdot m$, the mapping $L_{\mathcal{S} \rightarrow [q \cdot m]}$ is a permutation by Proposition 16, and

$$\begin{aligned} \sum_{z \in (\mathbb{F}_2^n)^q} \Pr_{y \sim \mu^{\subseteq \mathcal{S}}} [L_{\mathcal{S} \rightarrow [q \cdot m]}(y) = (x, z)] &= \sum_{z \in (\mathbb{F}_2^n)^q} \Pr_{y \sim \mu^{\subseteq \mathcal{S}}} [y = (L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, z)] \\ &= \sum_{z \in (\mathbb{F}_2^n)^q} \mu^{\subseteq \mathcal{S}}((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, z)). \end{aligned}$$

To simplify notation, denote $v(x, z) = (L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, z) \in (\mathbb{F}_2^n)^{q \cdot m}$. Partition $\mathcal{S}$ arbitrarily into

subsets $\mathcal{S}_1, \dots, \mathcal{S}_m$ such that $|\mathcal{S}_i| = q$ for all $i \in [m]$. Since μ is a product distribution,

$$\begin{aligned} \nu_{\mathcal{S} \rightarrow \mathcal{T}}(x) &= \sum_{z \in (\mathbb{F}_2^n)^q} \mu^{\subseteq \mathcal{S}}(v(x, z)) = \sum_{z \in (\mathbb{F}_2^n)^q} \prod_{i \in [m]} \mu^{\subseteq \mathcal{S}_i}(v(x, z)_{\mathcal{S}_i}) \\ &\leq \prod_{i \in [m]} \left(\sum_{z \in (\mathbb{F}_2^n)^q} \mu^{\subseteq \mathcal{S}_i}(v(x, z)_{\mathcal{S}_i})^m \right)^{1/m}, \end{aligned} \quad (31)$$

where we have used Hölder's inequality. Next, by linearity, for every $i \in [m]$,

$$\begin{aligned} v(x, z)_{\mathcal{S}_i} &= ((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, z))_{\mathcal{S}_i} \\ &= ((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, \mathbf{0}_{n \cdot q}))_{\mathcal{S}_i} \oplus ((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(\mathbf{0}_{n \cdot q(m-1)}, z))_{\mathcal{S}_i}. \end{aligned}$$

The value $((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(x, \mathbf{0}_{n \cdot q}))_{\mathcal{S}_i}$ does not depend on z . On the other hand,

$$((L_{\mathcal{S} \rightarrow [q \cdot m]})^{-1}(\mathbf{0}_{n \cdot q(m-1)}, z))_{\mathcal{S}_i} = (L^{-1})_{[q \cdot m - q + 1, q \cdot m] \rightarrow \mathcal{S}_i}(z).$$

Since L^{-1} is an MDS mapping and $|[q \cdot m - q + 1, q \cdot m]| = |\mathcal{S}_i| = q$ then

$$(L^{-1})_{[q \cdot m - q + 1, q \cdot m] \rightarrow \mathcal{S}_i}: (\mathbb{F}_2^n)^q \rightarrow (\mathbb{F}_2^n)^q$$

is a permutation, and so is the mapping $z \rightarrow v(x, z)_{\mathcal{S}_i}$ (when x is fixed). Therefore,

$$\sum_{z \in (\mathbb{F}_2^n)^q} \mu^{\subseteq \mathcal{S}_i}(v(x, z)_{\mathcal{S}_i})^m = \sum_{u \in (\mathbb{F}_2^n)^q} \mu^{\subseteq \mathcal{S}_i}(u)^m = \|\mu^{\subseteq \{\mathcal{S}_i\}}\|_m^m.$$

Plugging this into (31), we conclude

$$\nu_{\mathcal{S} \rightarrow \mathcal{T}}(x) \leq \prod_{i \in [m]} \|\mu^{\subseteq \{\mathcal{S}_i\}}\|_m = \prod_{i \in [m]} \prod_{j \in \mathcal{S}_i} \|\mu^{\subseteq \{j\}}\|_m = \prod_{i \in \mathcal{S}} \|\mu^{\subseteq \{i\}}\|_m,$$

where the first equality is by Proposition 9. ■

We now prove bounds via higher Fourier norms of the input distribution.

Proposition 17. *Assume Setting 1. Then, for any subsets $\mathcal{T}, \mathcal{S}_1, \dots, \mathcal{S}_m \subseteq [k]$, such that $|\mathcal{T}| = |\mathcal{S}_1| = \dots = |\mathcal{S}_m| > 0$ and $\mathcal{S}_1, \dots, \mathcal{S}_m$ that are pairwise disjoint,*

$$\text{Col}[\nu^{\subseteq \mathcal{T}}] \leq 2^{-|\mathcal{T}|n} + 2^{-|\mathcal{T}|n} \prod_{i \in [m]} (\|\widetilde{\mu^{\subseteq \mathcal{S}_i}}\|_{2m}^{2m} - 1)^{1/m} \quad (32)$$

$$= 2^{-|\mathcal{T}|n} + 2^{-|\mathcal{T}|n} \prod_{i \in [m]} \left(\prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m} - 1 \right)^{1/m},$$

$$\text{and} \quad \text{DV}[\nu^{\subseteq \mathcal{T}}] \leq 2^{-|\mathcal{T}|n} \prod_{i \in [m]} (\|\widetilde{\mu^{\subseteq \mathcal{S}_i}}\|_m^m - 1)^{1/m} \quad (33)$$

$$= 2^{-|\mathcal{T}|n} \prod_{i \in [m]} \left(\prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_m^m - 1 \right)^{1/m}.$$

Proof. Denote $\mathcal{S} = \mathcal{S}_1 \cup \dots \cup \mathcal{S}_m$. By Fact 6, $\nu^{\subseteq \mathcal{T}} = \nu_{\mathcal{S}_1 \rightarrow \mathcal{T}} * \dots * \nu_{\mathcal{S}_m \rightarrow \mathcal{T}} * \nu_{\overline{\mathcal{S}} \rightarrow \mathcal{T}}$.

Denote $\rho = \nu_{\mathcal{S}_1 \rightarrow \mathcal{T}} * \dots * \nu_{\mathcal{S}_m \rightarrow \mathcal{T}}$. Further, denote $n' = |\mathcal{T}|n$. By Proposition 10 and Proposition 13, $\text{Col}[\nu^{\subseteq \mathcal{T}}] \leq \text{Col}[\rho] \leq 2^{-n'} + 2^{-n'} \prod_{i \in [m]} (\|\widetilde{\nu_{\mathcal{S}_i \rightarrow \mathcal{T}}}\|_{2m}^{2m} - 1)^{1/m}$.

Since for every $i \in [m]$, $|\mathcal{S}_i| = |\mathcal{T}|$, by Proposition 16 the mapping $L_{\mathcal{S}_i \rightarrow \mathcal{T}}$ is a linear permutation over $\mathbb{F}_2^{n'}$. Hence, by Proposition 15, the Fourier coefficients of $\nu_{\mathcal{S}_i \rightarrow \mathcal{T}}$ are a permutation of the Fourier coefficients of $\mu^{\subseteq \mathcal{S}_i}$, implying that

$$\|\widetilde{\nu_{\mathcal{S}_i \rightarrow \mathcal{T}}}\|_{2m}^{2m} = \|\widetilde{\mu^{\subseteq \mathcal{S}_i}}\|_{2m}^{2m} = \prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m},$$

where the second equality is by Proposition 14. Plugging this into the above inequality for $\text{Col}[\nu^{\subseteq \mathcal{T}}]$ completes the proof of (32).

The proof of (33) is similar, where the main step again uses Proposition 13. ■

We restate and prove Lemma 3, which is a simple consequence of Proposition 17.

Lemma 3 (MDS Fourier contraction). *Assume Setting 1. Let $t, m \in [k]$. Let $\mathcal{S} \subseteq [k]$ be a subset such that $|\mathcal{S}| = t \cdot m$. Then,*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} \max_{j \in \mathcal{S}} \{\|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m \cdot t}\}, \text{ and} \quad (12)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} + 2^{-t \cdot n} (2^{t \cdot n} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \{j\}}]^t\} - 1)^m. \quad (13)$$

Moreover, assuming $m \geq 2$,

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} (\max_{j \in \mathcal{S}} \{\|\widetilde{\mu^{\subseteq \{j\}}}\|_m^{m \cdot t}\} - 1), \text{ and} \quad (14)$$

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{T}}]\} \leq 2^{-t \cdot n} (2^{t \cdot n} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \{j\}}]^t\} - 1)^{m/2}. \quad (15)$$

Proof. Fix $\mathcal{T} \subseteq [k]$ with $|\mathcal{T}| = t$. Partition $\mathcal{S}$ arbitrarily into pairwise disjoint subsets $\mathcal{S}_1, \dots, \mathcal{S}_m$, each of size t . Then, Proposition 17 is applicable. Inequality (12) holds since

$$\begin{aligned} \max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} &\leq 2^{-t \cdot n} + 2^{-t \cdot n} \prod_{i \in [m]} \left(\prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m} - 1 \right)^{1/m} \\ &\leq 2^{-t \cdot n} + 2^{-t \cdot n} \max_{i \in [m]} \left\{ \prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m} - 1 \right\} \leq 2^{-t \cdot n} \max_{j \in \mathcal{S}} \{\|\widetilde{\mu^{\subseteq \{j\}}}\|_{2m}^{2m \cdot t}\}, \end{aligned}$$

where the first inequality follows from (32) of Proposition 17 and the final one uses Proposition 14. Inequality (14) follows similarly from (33) as

$$\begin{aligned} \max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{DV}[\nu^{\subseteq \mathcal{T}}]\} &\leq 2^{-t \cdot n} \max_{i \in [m]} \left\{ \prod_{j \in \mathcal{S}_i} \|\widetilde{\mu^{\subseteq \{j\}}}\|_m^m - 1 \right\} \\ &\leq 2^{-t \cdot n} (\max_{j \in \mathcal{S}} \{\|\widetilde{\mu^{\subseteq \{j\}}}\|_m^{m \cdot t}\} - 1). \end{aligned}$$

Inequality (13) holds since

$$\begin{aligned}
\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} &\leq 2^{-t \cdot n} + 2^{-t \cdot n} \prod_{i \in [m]} (\|\widetilde{\mu^{\subseteq \mathcal{S}_i}}\|_{2m}^{2m} - 1)^{1/m} \\
&\leq 2^{-t \cdot n} + 2^{-t \cdot n} \prod_{i \in [m]} (2^{t \cdot n} \text{Col}[\mu^{\subseteq \mathcal{S}_i}] - 1) \\
&= 2^{-t \cdot n} + 2^{-t \cdot n} \prod_{i \in [m]} \left(2^{t \cdot n} \prod_{j \in \mathcal{S}_i} \text{Col}[\mu^{\subseteq \{j\}}] - 1 \right) \\
&\leq 2^{-t \cdot n} + 2^{-t \cdot n} (2^{t \cdot n} \max_{j \in \mathcal{S}} \{\text{Col}[\mu^{\subseteq \{j\}}]\}^t - 1)^m,
\end{aligned}$$

where the first inequality is by (32) of Proposition 17, the second inequality is by Proposition 12 and the equality uses the product structure of μ . Inequality (15) follows similarly from (33). ■

6 Norm Bounds for the Kronecker Product

In this section, we prove the following lemma.

Lemma 12 (2-norm bound for Kronecker product). *Let $A \in \mathbb{R}^{\Pi \times \Pi}$ be a doubly stochastic matrix where the second largest eigenvalue of $A^T A$ is at most λ . Let $A^{\otimes k} \in \mathbb{R}^{\Pi^k \times \Pi^k}$ be its k -wise Kronecker product, and $\nu \in \mathbb{R}^{\Pi^k}$ a distribution. Then,*

$$\text{Col}[A^{\otimes k} \nu] \leq |\Pi|^{-k} \left(1 + \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} (|\Pi|^{|\mathcal{J}|} \text{Col}[\nu^{\subseteq \mathcal{J}}] - 1) \right).$$

We first describe properties of the Kronecker product (see [HJ91] for more details).

6.1 Kronecker Product

Let Π_1, Π_2 be finite domains. Let $A \in \mathbb{R}^{\Pi_1 \times \Pi_1}, B \in \mathbb{R}^{\Pi_2 \times \Pi_2}$. Then, the following properties hold.

1. $(A \otimes B)^T = A^T \otimes B^T$.
2. If A, B are doubly stochastic, then $A \otimes B$ is doubly stochastic.
3. Assume that A, B are symmetric, hence they are orthogonally diagonalizable. Let $\{v_i\}_{i \in [\Pi_1]}, \{u_i\}_{i \in [\Pi_2]}$ be orthonormal bases of eigenvectors for A and B , respectively, and denote the corresponding eigenvalues by $\{\lambda_i\}_{i \in [\Pi_1]}, \{\gamma_i\}_{i \in [\Pi_2]}$. Then, $\{v_i \otimes u_j\}_{(i,j) \in [\Pi_1] \times [\Pi_2]}$ is an orthonormal basis of eigenvectors for $A \otimes B$ with corresponding eigenvalues $\{\lambda_i \gamma_j\}_{(i,j) \in [\Pi_1] \times [\Pi_2]}$.

6.2 Proof of Lemma 12

Throughout this section, let $A \in \mathbb{R}^{\Pi \times \Pi}$ be a doubly stochastic matrix, where the second largest eigenvalue of $A^T A$ is λ .

We begin by introducing additional notation. Denote the sorted eigenvalues of $A^T A$ by $\lambda_1 = 1, \lambda_2 = \lambda, \dots, \lambda_{|\Pi|}$, and denote a corresponding basis of orthonormal eigenvectors by $v_1 = \tilde{\mathbf{1}}_\Pi, v_2, \dots, v_{|\Pi|}$ (eigenvectors with the same eigenvalue are sorted arbitrarily).

Thus, $(A^T A)^{\otimes k}$ has an eigenbasis of orthonormal vectors of the form

$$\{v_{i_1} \otimes \dots \otimes v_{i_k}\}_{(i_1, \dots, i_k) \in [\Pi]^k},$$

with corresponding eigenvalues $\{\prod_{j \in [k]} \lambda_{i_j}\}_{(i_1, \dots, i_k) \in [\Pi]^k}$.

For $(i_1, \dots, i_k) \in [\Pi]^k$, denote $u_{(i_1, \dots, i_k)} = v_{i_1} \otimes \dots \otimes v_{i_k}$. Note that for any $(x_1, \dots, x_k) \in \Pi^k$, $u_{(i_1, \dots, i_k)}(x_1, \dots, x_k) = \prod_{j \in [k]} v_{i_j}(x_j)$. More generally, for any $t \in [k]$ and $(i_1, \dots, i_t) \in [\Pi]^t$, denote $u_{(i_1, \dots, i_t)} = v_{i_1} \otimes \dots \otimes v_{i_t}$.

For a set $\mathcal{J} \subseteq [k]$, denote by $\mathcal{M}_{=\mathcal{J}} = \{(i_1, \dots, i_k) \in [\Pi]^k, i_j \neq 1 \Leftrightarrow j \in \mathcal{J}\}$ the set of all multi-indices where the non-1 values are indexed by $\mathcal{J}$. Note that for any $(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}$, the eigenvalue that corresponds to $u_{(i_1, \dots, i_k)}$ satisfies

$$\prod_{j \in [k]} \lambda_{i_j} = \prod_{j \in \mathcal{J}} \lambda_{i_j} \leq \lambda^{|\mathcal{J}|}. \quad (34)$$

Let $f \in \mathbb{R}^{\Pi^k}$ and $\mathcal{J} \subseteq [k]$. Consider the decomposition of $f^{\subseteq \mathcal{J}}$ using the orthonormal eigenbasis $\{u_{(i_1, \dots, i_{|\mathcal{J}|})}\}_{(i_1, \dots, i_{|\mathcal{J}|}) \in [\Pi]^{|\mathcal{J}|}}$, namely, write

$$f^{\subseteq \mathcal{J}} = \sum_{(i_1, \dots, i_{|\mathcal{J}|}) \in [\Pi]^{|\mathcal{J}|}} \alpha_{(i_1, \dots, i_{|\mathcal{J}|})}^{\mathcal{J}} u_{(i_1, \dots, i_{|\mathcal{J}|})},$$

where $\alpha_{(i_1, \dots, i_{|\mathcal{J}|})}^{\mathcal{J}} = \langle f^{\subseteq \mathcal{J}}, u_{(i_1, \dots, i_{|\mathcal{J}|})} \rangle$. By orthonormality of the eigenvectors,

$\|f^{\subseteq \mathcal{J}}\|_2^2 = \sum_{(i_1, \dots, i_{|\mathcal{J}|}) \in [\Pi]^{|\mathcal{J}|}} (\alpha_{(i_1, \dots, i_{|\mathcal{J}|})}^{\mathcal{J}})^2$. For $\mathcal{J} = [k]$, we have $f = f^{\subseteq [k]}$, and we simply write $f = \sum_{(i_1, \dots, i_k) \in [\Pi]^k} \alpha_{(i_1, \dots, i_k)} u_{(i_1, \dots, i_k)}$.

We first relate the coefficients in the decomposition of $f^{\subseteq \mathcal{J}}$ to those in the decomposition of f .

Proposition 18. *For any $\mathcal{J} \subseteq [k]$ and $(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}$ we have $\alpha_{(i_1, \dots, i_k)} = |\Pi|^{-|\mathcal{J}|/2} \alpha_{(i_1, \dots, i_k)_{\mathcal{J}}}^{\mathcal{J}}$, where $(i_1, \dots, i_k)_{\mathcal{J}} \in [\Pi]^{|\mathcal{J}|}$ is the projection of $(i_1, \dots, i_k)$ on the $\mathcal{J}$ -coordinates.*

Proof. Denote $(i_1, \dots, i_k)_{\mathcal{J}} = (j_1, \dots, j_{|\mathcal{J}|})$. We have

$$\begin{aligned} \alpha_{(i_1, \dots, i_k)} &= \langle f, u_{(i_1, \dots, i_k)} \rangle = \langle f, v_{i_1} \otimes \dots \otimes v_{i_k} \rangle \\ &= \sum_{x \in \Pi^k} f(x) \left(\prod_{j \in [k]} v_{i_j}(x_j) \right) = |\Pi|^{-|\mathcal{J}|/2} \sum_{x \in \Pi^k} f(x) \left(\prod_{j \in \mathcal{J}} v_{i_j}(x_j) \right) \\ &= |\Pi|^{-|\mathcal{J}|/2} \sum_{x \in \Pi^{|\mathcal{J}|}} \left(\prod_{\ell \in [\mathcal{J}]} v_{j_\ell}(x_\ell) \right) \sum_{x' \in \Pi^{|\mathcal{J}|}} f(x, x') \\ &= |\Pi|^{-|\mathcal{J}|/2} \sum_{x \in \Pi^{|\mathcal{J}|}} \left(\prod_{\ell \in [\mathcal{J}]} v_{j_\ell}(x_\ell) \right) f^{\subseteq \mathcal{J}}(x) \\ &= |\Pi|^{-|\mathcal{J}|/2} \langle f^{\subseteq \mathcal{J}}, u_{(i_1, \dots, i_k)_{\mathcal{J}}} \rangle = |\Pi|^{-|\mathcal{J}|/2} \alpha_{(i_1, \dots, i_k)_{\mathcal{J}}}^{\mathcal{J}}. \end{aligned}$$

The fourth equality holds as $(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}$ and $v_1 = \tilde{\mathbf{1}}_{\Pi} = \frac{1}{|\Pi|^{1/2}} \mathbf{1}_{\Pi}$. ■

Lemma 12 in a special case of the following (more general) result.

Proposition 19.

$$\|A^{\otimes k} f\|_2^2 \leq |\Pi|^k \mathbb{E}[f]^2 + |\Pi|^{-k} \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} |\Pi|^{|\mathcal{J}|} \left(\|f^{\subseteq \mathcal{J}}\|_2^2 - |\Pi|^{|\mathcal{J}|} \mathbb{E}[f^{\subseteq \mathcal{J}}]^2 \right).$$

Proof. We have

$$\begin{aligned}
(A^T A)^{\otimes k} f &= (A^T A)^{\otimes k} \sum_{(i_1, \dots, i_k) \in [\Pi]^k} \alpha_{(i_1, \dots, i_k)} u_{(i_1, \dots, i_k)} \\
&= \sum_{(i_1, \dots, i_k) \in [\Pi]^k} \alpha_{(i_1, \dots, i_k)} (A^T A)^{\otimes k} u_{(i_1, \dots, i_k)} \\
&= \sum_{(i_1, \dots, i_k) \in [\Pi]^k} \alpha_{(i_1, \dots, i_k)} \left(\prod_{j \in [k]} \lambda_{i_j} \right) u_{(i_1, \dots, i_k)}.
\end{aligned}$$

Hence,

$$\begin{aligned}
\|A^{\otimes k} f\|_2^2 &= \langle A^{\otimes k} f, A^{\otimes k} f \rangle = \langle f, (A^{\otimes k})^T A^{\otimes k} f \rangle = \langle f, (A^T A)^{\otimes k} f \rangle \\
&= \sum_{(i_1, \dots, i_k) \in [\Pi]^k} (\alpha_{(i_1, \dots, i_k)})^2 \left(\prod_{j \in [k]} \lambda_{i_j} \right) \\
&= \sum_{\mathcal{J} \subseteq [k]} \sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}} (\alpha_{(i_1, \dots, i_k)})^2 \left(\prod_{j \in [k]} \lambda_{i_j} \right) \\
&\leq \sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\emptyset}} (\alpha_{(i_1, \dots, i_k)})^2 + \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} \sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}} (\alpha_{(i_1, \dots, i_k)})^2,
\end{aligned} \tag{35}$$

where the fourth equality is by the orthonormality of the eigenvectors and the inequality is by (34). We consider each term in (35) separately. For every $\mathcal{J} \subseteq [k]$,

$$\alpha_{(1, \dots, 1)}^{\mathcal{J}} = \langle f^{\subseteq \mathcal{J}}, u_{(1, \dots, 1)} \rangle = |\Pi|^{-|\mathcal{J}|/2} \sum_{x \in \Pi^{|\mathcal{J}|}} f^{\subseteq \mathcal{J}}(x) = |\Pi|^{|\mathcal{J}|/2} \mathbb{E}[f^{\subseteq \mathcal{J}}]. \tag{36}$$

In particular, since $\mathcal{M}_{=\emptyset} = \{(1, \dots, 1)\}$,

$$\sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\emptyset}} (\alpha_{(i_1, \dots, i_k)})^2 = (\alpha_{(1, \dots, 1)})^2 = |\Pi|^k \mathbb{E}[f]^2. \tag{37}$$

In addition, by Proposition 18,

$$\begin{aligned}
&\sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}} (\alpha_{(i_1, \dots, i_k)})^2 = |\Pi|^{-|\overline{\mathcal{J}}|} \sum_{(i_1, \dots, i_k) \in \mathcal{M}_{=\mathcal{J}}} (\alpha_{(i_1, \dots, i_k)}^{\mathcal{J}})^2 \\
&= |\Pi|^{-|\overline{\mathcal{J}}|} \sum_{(i_1, \dots, i_{|\mathcal{J}|}) \in [2, |\Pi|]^{|\mathcal{J}|}} (\alpha_{(i_1, \dots, i_{|\mathcal{J}|})}^{\mathcal{J}})^2 \\
&\leq |\Pi|^{-|\overline{\mathcal{J}}|} \sum_{(i_1, \dots, i_{|\mathcal{J}|}) \in [\Pi]^{|\mathcal{J}|} \setminus \{(1, \dots, 1)\}} (\alpha_{(i_1, \dots, i_{|\mathcal{J}|})}^{\mathcal{J}})^2 \\
&= |\Pi|^{-|\overline{\mathcal{J}}|} \left(\|f^{\subseteq \mathcal{J}}\|_2^2 - |\Pi|^{|\mathcal{J}|} \mathbb{E}[f^{\subseteq \mathcal{J}}]^2 \right),
\end{aligned} \tag{38}$$

where the final equality is by orthonormality of the eigenvectors and by (36).

Plugging (37) and (38) into (35) completes the proof. ■

Proof (of Lemma 12). By Proposition 19, $\text{Col}[A^{\otimes k}\nu]$ is bounded by

$$|\Pi|^k \mathbb{E}[\nu]^2 + |\Pi|^{-k} \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} |\Pi|^{|\mathcal{J}|} \left(\|\nu^{\subseteq \mathcal{J}}\|_2^2 - |\Pi|^{|\mathcal{J}|} \mathbb{E}[\nu^{\subseteq \mathcal{J}}]^2 \right).$$

Since ν is a distribution, $\mathbb{E}[\nu] = |\Pi|^{-k}$ and $\mathbb{E}[\nu^{\subseteq \mathcal{J}}] = |\Pi|^{-|\mathcal{J}|}$ for $\mathcal{J} \subseteq [k]$. Plugging these into the above inequality completes the proof. $\blacksquare$

7 Norm Bounds for an Sbox Layer

We prove Lemma 10 and Lemma 4. These lemmas bound the 2-norm for a (masked) Sbox layer, taking activity patterns into account. Recall that for every $\mathcal{D} \subseteq [k]$ the corresponding activity pattern is the set

$$\mathcal{Z}_{\mathcal{D}} = \{x \in (\mathbb{F}_2^n)^k : x_i = \mathbf{0}_n \iff i \in \mathcal{D}\}.$$

We begin by defining conditional distributions on activity patterns.

Definition 17 (Conditional distributions on activity patterns). *Consider an Sbox (permutation) layer composed of k instances of an arbitrary Sbox with the input and output space $\mathbb{F}_2^n$. Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to the Sbox layer. Assuming that $\Pr_{y \sim \nu}[y \in \mathcal{Z}_{\mathcal{D}}] > 0$, define the conditional distribution on the active Sboxes $\nu_{|\mathcal{D}} \in \mathbb{R}^{(\mathbb{F}_2^n)^{k-|\mathcal{D}|}}$ by*

$$\nu_{|\mathcal{D}}(x) = \frac{\Pr_{y \sim \nu}[y_{\overline{\mathcal{D}}} = x \wedge y \in \mathcal{Z}_{\mathcal{D}}]}{\Pr_{y \sim \nu}[y \in \mathcal{Z}_{\mathcal{D}}]}.$$

In addition, define $\nu_{|\mathcal{D}^-} \in \mathbb{R}^{(\mathbb{F}_2^{n(-)})^{k-|\mathcal{D}|}}$ as a distribution that is identical to $\nu_{|\mathcal{D}}$, but restricted to the domain $(\mathbb{F}_2^{n(-)})^{k-|\mathcal{D}|}$.

Note that for $\mathcal{D} \subseteq [k]$, $\nu_{|\mathcal{D}}$ is supported on $(\mathbb{F}_2^{n(-)})^{k-|\mathcal{D}|}$, hence $\nu_{|\mathcal{D}^-}$ is indeed a valid distribution.

We start by proving a basic result for the collision probability of the output difference distribution of an Sbox layer, considering activity patterns.

Proposition 20. *Let F be a keyed permutation over $\mathbb{F}_2^n$. Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a layer composed of k instances of F , such that $\nu(\mathbf{0}_{n \cdot k}) = 0$ (i.e., $\Pr_{x \sim \nu}[x \in \mathcal{Z}_{[k]}] = 0$). Denote by $\xi \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the keyed Sbox layer. Then,*

$$\text{Col}[\xi] = \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu}[x \in \mathcal{Z}_{\mathcal{D}}]^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}].$$

Proof. Observe that the Sbox layer acts independently on each set $\mathcal{Z}_{\mathcal{D}}$, namely, $x \in \mathcal{Z}_{\mathcal{D}}$ if and only if $(\mathfrak{D}_F)^{\otimes k} e_x$ is supported on $\mathcal{Z}_{\mathcal{D}}$.

Consider drawing $x, x' \sim \nu$ independently. Denote by Col the event that x, x' collide after the Sbox layer. By the observation above, Col can occur if and only if $x, x' \in \mathcal{Z}_{\mathcal{D}}$ for some $\mathcal{D} \subsetneq [k]$

(they cannot be in $\mathcal{Z}_{[k]}$ by assumption). Therefore, by Fact 2,

$$\begin{aligned}
\text{Col}[\xi] &= \Pr[\text{Col}] = \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x, x' \sim \nu} [x, x' \in \mathcal{Z}_{\mathcal{D}}] \Pr[\text{Col} \mid x, x' \in \mathcal{Z}_{\mathcal{D}}] \\
&= \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x, x' \sim \nu} [x, x' \in \mathcal{Z}_{\mathcal{D}}] \text{Col}[(\mathfrak{D}_F)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}}] \\
&= \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x, x' \sim \nu} [x, x' \in \mathcal{Z}_{\mathcal{D}}] \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}] \\
&= \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}],
\end{aligned}$$

where the equality $\text{Col}[(\mathfrak{D}_F)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}}] = \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}]$ again follows from the fact that the Sbox layer acts independently on each set $\mathcal{Z}_{\mathcal{D}}$. $\blacksquare$

7.1 Bound by Expected Collision Probability

We consider an input difference distribution ν to an Sbox layer, and prove a simple bound on the output difference collision probability by taking the expectation of probabilities over elements drawn from ν .

Proposition 21 (Basic bound by expected collision probability). *Let F be a keyed permutation over $\mathbb{F}_2^n$. Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a keyed Sbox layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$. Denote by $\xi \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the keyed Sbox layer. Then*

$$\text{Col}[\xi] \leq \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \mathbb{E}_{z \sim \nu_{|\mathcal{D}^-}} \left[\prod_{\ell=1}^{k-|\mathcal{D}|} \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right].$$

Proof. By Proposition 20,

$$\text{Col}[\xi] = \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}]. \quad (39)$$

Fix $\mathcal{D} \subsetneq [k]$. By Fact 2,

$$\begin{aligned}
\text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}] &= \sum_{y \in (\mathbb{F}_2^{n(-)})^{k-|\mathcal{D}|}} \left(\mathbb{E}_{z \sim \nu_{|\mathcal{D}^-}} [(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} [y, z]] \right)^2 \\
&\leq \mathbb{E}_{z \sim \nu_{|\mathcal{D}^-}} \left[\sum_{y \in (\mathbb{F}_2^{n(-)})^{k-|\mathcal{D}|}} (\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} [y, z]^2 \right] \\
&= \mathbb{E}_{z \sim \nu_{|\mathcal{D}^-}} \left[\text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} e_z] \right] \\
&= \mathbb{E}_{z \sim \nu_{|\mathcal{D}^-}} \left[\prod_{\ell=1}^{k-|\mathcal{D}|} \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right].
\end{aligned}$$

where we have applied Jensen's inequality to the convex function $t \rightarrow t^2$.

Combining with (39) we obtain

$$\text{Col}[\xi] \leq \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \mathbb{E}_{z \sim \nu|_{\mathcal{D}^c}} \left[\prod_{\ell=1}^{k-|\mathcal{D}|} \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right].$$

■

We restate and prove Lemma 10, which is a refinement of the previous bound that is slightly easier to use.

Lemma 10 (Bound by expected collision probability). *Let F be a keyed permutation over the space $\mathbb{F}_2^n$ such that*

$$\|\mathfrak{D}_F^-\|_{1,2}^2 = \max_{y \in \mathbb{F}_2^{n(-)}} \{\text{Col}[\mathfrak{D}_F^- e_y]\} \leq \mathbf{C}$$

for some constant $\mathbf{C} \geq 0$.

Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$. For every $1 \leq t \leq k$, let $\mathbf{M}_t$ be a constant such that

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{M}[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{M}_t.$$

Moreover, for $1 \leq d \leq k-1$, let $\mathbf{Z}_d \in \{0,1\}$ be an indicator that satisfies the following constraint: if $\mathbf{Z}_d = 0$, then for any $\mathcal{D} \subseteq [k], |\mathcal{D}| = d$

$$\Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}] = 0.$$

Let $\text{ColF}: \mathbb{R} \rightarrow \mathbb{R}$ be a function (that depends on $\mathfrak{D}_F$) with the following property: if $\rho \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ is a distribution that is supported only on $\mathcal{Z}_0$ such that $\mathbf{M}[\rho^{\subseteq \{i\}}] \leq \mathbf{M}$ for all $i \in [k]$, then the output difference distribution satisfies

$$\mathbb{E}_{z \sim \rho} [\text{Col}[(\mathfrak{D}_F)^{\otimes k} e_z]] \leq \text{ColF}(\mathbf{M}).$$

Denote by $\xi = (\mathfrak{D}_F)^{\otimes k} \nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the Sbox layer. Then,

$$\text{Col}[\xi] \leq \text{ColF}\left(\frac{\mathbf{M}_1}{1 - k \cdot \mathbf{M}_1}\right) + \sum_{d=1}^{k-1} \binom{k}{d} \mathbf{Z}_d (\mathbf{M}_d)^2 \cdot \mathbf{C}^{k-d}, \quad (25)$$

assuming $k \cdot \mathbf{M}_1 < 1$.

Proof. By Proposition 21,

$$\text{Col}[\xi] \leq \sum_{\mathcal{D} \subsetneq [k]} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \mathbb{E}_{z \sim \nu|_{\mathcal{D}^c}} \left[\prod_{\ell=1}^{k-|\mathcal{D}|} \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right]. \quad (40)$$

By assumption for every $y \in \mathbb{F}_2^{n(-)}$, $\text{Col}[\mathfrak{D}_F^- e_y] \leq \max_{y \in \mathbb{F}_2^{n(-)}} \{\text{Col}[\mathfrak{D}_F^- e_y]\} \leq \mathbf{C}$. Moreover, $\Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}] \leq \mathbf{Z}_{|\mathcal{D}|} \Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}] \leq \mathbf{Z}_{|\mathcal{D}|} \mathbf{M}_{|\mathcal{D}|}$, Therefore, for every $\mathcal{D} \subsetneq [k]$ with $|\mathcal{D}| = d$,

$$\Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}]^2 \mathbb{E}_{z \sim \nu|_{\mathcal{D}^c}} \left[\prod_{\ell=1}^{k-|\mathcal{D}|} \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right] \leq \mathbf{Z}_d (\mathbf{M}_d)^2 \mathbf{C}^{k-d}.$$

We apply this inequality to each term in (40), but keep the term for $\mathcal{D} = \emptyset$. We obtain

$$\begin{aligned} \text{Col}[\xi] &\leq \Pr[x \in \mathcal{Z}_\emptyset]^2 \mathbb{E}_{z \sim \nu_{|\emptyset-}} \left[\prod_{\ell=1}^k \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right] + \sum_{d=1}^{k-1} \binom{k}{d} \mathbb{Z}_d(\mathbb{M}_d)^2 \cdot \mathfrak{C}^{k-d} \\ &\leq \mathbb{E}_{z \sim \nu_{|\emptyset-}} \left[\prod_{\ell=1}^k \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right] + \sum_{d=1}^{k-1} \binom{k}{d} \mathbb{Z}_d(\mathbb{M}_d)^2 \cdot \mathfrak{C}^{k-d}. \end{aligned} \quad (41)$$

We now bound $\mathbb{M}[(\nu_{|\emptyset-})^{\subseteq \{\ell\}}]$ for every $\ell \in [k]$. Fix $\ell \in [k]$ and $v \in \mathbb{F}_2^{n(-)}$. We have

$$(\nu_{|\emptyset-})^{\subseteq \{\ell\}}(v) = \frac{\Pr_{u \sim \nu}[u_\ell = v \wedge u \in \mathcal{Z}_\emptyset]}{\Pr_{u \sim \nu}[u \in \mathcal{Z}_\emptyset]} \leq \frac{\Pr_{u \sim \nu}[u_\ell = v]}{\Pr_{u \sim \nu}[u \in \mathcal{Z}_\emptyset]} = \frac{\nu^{\subseteq \{\ell\}}(v)}{\Pr_{u \sim \nu}[u \in \mathcal{Z}_\emptyset]} \leq \frac{\mathbb{M}_1}{\Pr_{u \sim \nu}[u \in \mathcal{Z}_\emptyset]}.$$

Moreover, a union bound over $[k]$ gives

$$\Pr_{u \sim \nu}[u \in \mathcal{Z}_\emptyset] \geq 1 - k \cdot \mathbb{M}_1.$$

Hence, for every $\ell \in [k]$ and $v \in \mathbb{F}_2^{n(-)}$,

$$(\nu_{|\emptyset-})^{\subseteq \{\ell\}}(v) \leq \frac{\mathbb{M}_1}{1 - k \cdot \mathbb{M}_1},$$

implying

$$\mathbb{M}[(\nu_{|\emptyset-})^{\subseteq \{\ell\}}] \leq \frac{\mathbb{M}_1}{1 - k \cdot \mathbb{M}_1}.$$

Hence, by definition of ColF ,

$$\mathbb{E}_{z \sim \nu_{|\emptyset-}} \left[\prod_{\ell=1}^k \text{Col}[\mathfrak{D}_F^- e_{z_\ell}] \right] = \mathbb{E}_{z \sim \nu_{|\emptyset}} \left[\prod_{\ell=1}^k \text{Col}[\mathfrak{D}_F e_{z_\ell}] \right] = \mathbb{E}_{z \sim \nu_{|\emptyset}} [\text{Col}[(\mathfrak{D}_F)^{\otimes k} e_z]] \leq \text{ColF}\left(\frac{\mathbb{M}_1}{1 - k \cdot \mathbb{M}_1}\right).$$

Plugging this into (41) completes the proof. ■

7.2 Spectral Bound

We now bound the collision probability of the output difference of the Sbox layer using the spectral technique of Lemma 12. We start by bounding the term for a single activity pattern in the expression of Proposition 20.

Proposition 22 (Activity pattern spectral bound). *Let F be a keyed permutation over $\mathbb{F}_2^n$ such that the second largest eigenvalue of $(\mathfrak{D}_F^-)^T \mathfrak{D}_F^-$ is at most λ .*

Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a keyed Sbox layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$. For any $\mathcal{D} \subseteq [k]$, denote

$$\mathbb{P}_\mathcal{D} = \Pr_{x \sim \nu}[x \in \mathcal{Z}_\mathcal{D}], \mathbb{M}_\mathcal{D} = \mathbb{M}[\nu^{\subseteq \mathcal{D}}], \text{ and } \mathfrak{C}_\mathcal{D} = \text{Col}[\nu^{\subseteq \mathcal{D}}].$$

Then, for $\mathcal{D} = \emptyset$,

$$(\mathbb{P}_\emptyset)^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k} \nu_{|\emptyset-}] \leq (2^n - 1)^{-k} \left((\mathbb{P}_\emptyset)^2 + \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} ((2^n - 1)^{|\mathcal{J}|} \mathfrak{C}_\mathcal{J} - (\mathbb{P}_\emptyset)^2) \right),$$

while for $\mathcal{D} \neq \emptyset$,

$$\begin{aligned} & (\mathbb{P}_{\mathcal{D}})^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}] \\ & \leq (2^n - 1)^{|\mathcal{D}|-k} \left((\mathbb{P}_{\mathcal{D}})^2 + \sum_{\mathcal{J} \subseteq [k] \setminus \mathcal{D}, \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} \left((2^n - 1)^{|\mathcal{J}|} \mathbb{M}_{\mathcal{J} \cup \mathcal{D}} \cdot \mathbb{M}_{\mathcal{D}} - (\mathbb{P}_{\mathcal{D}})^2 \right) \right). \end{aligned}$$

Proof. Fix $\mathcal{D} \subseteq [k]$. We apply Lemma 12 to the distribution $\nu_{|\mathcal{D}^-}$ which is defined only on the $k - |\mathcal{D}|$ active Sboxes. We obtain

$$(2^n - 1)^{k-|\mathcal{D}|} \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}] \leq 1 + \sum_{\mathcal{J} \subseteq [k] \setminus \mathcal{D}, \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} \left((2^n - 1)^{|\mathcal{J}|} \text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}] - 1 \right). \quad (42)$$

We now simplify the term we need to bound. Applying (42), we have

$$\begin{aligned} & (2^n - 1)^{k-|\mathcal{D}|} (\mathbb{P}_{\mathcal{D}})^2 \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}^-}] \\ & \leq (\mathbb{P}_{\mathcal{D}})^2 + \sum_{\mathcal{J} \subseteq [k] \setminus \mathcal{D}, \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} \left((2^n - 1)^{|\mathcal{J}|} (\mathbb{P}_{\mathcal{D}})^2 \text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}] - (\mathbb{P}_{\mathcal{D}})^2 \right). \end{aligned} \quad (43)$$

For $\mathcal{J} \subseteq [k] \setminus \mathcal{D}, \mathcal{J} \neq \emptyset$, observe that $\text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}]$ measures the collision probability of ν on the indices of $\mathcal{J}$, inside the set $\mathcal{Z}_{\mathcal{D}}$. Note that $\text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}] = \text{Col}[(\nu_{|\mathcal{D}})^{\subseteq \mathcal{J}}]$, as $\nu_{|\mathcal{D}^-}$ and $\nu_{|\mathcal{D}}$ have the same support, and are identical over this support. Therefore,

$$\text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}] = \text{Col}[(\nu_{|\mathcal{D}})^{\subseteq \mathcal{J}}] = \frac{\Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}} \wedge x, x' \in \mathcal{Z}_{\mathcal{D}}]}{\Pr_{x, x' \sim \nu}[x, x' \in \mathcal{Z}_{\mathcal{D}}]} = \frac{\Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}} \wedge x, x' \in \mathcal{Z}_{\mathcal{D}}]}{(\mathbb{P}_{\mathcal{D}})^2}$$

Hence,

$$(\mathbb{P}_{\mathcal{D}})^2 \text{Col}[(\nu_{|\mathcal{D}^-})^{\subseteq \mathcal{J}}] = \Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}} \wedge x, x' \in \mathcal{Z}_{\mathcal{D}}]. \quad (44)$$

First, observe that

$$\Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}} \wedge x, x' \in \mathcal{Z}_{\mathcal{D}}] \leq \Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}}] = \text{Col}[\nu^{\subseteq \mathcal{J}}] = \mathbb{C}_{\mathcal{J}}.$$

For $\mathcal{D} = \emptyset$, we plug this result into (44) and then into (43). We obtain,

$$(2^n - 1)^k \Pr_{x, x' \sim \nu}[x, x' \in \mathcal{Z}_{\emptyset}] \text{Col}[(\mathfrak{D}_F^-)^{\otimes k} \nu_{|\emptyset^-}] \leq (\mathbb{P}_{\emptyset})^2 + \sum_{\mathcal{J} \subseteq [k], \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} \left((2^n - 1)^{|\mathcal{J}|} \mathbb{C}_{\mathcal{J}} - (\mathbb{P}_{\emptyset})^2 \right),$$

proving the desired bound for $\mathcal{D} = \emptyset$.

For $\mathcal{D} \neq \emptyset$,

$$\begin{aligned} \Pr_{x, x' \sim \nu}[x_{\mathcal{J}} = x'_{\mathcal{J}} \wedge x, x' \in \mathcal{Z}_{\mathcal{D}}] & \leq \Pr_{x, x' \sim \nu}[x_{\mathcal{J} \cup \mathcal{D}} = x'_{\mathcal{J} \cup \mathcal{D}} \wedge x \in \mathcal{Z}_{\mathcal{D}}] \\ & \leq \Pr_{x \sim \nu}[x \in \mathcal{Z}_{\mathcal{D}}] \max_{x \in \mathcal{Z}_{\mathcal{D}}} \{ \Pr_{x' \sim \nu}[x'_{\mathcal{J} \cup \mathcal{D}} = x_{\mathcal{J} \cup \mathcal{D}}] \} \\ & \leq \mathbb{M}_{\mathcal{D}} \cdot \mathbb{M}_{\mathcal{J} \cup \mathcal{D}}. \end{aligned}$$

We plug this result into (44) and then into (43). We obtain,

$$\begin{aligned} & (2^n - 1)^{k-|\mathcal{D}|} \Pr_{x, x' \sim \nu} [x, x' \in \mathcal{Z}_{\mathcal{D}}] \text{Col}[(\mathfrak{D}_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}|-}] \\ & \leq (\mathbf{P}_{\mathcal{D}})^2 + \sum_{\mathcal{J} \subseteq [k] \setminus \mathcal{D}, \mathcal{J} \neq \emptyset} \lambda^{|\mathcal{J}|} ((2^n - 1)^{|\mathcal{J}|} \mathbf{M}_{\mathcal{J} \cup \mathcal{D}} \cdot \mathbf{M}_{\mathcal{D}} - (\mathbf{P}_{\mathcal{D}})^2). \end{aligned}$$

This completes the proof. ■

We now restate and prove Lemma 4.

Lemma 4 (Spectral bound). *Let F be a keyed permutation over $\mathbb{F}_2^n$ such that the second largest eigenvalue of $(\mathfrak{D}_F^-)^T \mathfrak{D}_F^-$ is at most λ . Let $\nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ be an input difference distribution to a keyed Sbox layer composed of k instances of F . Assume that $\nu(\mathbf{0}_{n \cdot k}) = 0$ (i.e., the 0 input difference has 0 probability). Moreover, assume that $\sum_{j=1}^k \binom{k}{j} \lambda^j \leq 1$. For every $1 \leq t \leq k$, let $\mathbf{C}_t, \mathbf{M}_t$ be constants such that:*

$$\max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\text{Col}[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{C}_t \text{ and } \max_{\mathcal{T} \subseteq [k], |\mathcal{T}|=t} \{\mathbf{M}[\nu^{\subseteq \mathcal{T}}]\} \leq \mathbf{M}_t,$$

and $\mathbf{MN}_1$ be a constant such that $\min_{\mathcal{T} \subseteq [k], |\mathcal{T}|=1} \{\mathbf{MN}[\nu^{\subseteq \mathcal{T}}]\} \geq \mathbf{MN}_1$.

For $1 \leq d \leq k-1$, let $\mathbf{Z}_d \in \{0, 1\}$ be an indicator such that if $\mathbf{Z}_d = 0$, then for any $\mathcal{D} \subset [k]$ with $|\mathcal{D}| = d$, $\Pr_{x \sim \nu} [x \in \mathcal{Z}_{\mathcal{D}}] = 0$.

Denote by $\xi = (\mathfrak{D}_F^-)^{\otimes k} \nu \in \mathbb{R}^{(\mathbb{F}_2^n)^k}$ the output difference distribution of the keyed Sbox layer. Then,

$$\begin{aligned} \text{Col}[\xi] & \leq (2^n - 1)^{-k} \left((\mathbf{P}_{\emptyset})^2 + \sum_{j=1}^k \binom{k}{j} \lambda^j ((2^n - 1)^j \mathbf{C}_j - (\mathbf{P}_{\emptyset})^2) \right) \\ & + \sum_{\mathcal{D} \subsetneq [k], \mathcal{D} \neq \emptyset} \mathbf{Z}_{|\mathcal{D}|} (2^n - 1)^{|\mathcal{D}|-k} \left((\mathbf{P}_{\mathcal{D}})^2 + \sum_{j=1}^{k-|\mathcal{D}|} \binom{k-|\mathcal{D}|}{j} \lambda^j ((2^n - 1)^j \mathbf{M}_{j+|\mathcal{D}|} \cdot \mathbf{M}_{|\mathcal{D}|} - (\mathbf{P}_{\mathcal{D}})^2) \right), \end{aligned} \tag{16}$$

where $\{\mathbf{P}_{\mathcal{D}}\}_{\mathcal{D} \subsetneq [k]}$ are variables that satisfy the following constraints:

$$\mathbf{P}_{\emptyset} + \sum_{\mathcal{D} \subsetneq [k], \mathcal{D} \neq \emptyset} \mathbf{Z}_{|\mathcal{D}|} \mathbf{P}_{\mathcal{D}} = 1, \tag{17}$$

$$0 \leq \mathbf{P}_{\emptyset} \leq \min\{1 - k \cdot \mathbf{MN}_1 + \binom{k}{2} \mathbf{M}_2, 1\}, \tag{18}$$

for every $\mathcal{D} \neq \emptyset$,

$$0 \leq \mathbf{P}_{\mathcal{D}} \leq \mathbf{M}_{|\mathcal{D}|}, \tag{19}$$

and moreover for every $d \in [k]$

$$\mathbf{P}_{\{d\}} \geq \mathbf{MN}_1 - \sum_{j=2}^{k-1} \binom{k-1}{j-1} \mathbf{M}_j. \tag{20}$$

Proof. By Proposition 20,

$$\text{Col}[\xi] = \sum_{D \subsetneq [k]} \Pr_{x \sim \nu}[x \in \mathcal{Z}_D]^2 \text{Col}[(D_F^-)^{\otimes k-|\mathcal{D}|} \nu_{|\mathcal{D}|-}].$$

For each $\mathcal{D} \subsetneq [k]$, we apply the bounds of Proposition 22, and further substitute the corresponding bounds $\mathbf{M}_t$ and $\mathbf{C}_t$ and indicators $\mathbf{Z}_d$. We obtain (16).

It remains to prove the constraints on the variables $\mathbf{P}_{\mathcal{D}} = \Pr_{x \sim \nu}[x \in \mathcal{Z}_{\mathcal{D}}]$. The first constraint (17) holds as the activity patterns partition the space.

For $\mathcal{D} = \emptyset$, we have

$$\begin{aligned} \mathbf{P}_{\emptyset} &= \Pr_{x \sim \nu}[\forall i \in [k]: x_i \neq \mathbf{0}_n] = 1 - \Pr_{x \sim \nu}[\exists i \in [k]: x_i = \mathbf{0}_n] \\ &\leq 1 - \sum_{i \in [k]} \Pr_{x \sim \nu}[x_i = \mathbf{0}_n] + \sum_{1 \leq i < j \leq k} \Pr_{x \sim \nu}[x_i = \mathbf{0}_n, x_j = \mathbf{0}_n] \\ &\leq 1 - k \cdot \mathbf{MN}_1 + \binom{k}{2} \mathbf{M}_2, \end{aligned}$$

where the first inequality is by the Bonferroni inequality of order 2. Since $0 \leq \mathbf{P}_{\emptyset} \leq 1$ holds trivially, this proves (18).

For $\mathcal{D} \neq \emptyset$,

$$0 \leq \mathbf{P}_{\mathcal{D}} = \Pr_{x \sim \nu}[x_i = \mathbf{0}_n \iff i \in \mathcal{D}] \leq \Pr_{x \sim \nu}[i \in \mathcal{D} \Rightarrow x_i = \mathbf{0}_n] \leq \mathbf{M}_{|\mathcal{D}|},$$

proving (19).

Finally, for $d \in [k]$, we have $\Pr_{x \sim \nu}[x_d = \mathbf{0}_n] = \sum_{\mathcal{D} \ni d, \mathcal{D} \neq [k]} \mathbf{P}_{\mathcal{D}}$. Hence,

$$\mathbf{P}_{\{d\}} = \Pr_{x \sim \nu}[x_d = \mathbf{0}_n] - \sum_{\mathcal{D} \ni d, 2 \leq |\mathcal{D}| < k} \mathbf{P}_{\mathcal{D}} \geq \mathbf{MN}_1 - \sum_{j=2}^{k-1} \binom{k-1}{j-1} \mathbf{M}_j,$$

proving (20). ■

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References

- [AK18] Ralph Ankele and Stefan Kölbl. Mind the Gap - A Closer Look at the Security of Block Ciphers against Differential Cryptanalysis. In Carlos Cid and Michael J. Jacobson Jr., editors, *SAC 2018*, volume 11349 of *Lecture Notes in Computer Science*, pages 163–190. Springer, 2018.
- [BBS05] Eli Biham, Alex Biryukov, and Adi Shamir. Cryptanalysis of Skipjack Reduced to 31 Rounds Using Impossible Differentials. *J. Cryptol.*, 18(4):291–311, 2005.
- [BFL⁺23] Christof Beierle, Patrick Felke, Gregor Leander, Patrick Neumann, and Lukas Stennes. On Perfect Linear Approximations and Differentials over Two-Round SPNs. In Helena

- Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023*, volume 14083 of *Lecture Notes in Computer Science*, pages 209–239. Springer, 2023.
- [BHLP25] Christof Beierle, Phil Hebborn, Gregor Leander, and Yevhen Perekhuda. Integral Resistance of Block Ciphers with Key Whitening by Modular Addition. In Yael Tauman Kalai and Seny F. Kamara, editors, *CRYPTO 2025*, volume 16004 of *Lecture Notes in Computer Science*, pages 497–529. Springer, 2025.
 - [BLNS18] Christina Boura, Virginie Lallemand, María Naya-Plasencia, and Valentin Suder. Making the Impossible Possible. *J. Cryptol.*, 31(1):101–133, 2018.
 - [BLS25] Tim Beyne, Gregor Leander, and Immo Schütt. Pairwise independence of AES-like block ciphers. Cryptology ePrint Archive, Paper 2025/1495, 2025.
 - [BR22] Tim Beyne and Vincent Rijmen. Differential Cryptanalysis in the Fixed-Key Model. In Yevgeniy Dodis and Thomas Shrimpton, editors, *CRYPTO 2022*, volume 13509 of *Lecture Notes in Computer Science*, pages 687–716. Springer, 2022.
 - [BS91] Eli Biham and Adi Shamir. Differential Cryptanalysis of DES-like Cryptosystems. *J. Cryptol.*, 4(1):3–72, 1991.
 - [CKL⁺03] Kilsoo Chun, Seungjoo Kim, Sangjin Lee, Soo Hak Sung, and Seonhee Yoon. Differential and linear cryptanalysis for 2-round SPNs. *Inf. Process. Lett.*, 87(5):277–282, 2003.
 - [CLL⁺19] Anne Canteaut, Virginie Lallemand, Gregor Leander, Patrick Neumann, and Friedrich Wiemer. BISON - Instantiating the Whitened Swap-Or-Not Construction. In Yuval Ishai and Vincent Rijmen, editors, *EUROCRYPT 2019*, volume 11478 of *Lecture Notes in Computer Science*, pages 585–616. Springer, 2019.
 - [CR15] Anne Canteaut and Joëlle Roué. On the Behaviors of Affine Equivalent Sboxes Regarding Differential and Linear Attacks. In Elisabeth Oswald and Marc Fischlin, editors, *EUROCRYPT 2015*, volume 9056 of *Lecture Notes in Computer Science*, pages 45–74. Springer, 2015.
 - [CSK⁺04] Hong-Su Cho, Soo Hak Sung, Daesung Kwon, Jung-Keun Lee, Jung Hwan Song, and Jongin Lim. New Method for Bounding the Maximum Differential Probability for SPNs and ARIA. In Choonsik Park and Seongtaek Chee, editors, *ICISC 2004*, volume 3506 of *Lecture Notes in Computer Science*, pages 21–32. Springer, 2004.
 - [DGV93] Joan Daemen, René Govaerts, and Joos Vandewalle. Weak Keys for IDEA. In Douglas R. Stinson, editor, *CRYPTO 1993*, volume 773 of *Lecture Notes in Computer Science*, pages 224–231. Springer, 1993.
 - [DGV94] Joan Daemen, René Govaerts, and Joos Vandewalle. Correlation Matrices. In Bart Preneel, editor, *Fast Software Encryption*, volume 1008 of *Lecture Notes in Computer Science*, pages 275–285. Springer, 1994.
 - [DLP⁺09] Joan Daemen, Mario Lamberger, Norbert Pramstaller, Vincent Rijmen, and Frederik Vercauteren. Computational aspects of the expected differential probability of 4-round AES and AES-like ciphers. *Computing*, 85(1-2):85–104, 2009.
 - [DR99] Joan Daemen and Vincent Rijmen. AES Proposal: RIJNDael. AES submission, 1999.

- [DR06] Joan Daemen and Vincent Rijmen. Understanding Two-Round Differentials in AES. In Roberto De Prisco and Moti Yung, editors, *Security and Cryptography for Networks*, volume 4116 of *Lecture Notes in Computer Science*, pages 78–94. Springer, 2006.
- [DR07] Joan Daemen and Vincent Rijmen. Plateau characteristics. *IET Inf. Secur.*, 1(1):11–17, 2007.
- [DR20] Joan Daemen and Vincent Rijmen. *The Design of Rijndael - The Advanced Encryption Standard (AES), Second Edition*. Information Security and Cryptography. Springer, 2020.
- [HJ91] Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.
- [HLL⁺00] Seokhie Hong, Sangjin Lee, Jongin Lim, Jaechul Sung, Dong Hyeon Cheon, and Inho Cho. Provable Security against Differential and Linear Cryptanalysis for the SPN Structure. In Bruce Schneier, editor, *Fast Software Encryption*, volume 1978 of *Lecture Notes in Computer Science*, pages 273–283. Springer, 2000.
- [HLLT21] Phil Hebborn, Baptiste Lambin, Gregor Leander, and Yosuke Todo. Strong and Tight Security Guarantees Against Integral Distinguishers. In Mehdi Tibouchi and Huaxiong Wang, editors, *ASIACRYPT 2021*, volume 13090 of *Lecture Notes in Computer Science*, pages 362–391. Springer, 2021.
- [KMT01a] Liam Keliher, Henk Meijer, and Stafford E. Tavares. Improving the Upper Bound on the Maximum Average Linear Hull Probability for Rijndael. In Serge Vaudenay and Amr M. Youssef, editors, *Selected Areas in Cryptography*, volume 2259 of *Lecture Notes in Computer Science*, pages 112–128. Springer, 2001.
- [KMT01b] Liam Keliher, Henk Meijer, and Stafford E. Tavares. New Method for Upper Bounding the Maximum Average Linear Hull Probability for SPNs. In Birgit Pfitzmann, editor, *EUROCRYPT 2001*, volume 2045 of *Lecture Notes in Computer Science*, pages 420–436. Springer, 2001.
- [KS07] Liam Keliher and Jiayuan Sui. Exact maximum expected differential and linear probability for two-round Advanced Encryption Standard. *IET Inf. Secur.*, 1(2):53–57, 2007.
- [LDKK08] Jiqiang Lu, Orr Dunkelman, Nathan Keller, and Jongsung Kim. New Impossible Differential Attacks on AES. In Dipanwita Roy Chowdhury, Vincent Rijmen, and Abhijit Das, editors, *INDOCRYPT 2008*, volume 5365 of *Lecture Notes in Computer Science*, pages 279–293. Springer, 2008.
- [LMM91] Xuejia Lai, James L. Massey, and Sean Murphy. Markov Ciphers and Differential Cryptanalysis. In Donald W. Davies, editor, *EUROCRYPT 1991*, volume 547 of *Lecture Notes in Computer Science*, pages 17–38. Springer, 1991.
- [LPS21] Gaëtan Leurent, Clara Pernot, and André Schrottenloher. Clustering Effect in Simon and Simeck. In Mehdi Tibouchi and Huaxiong Wang, editors, *ASIACRYPT 2021*, volume 13090 of *Lecture Notes in Computer Science*, pages 272–302. Springer, 2021.

- [LPTV23] Tianren Liu, Angelos Pelecanos, Stefano Tessaro, and Vinod Vaikuntanathan. Layout Graphs, Random Walks and the t-Wise Independence of SPN Block Ciphers. In Helena Handschuh and Anna Lysyanskaya, editors, *CRYPTO 2023*, volume 14083 of *Lecture Notes in Computer Science*, pages 694–726. Springer, 2023.
- [LTV21] Tianren Liu, Stefano Tessaro, and Vinod Vaikuntanathan. The t-wise Independence of Substitution-Permutation Networks. In Tal Malkin and Chris Peikert, editors, *CRYPTO 2021*, volume 12828 of *Lecture Notes in Computer Science*, pages 454–483. Springer, 2021.
- [MDRM10] Hamid Mala, Mohammad Dakhilalian, Vincent Rijmen, and Mahmoud Modarres-Hashemi. Improved Impossible Differential Cryptanalysis of 7-Round AES-128. In Guang Gong and Kishan Chand Gupta, editors, *INDOCRYPT 2010*, volume 6498 of *Lecture Notes in Computer Science*, pages 282–291. Springer, 2010.
- [MT05] Ravi Montenegro and Prasad Tetali. Mathematical Aspects of Mixing Times in Markov Chains. *Found. Trends Theor. Comput. Sci.*, 1(3), 2005.
- [MV15] Eric Miles and Emanuele Viola. Substitution-Permutation Networks, Pseudorandom Functions, and Natural Proofs. *J. ACM*, 62(6):46:1–46:29, 2015.
- [Nat77] National Bureau of Standards. Data Encryption Standard. Federal Information Processing Standard 46, 1977.
- [NK92] Kaisa Nyberg and Lars R. Knudsen. Provable Security Against Differential Cryptanalysis. In Ernest F. Brickell, editor, *CRYPTO 1992*, volume 740, pages 566–574. Springer, 1992.
- [NK95] Kaisa Nyberg and Lars R. Knudsen. Provable Security Against a Differential Attack. *J. Cryptol.*, 8(1):27–37, 1995.
- [O’D14] Ryan O’Donnell. *Analysis of Boolean Functions*. Cambridge University Press, 2014.
- [PSC⁺02] Sangwoo Park, Soo Hak Sung, Seongtaek Chee, E-Joong Yoon, and Jongin Lim. On the Security of Rijndael-Like Structures against Differential and Linear Cryptanalysis. In Yuliang Zheng, editor, *ASIACRYPT 2002*, volume 2501 of *Lecture Notes in Computer Science*, pages 176–191. Springer, 2002.
- [PSLL03] Sangwoo Park, Soo Hak Sung, Sangjin Lee, and Jongin Lim. Improving the Upper Bound on the Maximum Differential and the Maximum Linear Hull Probability for SPN Structures and AES. In Thomas Johansson, editor, *Fast Software Encryption*, volume 2887 of *Lecture Notes in Computer Science*, pages 247–260. Springer, 2003.
- [Vau03] Serge Vaudenay. Decorrelation: A Theory for Block Cipher Security. *J. Cryptol.*, 16(4):249–286, 2003.

A Efficiently Bounding $\text{Col}[\xi]$ in Lemma 4

Given the parameters of Lemma 4, $\text{Col}[\xi]$ is upper bounded by maximizing the right-hand-side of (16). We now explain how to compute the bound efficiently, assuming that k is small.

First, as already noted, the assumption $\sum_{j=1}^k \binom{k}{j} \lambda^j \leq 1$ assures that all variables $\{(\mathbf{P}_{\mathcal{D}})^2\}_{\mathcal{D} \subseteq [k]}$ have non-negative coefficients in (16). Thus, the maximization of the right-hand-side of (16) is a

maximization problem of a convex quadratic function. The feasible region defined by the constraints (17), (18), (19), (20) is a convex polytope which is formed by the intersection of a hyperrectangle (constraints (18), (19), (20)) and a hyperplane (constraint (17)).

By the Bauer maximum principle, the maximal value of the function is attained at an extreme point of the polytope. Denoting the number of variables by v , an extreme point of the polytope is defined by $v - 1$ of these variables attaining their maximal or minimal value, while the remaining variable is uniquely determined by these variables and the hyperplane (constraint (17)). Thus, assuming the number of variables is not too large, the maximization problem can be solved by exhaustively searching all extreme points of the polytope.

Since the constraints on variables $\{P_{\mathcal{D}}\}_{\mathcal{D} \subseteq [k]}$ with sets $\mathcal{D}$ of the same size are identical, the exhaustive search can be optimized. For example, in the relevant case of $k = 4$ where for $1 \leq d \leq k - 1$, $Z_d = 1$, we have $2^4 - 1 = 15$ variables, categorized by their types: $\binom{4}{0} = 1$ variable $P_{\emptyset}$, $\binom{4}{1} = 4$ variables $P_{\mathcal{D}}$ with $|\mathcal{D}| = 1$, $\binom{4}{2} = 6$ variables $P_{\mathcal{D}}$ with $|\mathcal{D}| = 2$ and $\binom{4}{3} = 4$ variables $P_{\mathcal{D}}$ with $|\mathcal{D}| = 3$. An extreme point of the polytope can be defined by specifying how many variables of each type are set to their maximal value (there are $2 \cdot 5 \cdot 7 \cdot 5 = 350$ possibilities), and which additional variable type is determined by the remaining ones (4 possibilities). Altogether, there are at most $350 \cdot 4 = 1400$ vertices to check.

B Code for Analyzing AES Differentials

B.1 Code for Deriving the AES Sbox DDT Properties

```
#!/bin/python
import numpy
import sys
from scipy.linalg import hadamard

def SBOX(inp):
    s=((0x63,0x7C,0x77,0x7B,0xF2,0x6B,0x6F,0xC5,0x30,0x01,0x67,0x2B,0xFE,0xD7,0xAB,0x76),
      (0xCA, 0x82, 0xC9, 0x7D, 0xFA, 0x59, 0x47, 0xF0, 0xAD, 0xD4, 0xA2, 0xAF, 0x9C, 0xA4, 0x72, 0xC0),
      (0xB7, 0xFD, 0x93, 0x26, 0x36, 0x3F, 0xF7, 0xCC, 0x34, 0xA5, 0xE5, 0xF1, 0x71, 0xD8, 0x31, 0x15),
      (0x04, 0xC7, 0x23, 0xC3, 0x18, 0x96, 0x05, 0x9A, 0x07, 0x12, 0x80, 0xE2, 0xEB, 0x27, 0xB2, 0x75),
      (0x09, 0x83, 0x2C, 0x1A, 0x1B, 0x6E, 0x5A, 0xA0, 0x52, 0x3B, 0xD6, 0xB3, 0x29, 0xE3, 0x2F, 0x84),
      (0x53, 0xD1, 0x00, 0xED, 0x20, 0xFC, 0xB1, 0x5B, 0x6A, 0xCB, 0xBE, 0x39, 0x4A, 0x4C, 0x58, 0xCF),
      (0xD0, 0xEF, 0xAA, 0xFB, 0x43, 0x4D, 0x33, 0x85, 0x45, 0xF9, 0x02, 0x7F, 0x50, 0x3C, 0x9F, 0xA8),
      (0x51, 0xA3, 0x40, 0x8F, 0x92, 0x9D, 0x38, 0xF5, 0xBC, 0xB6, 0xDA, 0x21, 0x10, 0xFF, 0xF3, 0xD2),
      (0xCD, 0x0C, 0x13, 0xEC, 0x5F, 0x97, 0x44, 0x17, 0xC4, 0xA7, 0x7E, 0x3D, 0x64, 0x5D, 0x19, 0x73),
      (0x60, 0x81, 0x4F, 0xDC, 0x22, 0x2A, 0x90, 0x88, 0x46, 0xEE, 0xB8, 0x14, 0xDE, 0x5E, 0x0B, 0xDB),
      (0xE0, 0x32, 0x3A, 0x0A, 0x49, 0x06, 0x24, 0x5C, 0xC2, 0xD3, 0xAC, 0x62, 0x91, 0x95, 0xE4, 0x79),
      (0xE7, 0xC8, 0x37, 0x6D, 0x8D, 0xD5, 0x4E, 0xA9, 0x6C, 0x56, 0xF4, 0xEA, 0x65, 0x7A, 0xAE, 0x08),
      (0xBA, 0x78, 0x25, 0x2E, 0x1C, 0xA6, 0xB4, 0xC6, 0xE8, 0xDD, 0x74, 0x1F, 0x4B, 0xBD, 0x8B, 0x8A),
      (0x70, 0x3E, 0xB5, 0x66, 0x48, 0x03, 0xF6, 0x0E, 0x61, 0x35, 0x57, 0xB9, 0x86, 0xC1, 0x1D, 0x9E),
      (0xE1, 0xF8, 0x98, 0x11, 0x69, 0xD9, 0x8E, 0x94, 0x9B, 0x1E, 0x87, 0xE9, 0xCE, 0x55, 0x28, 0xDF),
      (0x8C, 0xA1, 0x89, 0x0D, 0xBF, 0xE6, 0x42, 0x68, 0x41, 0x99, 0x2D, 0x0F, 0xB0, 0x54, 0xBB, 0x16))
    col = inp >> 4
    row = inp & 15
    return s[row][col]

#Calculate DDT
Table = numpy.array([[0 for row in range(256)] for col in range(256)]);

c2 = [0]*256
for p2 in range(256):
    c2[p2]=SBOX(p2)

for p1 in range(256):
    c1=SBOX(p1)
    for p2 in range(256):
        x = p1 ^ p2
        y = c1 ^ c2[p2]
        Table[x][y] += 1

#####

#uncomment to print properties of transpose
#Table = Table.transpose()

def printHists(table):
    for col in range(1, 256):
        printHist(table,col)
        print('')

def printHist(table,col):
    hist = [0] * 257
    for row in range(0, 256):
        val = table[row][col]
        val = abs(val)
        hist[val]+=1
    print('')
    for v in range(0, 257):
        if(hist[v] != 0):
            print('( ',v,', ',hist[v],') ',end='')

#####

#DDT histogram
print('Histograms of normalized (x256) table columns:')
printHists(Table)

#####
```

```
#histogram of DDT Fourier transform
print('Histograms of normalized (x256) Fourier transform of table columns:')
had = hadamard(256)
four = numpy.matmul(had, Table)
four = four.transpose()
printHists(four)
```

```
#####
```

```
#sorted eigenvalues of  $D^T D$ 
TableT = Table.transpose()
resTN = numpy.matmul(TableT, Table)
eigenvalues, eigenvectors = numpy.linalg.eig((resTN))
sortedE = numpy.sort(eigenvalues)
print("Normalized (x2562) eigenvalues:")
print(sortedE)
```

B.2 Code for Lemma 5 and Lemma 11

```
#!/bin/python
from decimal import Decimal, getcontext
import math
from math import comb
import itertools
```

```
getcontext().prec = 64
```

```
#####
```

```
#general code for bounding collision probability using spectral decomposition as in paper
```

```
def mainTermNoPattern(N, lambda_p, Lc):
    term = Decimal(0)
    for j in range(1, 5):
        term += Decimal(comb(4, j)) * (lambda_p ** j) * ((N ** j) * Lc[j])
    return (N ** (-4)) * (term)
```

```
def mainTermPattern(N, lambda_p, pp):
    term1 = pp ** 2
    term2 = Decimal(0)
    for j in range(1, 5):
        term2 += Decimal(comb(4, j)) * (lambda_p ** j) * (0 - term1)
    return (N ** (-4)) * (term1 + term2)
```

```
def mainTerm(N, lambda_p, Lc, pp):
    return mainTermPattern(N, lambda_p, pp) + mainTermNoPattern(N, lambda_p, Lc)
```

```
def secondaryTermNoPattern(N, lambda_p, d, Lc, Lm):
    term2 = Decimal(0)
    for j in range(1, 4 - d + 1):
        term2 += Decimal(comb(4 - d, j)) * (lambda_p ** j) * ((N ** j) * Lm[j + d] * Lm[d])
    return (N ** (d-4)) * (term2)
```

```
def secondaryTermPattern(N, lambda_p, d, pp):
    term1 = pp ** 2
    term2 = Decimal(0)
    for j in range(1, 4 - d + 1):
        term2 += Decimal(comb(4 - d, j)) * (lambda_p ** j) * (0 - term1)
    return (N ** (d-4)) * (term1 + term2)
```

```
def secondaryTerm(N, lambda_p, d, Lc, Lm, pp):
    return secondaryTermPattern(N, lambda_p, d, pp) + secondaryTermNoPattern(N, lambda_p, d, Lc, Lm)
```

```
def allPossibleTightMaxConstraints(maxCoeffs, active):
    possibleTightMaxConstraints = []
    currC = [0, 0, 0, 0]
    end = 0
```

```

while True:
    if(end == 1):
        return possibleTightMaxConstraints
    currCCopy = [currC[0], currC[1], currC[2], currC[3]]
    possibleTightMaxConstraints.append(currCCopy)
    for i in range(0, active):
        if(currC[i] < maxCoeffs[i]):
            currC[i] = currC[i] + 1
            break
        #currC[i] == maxCoeffs[i]
        if(i == active - 1):
            end = 1
            break
        #i < active - 1
        currC[i] = 0
        continue

```

```

def calculatePatternPrForVertex(N, lambda_p, active, LmaxP, LminP, maxCoeffs, tightMaxConstraints, determinedVar):
    if(tightMaxConstraints[determinedVar] == maxCoeffs[determinedVar]):
        return -1
    prSum = 0
    for i in range(0, active):
        prSum += tightMaxConstraints[i] * LmaxP[i]
        prSum += (maxCoeffs[i] - tightMaxConstraints[i] - 1) * LminP[i]
        if(i != determinedVar):
            prSum += LminP[i]
    remainingPr = 1 - prSum
    #allow for minor slack to avoid precision issues
    slack = Decimal(2) ** (-128)
    if(remainingPr < 0 - slack):
        return -1
    remainingPr = max(remainingPr, 0)
    determinedVarVal = min(remainingPr, LmaxP[determinedVar])
    if(determinedVarVal < LminP[determinedVar] - slack):
        return -1

    #calculate current value
    vertexPr = Decimal(0)

    vertexPr += tightMaxConstraints[0] * mainTermPattern(N, lambda_p, LmaxP[0])
    vertexPr += (maxCoeffs[0] - tightMaxConstraints[0] - 1) * mainTermPattern(N, lambda_p, LminP[0])
    if(determinedVar == 0):
        vertexPr += mainTermPattern(N, lambda_p, determinedVarVal)
    else:
        vertexPr += mainTermPattern(N, lambda_p, LminP[0])
        vertexPr += secondaryTermPattern(N, lambda_p, determinedVar, determinedVarVal)
    for i in range(1, active):
        vertexPr += tightMaxConstraints[i] * secondaryTermPattern(N, lambda_p, i, LmaxP[i])
        vertexPr += (maxCoeffs[i] - tightMaxConstraints[i] - 1) * secondaryTermPattern(N, lambda_p, i, LminP[i])
        if(i != determinedVar):
            vertexPr += secondaryTermPattern(N, lambda_p, i, LminP[i])
    return vertexPr

def bruteForceVertices(N, lambda_p, active, Lc, Lm, LmaxP, LminP):
    baseVal = mainTermNoPattern(N, lambda_p, Lc)
    for d in range(1, active):
        baseVal += Decimal(comb(4, d)) * secondaryTermNoPattern(N, lambda_p, d, Lc, Lm)

    opt = Decimal(0)
    maxCoeffs = [1, 4, 6, 4]
    possibleTightMaxConstraints = allPossibleTightMaxConstraints(maxCoeffs, active)
    for currTightMaxConstraints in possibleTightMaxConstraints:
        for determinedVar in range(0, active):
            currVal = calculatePatternPrForVertex(
                N, lambda_p, active, LmaxP, LminP, maxCoeffs, currTightMaxConstraints, determinedVar
            )
            if(currVal <= 0):
                continue
            currVal += baseVal

```

```

        opt = max(opt, currVal)

    return opt

#main function for calculating collision probability
def calcC(N, lambda_p, active, Lc, Lm, mn1, useBruteForce = 0):
    maxP0 = 1 - (4 * mn1) + (comb(4, 2) * Lm[2])
    maxP0 = min(1, maxP0)
    LmaxP = [maxP0, Lm[1], Lm[2], Lm[3]]
    if (useBruteForce == 1):
        minP1 = mn1 - (3 * Lm[2]) - (3 * Lm[3])
        minP1 = max(minP1, 0)
        LminP = [0, minP1, 0, 0]
        return bruteForceVertices(N, lambda_p, active, Lc, Lm, LmaxP, LminP)
    return calcCforParams(N, lambda_p, active, Lc, Lm, LmaxP)

def calcCforParams(N, lambda_p, active, Lc, Lm, Lpp):
    term = mainTerm(N, lambda_p, Lc, Lpp[0])
    for d in range(1, active):
        term += Decimal(comb(4, d)) * secondaryTerm(N, lambda_p, d, Lc, Lm, Lpp[d])
    return term

def normalize(N, v):
    v1 = (v * ((N + 1)** 4)) - 1
    return 1/v1

#####

#Sbox params

N1 = Decimal(255)
lambda1 = ((Decimal(18.0)/256) ** 2)
n_inf = (Decimal(4.0)/256)

#####

#code for calculating Sbox norms and generating parameters for analyzing the super-Sbox

def calcNorm(p):
    return (Decimal(126) + 2 ** p)/(Decimal(128) ** p)

def calcFNorm(p):
    val = 0
    val += Decimal(84) * (8 ** p)
    val += Decimal(74) * (16 ** p)
    val += Decimal(52) * (24 ** p)
    val += Decimal(13) * (32 ** p)
    val = val/(Decimal(256) ** p)
    val += 1
    return val

#M_t^(ac)
def generateSMParm(ac, t):
    s = min(ac, t)
    b = n_inf ** s
    if (t == ac - 1):
        b = min(b, calcNorm(ac))
    if (t == 1 and ac > 1):
        b = min(b, (Decimal(2) ** (-8)) * calcFNorm(ac))
    if (t == 2 and ac == 4):
        b = min(b, (Decimal(2) ** (-16)) * (calcFNorm(2) ** 2))
    return b

#C_t^(ac)
def generateSColParam(ac, t):
    s = min(ac, t)
    b = calcNorm(2) ** s
    if (t == 1 and ac > 1):
        b = min(b, (Decimal(2) ** (-8)) * calcFNorm(2 * ac))

```

```

    if(t == 2 and ac == 4):
        b = min(b, (Decimal(2) ** (-16)) * (calcFNorm(4) ** 2))
    return b

def generateSMN1Param(ac):
    b = Decimal(2) ** (-8) - ((Decimal(2) ** (-8)) * (calcFNorm(ac) - 1))
    return max(b, Decimal(0))

def generateSPParams(ac):
    _Lc = [None, None, None, None, None]
    for t in range(1, 5):
        _Lc[t] = generateSColParam(ac, t)
    _Lm = [None, None, None, None, None]
    for t in range(1, 5):
        _Lm[t] = generateSMPParam(ac, t)
    _mn1 = generateSMN1Param(ac)
    return (_Lc, _Lm, _mn1)

#####

print("Printing super-Sbox DDT collision bounds by number of active Sboxes.")
print("Result p is normalized as 1/((p * 2^{32}) - 1).")
for ac in range(1, 5):
    useBruteForce = 0
    if(ac > 1):
        useBruteForce = 1
    (_Lc, _Lm, _mn1) = generateSPParams(ac)
    pr = calcC(N1, lambda1, ac, _Lc, _Lm, _mn1, useBruteForce)
    prN = normalize(N1, pr)
    print("active = ", ac)
    print(prN)

#####

# super-Sbox params

N2 = (Decimal(256) ** 4) - 1
lambda2 = Decimal(426) ** (-1)

# super-Sbox norms

spSn_inf = Decimal(53) * (Decimal(2) ** (-34))

u1spSn2 = Decimal(1)/Decimal(N2+1) * (1 + Decimal(5) ** (-1))
u2spSn2 = Decimal(1)/Decimal(N2+1) * (1 + Decimal(24) ** (-1))
u3spSn2 = Decimal(1)/Decimal(N2+1) * (1 + Decimal(283.4) ** (-1))
u4spSn2 = Decimal(1)/Decimal(N2+1) * (1 + Decimal(438) ** (-1))

spSn2 = [None, u1spSn2, u2spSn2, u3spSn2, u4spSn2]

#####

#code for generating parameters for theorem for analyzing 4-round AES

def calcFNormspS(Np, col, p1, p2):
    main = ((Np * (col ** p1)) - 1) ** p2
    return (1/Np) + (1/Np) * main

def generatespSPParams(Np, n2val, ninfval):
    #C_t^(U) (only depends on 2-norm param)
    _Lc = [None, None, None, None, None]
    _Lc[1] = calcFNormspS(Np, n2val, 1, 4)
    _Lc[2] = calcFNormspS(Np ** 2, n2val, 2, 2)
    _Lc[3] = n2val ** 3
    _Lc[4] = n2val ** 4

```



```

#M_t^(U) (only depends on 2-norm and inf-norm params)
_Lm = [None, None, None, None, None]
_Lm[1] = calcFNormspS(Np, n2val, 1, 2)
_Lm[2] = n2val ** 2
_Lm[3] = ninfval ** 3
_Lm[4] = ninfval ** 4

temp = ((Np * n2val) - 1) ** 2
mn1 = (1/Np) - (1/Np) * temp

return (_Lc, _Lm, mn1)

#####

print("Printing 4-round AES DDT collision bounds by parameter U.")
print("Result p is normalized as the log2 of 1/((p * 2^{128}) - 1).")
useBruteForce = 1
for U in range(1, 5):
    (a_Lc, a_Lm, mn1) = generatespSPParams(N2 + 1, spSn2[U], spSn_inf)
    _a_pr = calcC(N2, lambda2, 4, a_Lc, a_Lm, mn1, useBruteForce)
    _a_prN = normalize(N2, _a_pr)
    print("U = ", U)
    print(math.log2(_a_prN))

```

B.3 Code for Theorem 1

```

#!/bin/python
from decimal import Decimal, getcontext
from math import comb
import sys

getcontext().prec = 64

#####

def colF(m):
    return (Decimal(2) ** (-128)) * (Decimal(1.0142) + (Decimal(2) ** (Decimal(20.83))) * m)

#general code for bounding collision probability as in paper

def calcLz(active, Lm):
    Lz = [Decimal(0), Decimal(0), Decimal(0), Decimal(0)]
    for d in range(1, active):
        Lz[d] = Decimal(1)
    Lz[0] = Decimal(1)
    return Lz

#main function of calculating collision probability
def calcC(active, Lm, M, C):
    Lz = calcLz(active, Lm)
    term = Decimal(0)
    colFparam = M/(1 - 4 * M)
    term += (colF(colFparam))
    for d in range(1, 4):
        term += Lz[d] * Decimal(comb(4, d)) * (Lm[d] ** 2) * (C ** (4 - d))
    return term

def normalize(N, v):
    v1 = (v * ((N + 1) ** 4)) - 1
    return 1/v1

#####

```

```

#super-Sbox parameters and code for generating parameters for analyzing 4-round AES

CspS = (Decimal(2) ** (-32)) * (1 + Decimal(1)/Decimal(5.05))
MspS = Decimal(53) * (Decimal(2) ** (-34))
N2 = (Decimal(256) ** 4) - 1

```

```

def generatespSMPParam(ac,t):
    s = min(ac,t)
    b = MspS ** s
    return b

def generatespSParams(ac):
    a_Lm = [None, None , None , None , None]
    for t in range(1, 5):
        a_Lm[t] = generatespSMPParam(ac,t)
    return a_Lm

#####

print("Printing 4-round AES DDT collision bounds by number of active diagonals.")
print("Result p is normalized as 1/((p * 2^{128}) - 1).")
for ac in range(1, 5):
    Lm = generatespSParams(ac)
    pr = calcC(ac, Lm ,MspS,CspS)
    prN = normalize(N2, pr)
    print("active = ", ac)
    print(prN)

```